



(12) **United States Patent**
Mujica et al.

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(54) **MOUSE COMPRISING A HUMANIZED TRKB LOCUS**

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A01K 67/0278 (2024.01)
A01K 67/0271 (2024.01)
A61K 49/00 (2006.01)
C12N 5/10 (2006.01)
C12N 9/12 (2006.01)

(52) **U.S. Cl.**
CPC **A01K 67/0278** (2013.01); **A01K 67/0271** (2013.01); **A61K 49/0008** (2013.01); **C12N 5/10** (2013.01); **C12N 9/12** (2013.01); **C12Y 207/10001** (2013.01); **A01K 2207/15** (2013.01); **A01K 2217/072** (2013.01); **A01K 2227/105** (2013.01); **A01K 2267/03** (2013.01); **A01K 2267/0318** (2013.01); **C07K 2319/02** (2013.01); **C07K 2319/03** (2013.01); **C07K 2319/033** (2013.01); **C07K 2319/035** (2013.01)

(58) **Field of Classification Search**
None

See application file for complete search history.

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(57) **ABSTRACT**

Non-human animal genomes, non-human animal cells, and non-human animals comprising a humanized TRKB locus and methods of making and using such non-human animal genomes, non-human animal cells, and non-human animals are provided. Non-human animal cells or non-human animals comprising a humanized TRKB locus express a human TRKB protein or a chimeric transthyretin protein, fragments of which are from human TRKB. Methods are provided for using such non-human animals comprising a humanized TRKB locus to assess in vivo efficacy of human-TRKB-targeting reagents such as nuclease agents designed to target human TRKB.

14 Claims, 16 Drawing Sheets

Specification includes a Sequence Listing.

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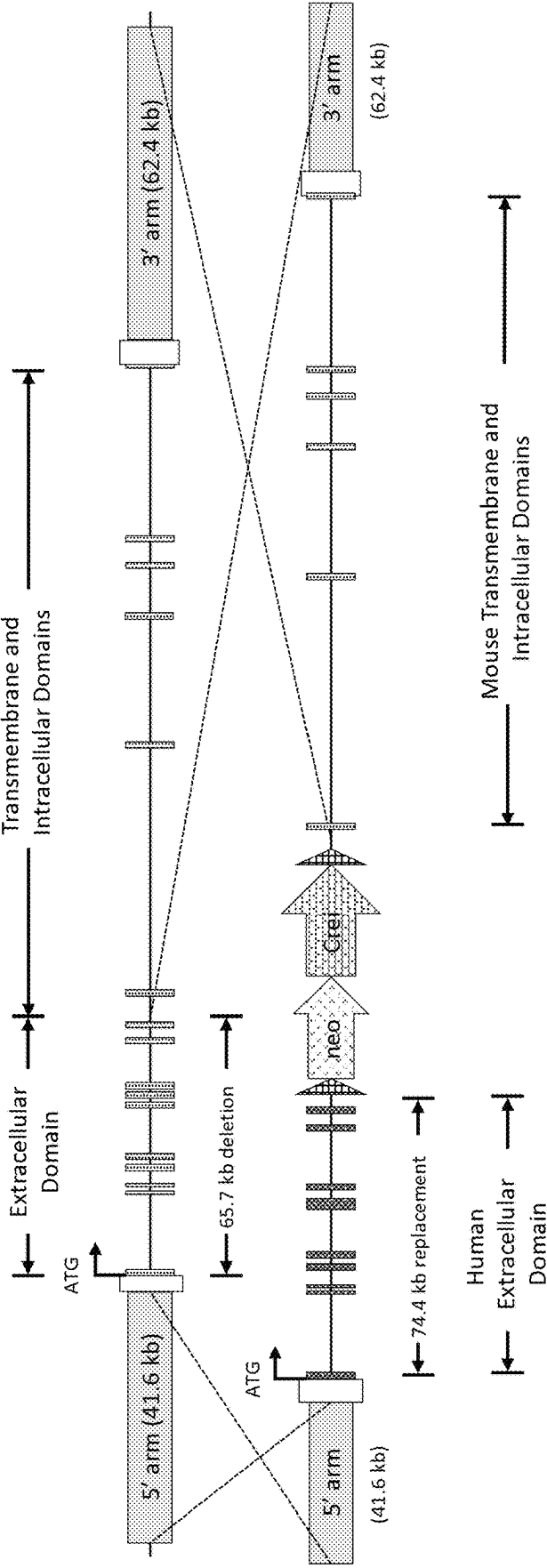
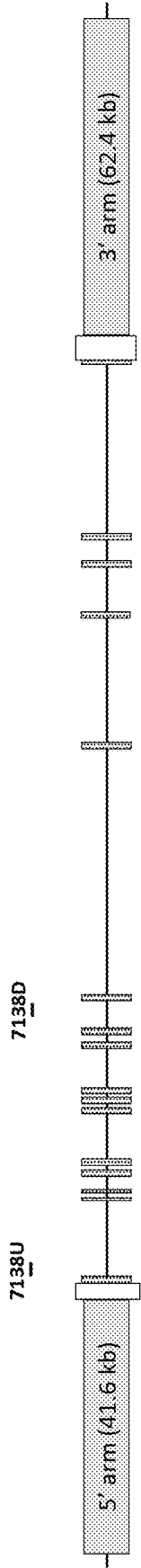


FIG. 1

Mouse *Ntrk2* Locus



Humanized Mouse *Ntrk2* Locus

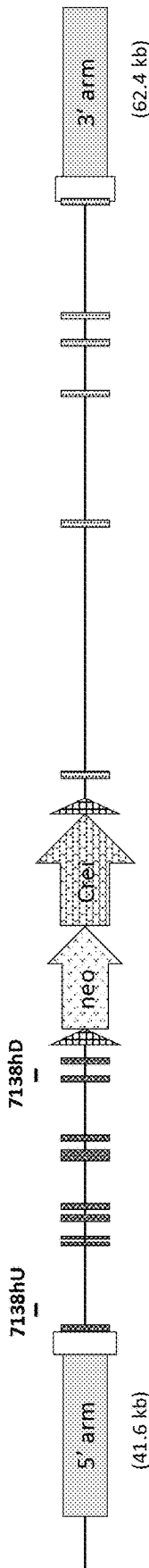


FIG. 2

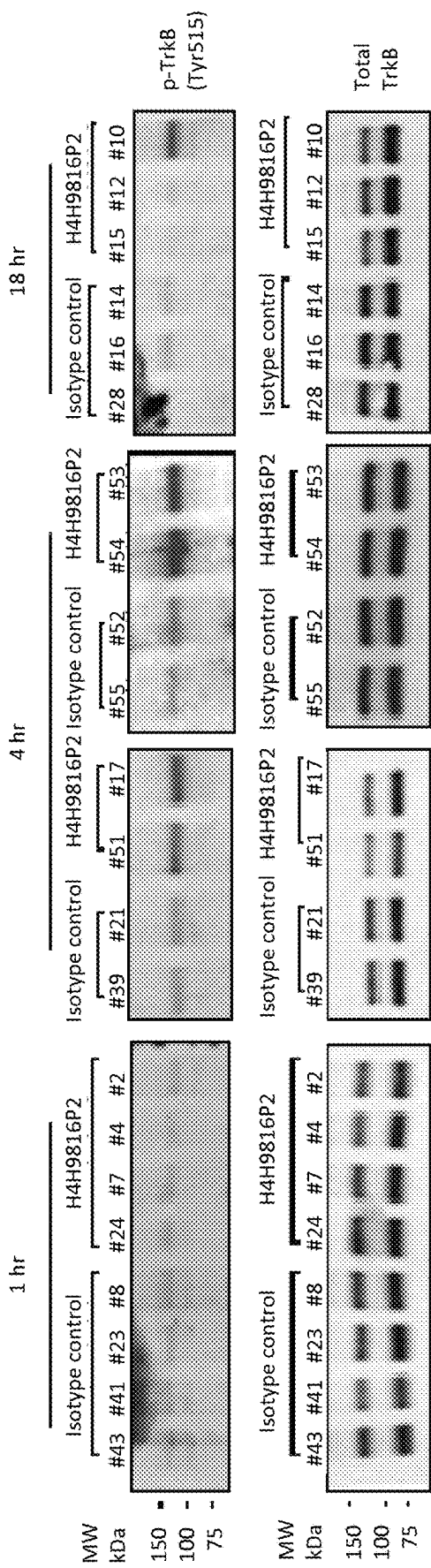


FIG. 3

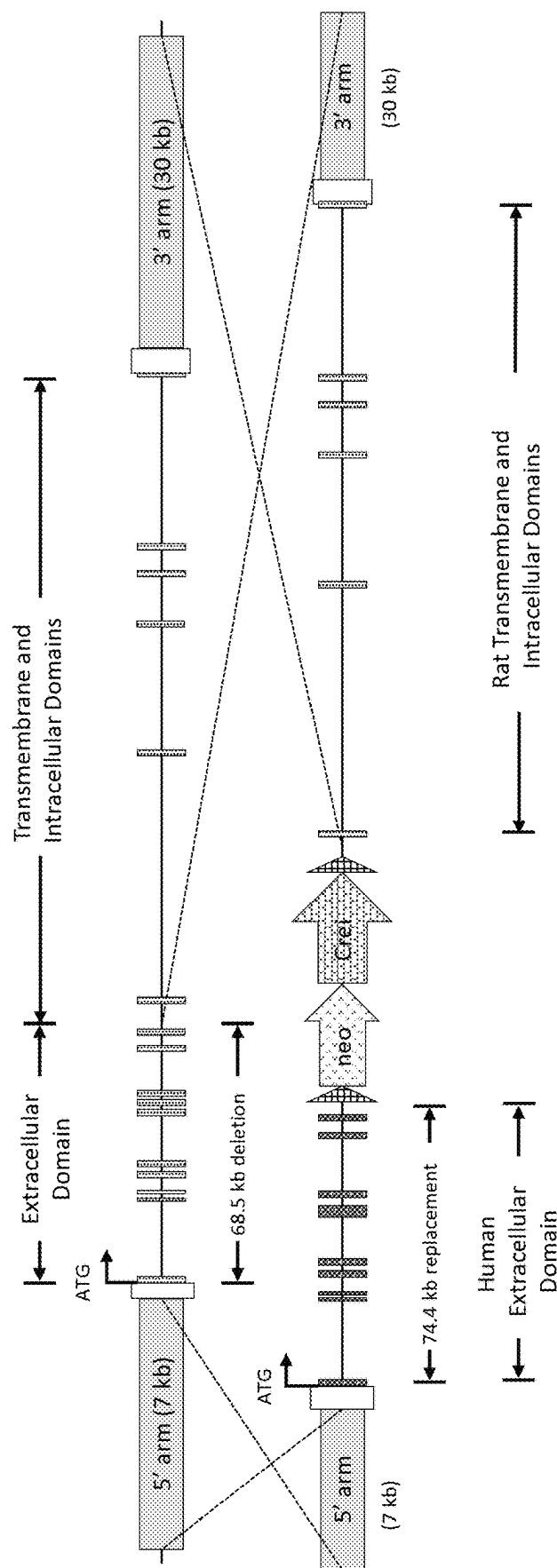
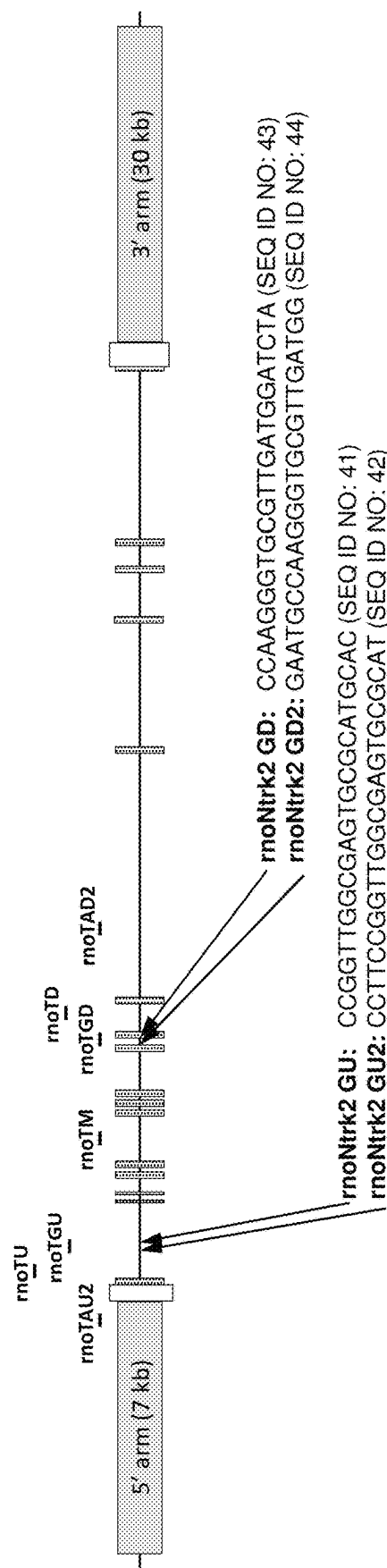


FIG. 4

Rat *Ntrk2* Locus



Humanized Rat *Ntrk2* Locus

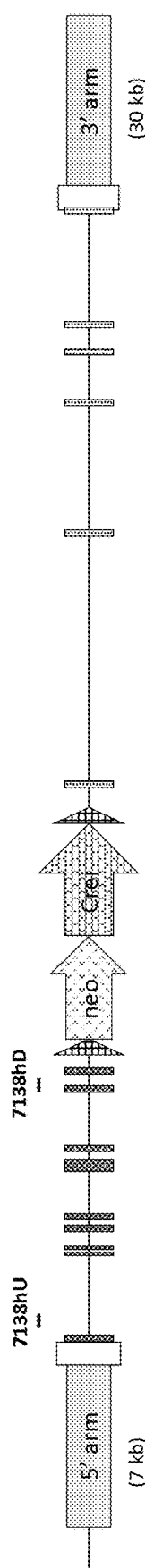
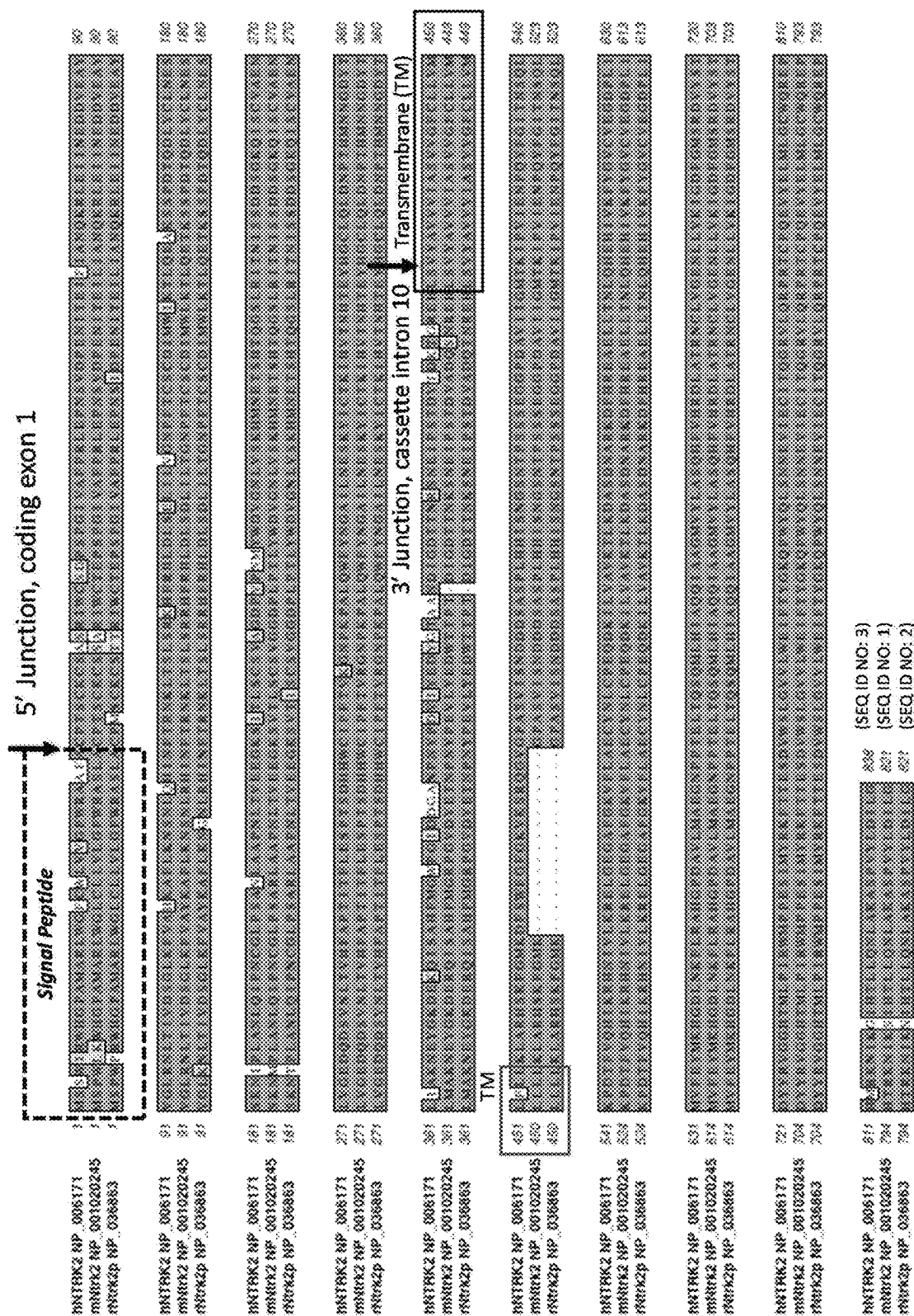


FIG. 5



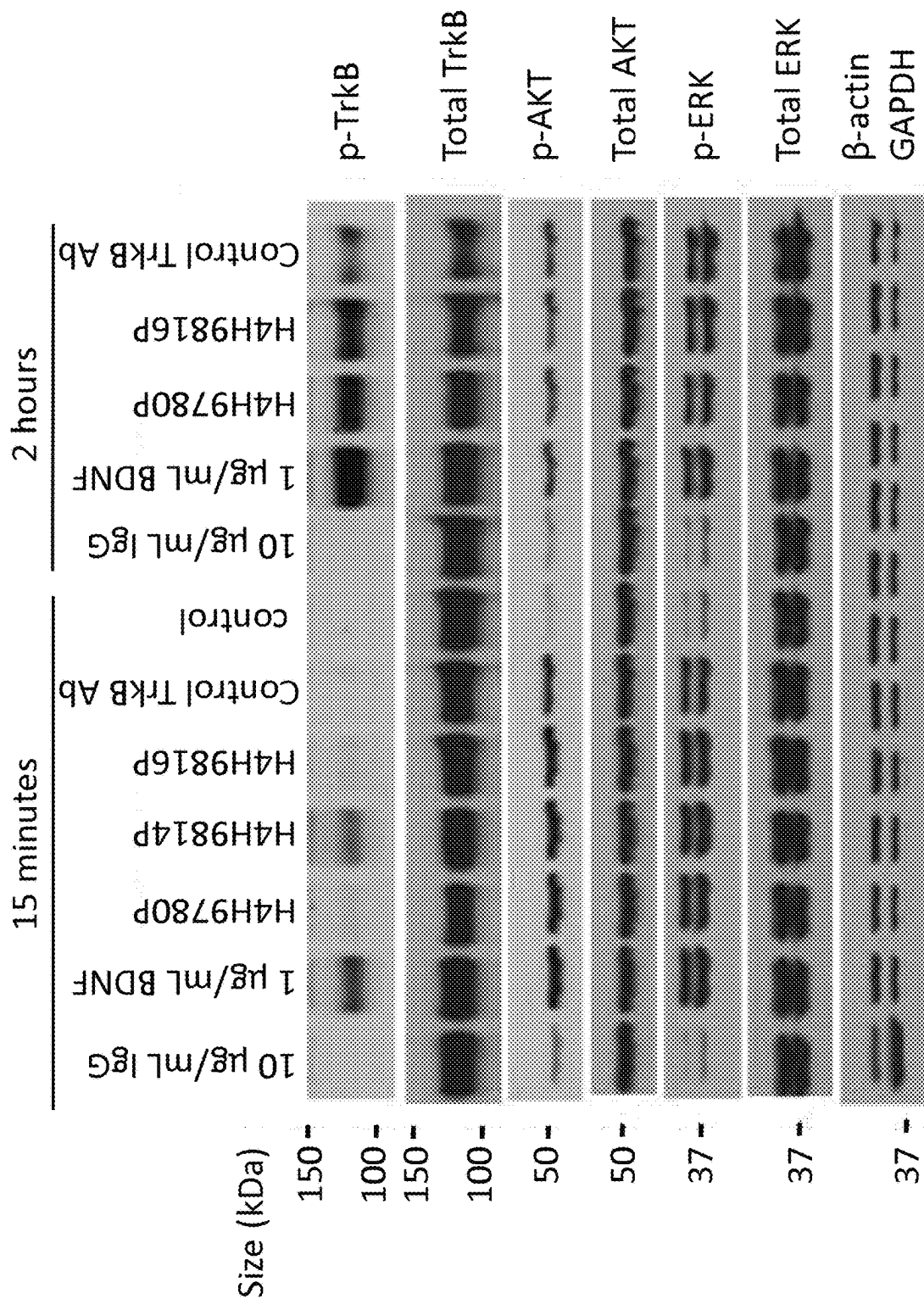


FIG. 7

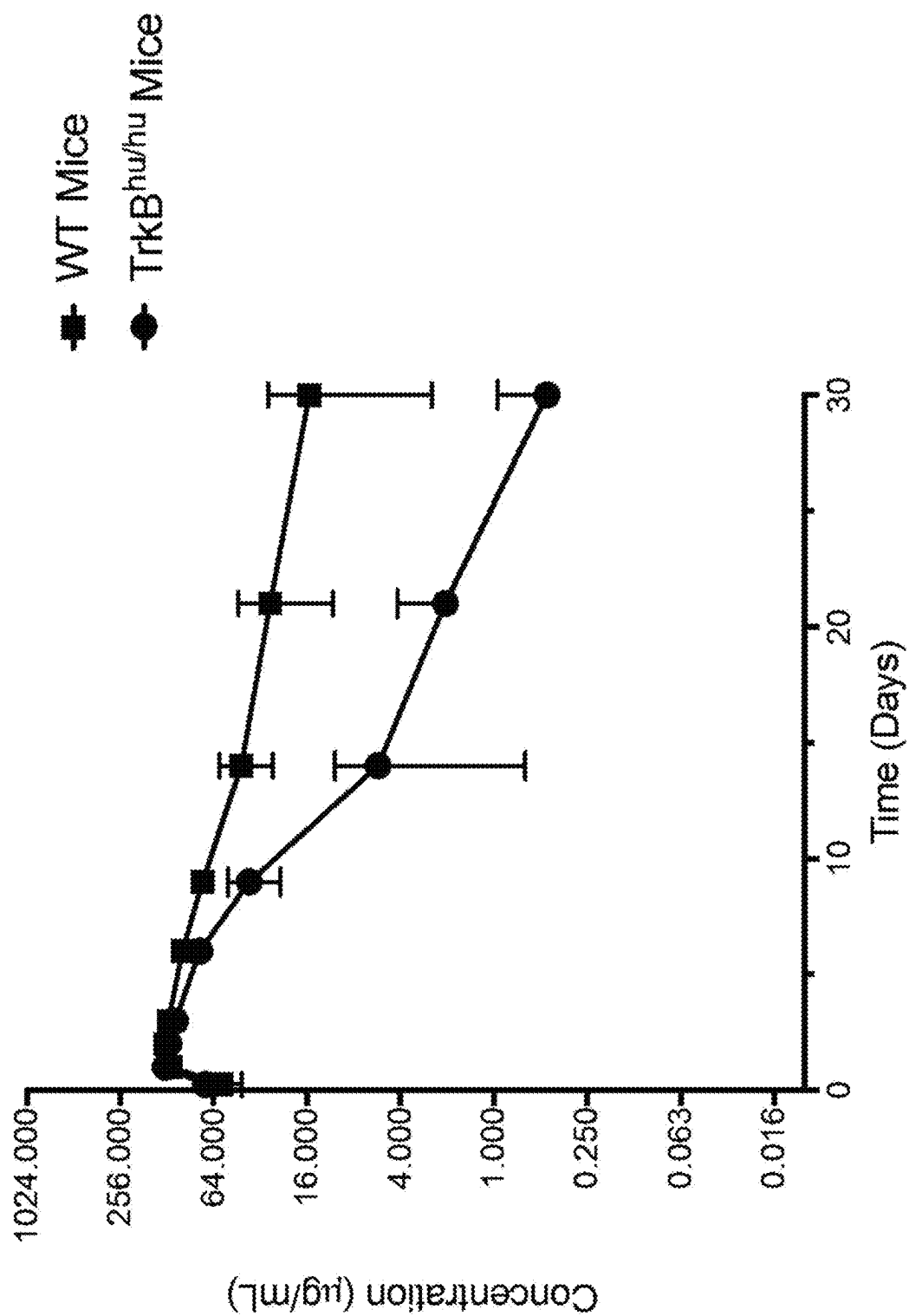


FIG. 8

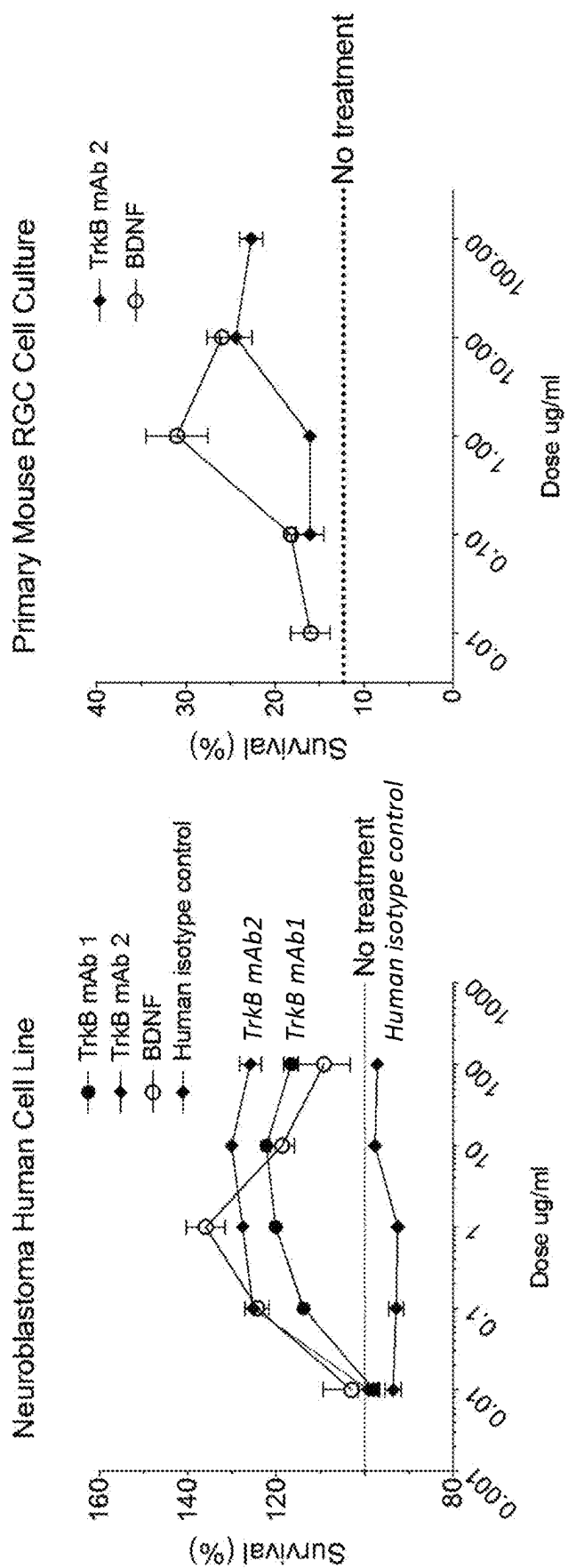


FIG. 9

FIG. 10

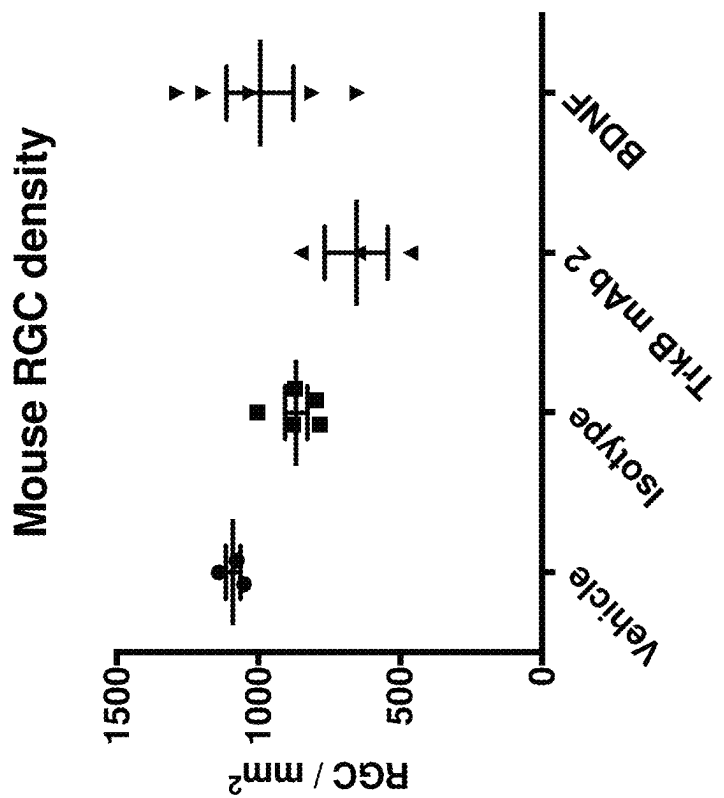


FIG. 11B

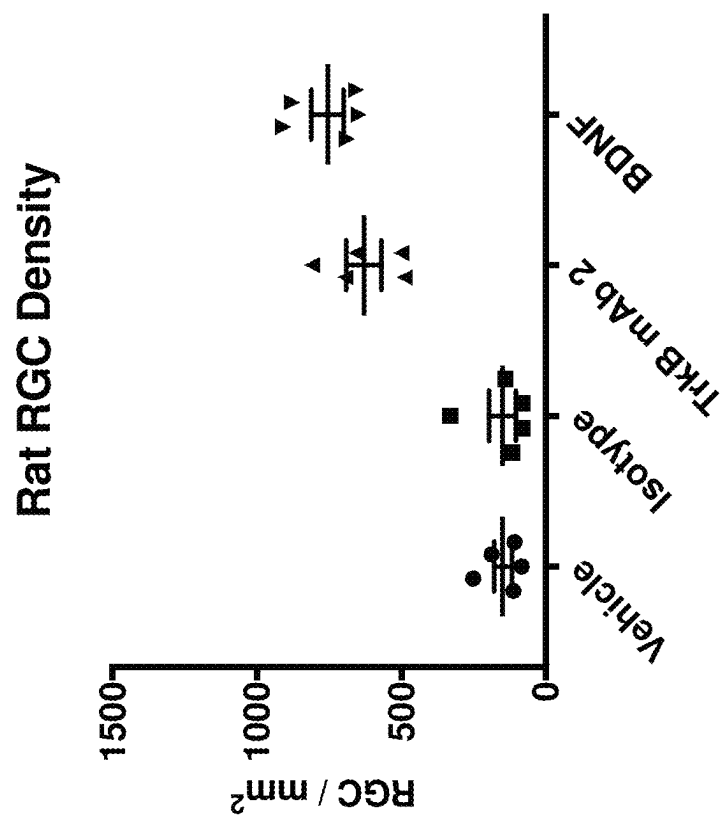


FIG. 11A

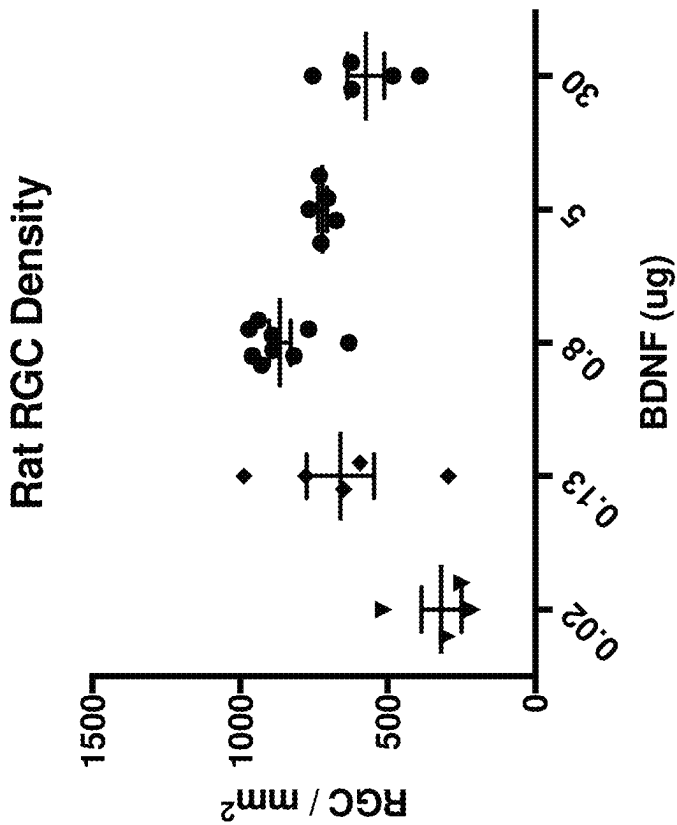


FIG. 12B

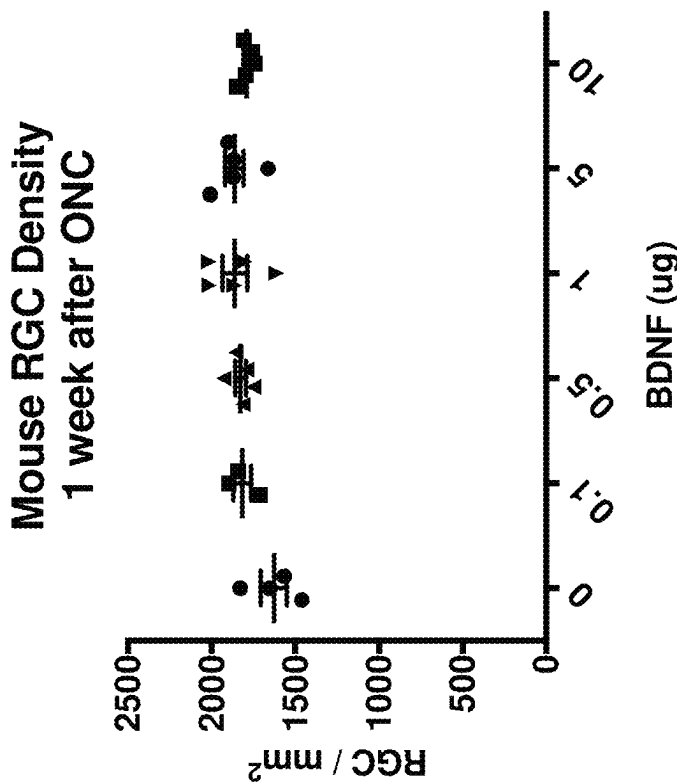


FIG. 12A

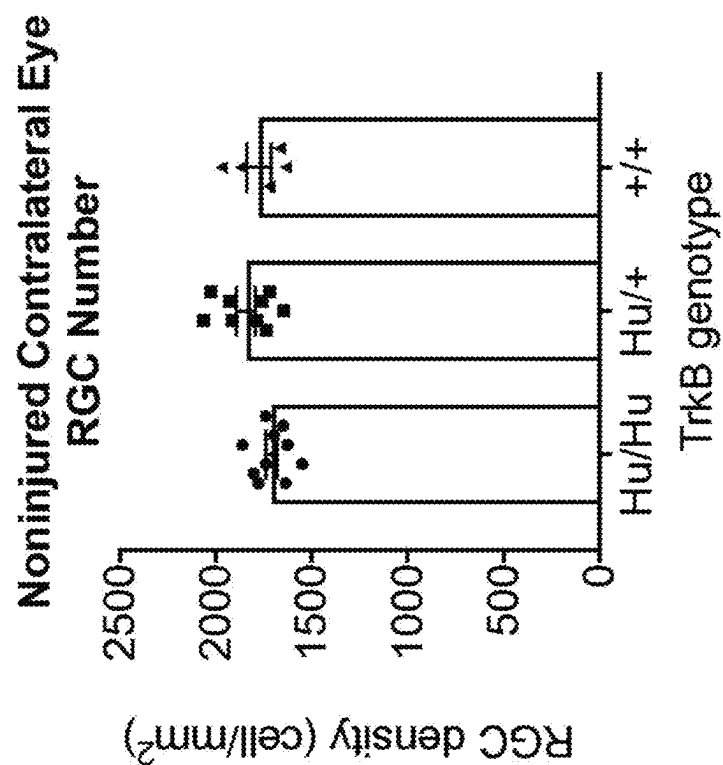


FIG. 13B

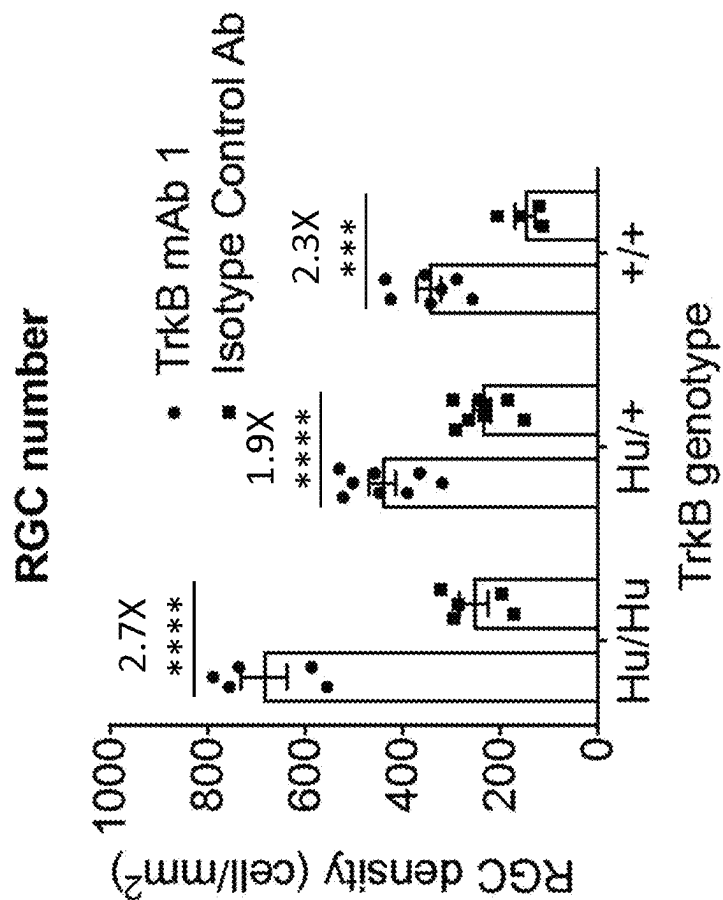
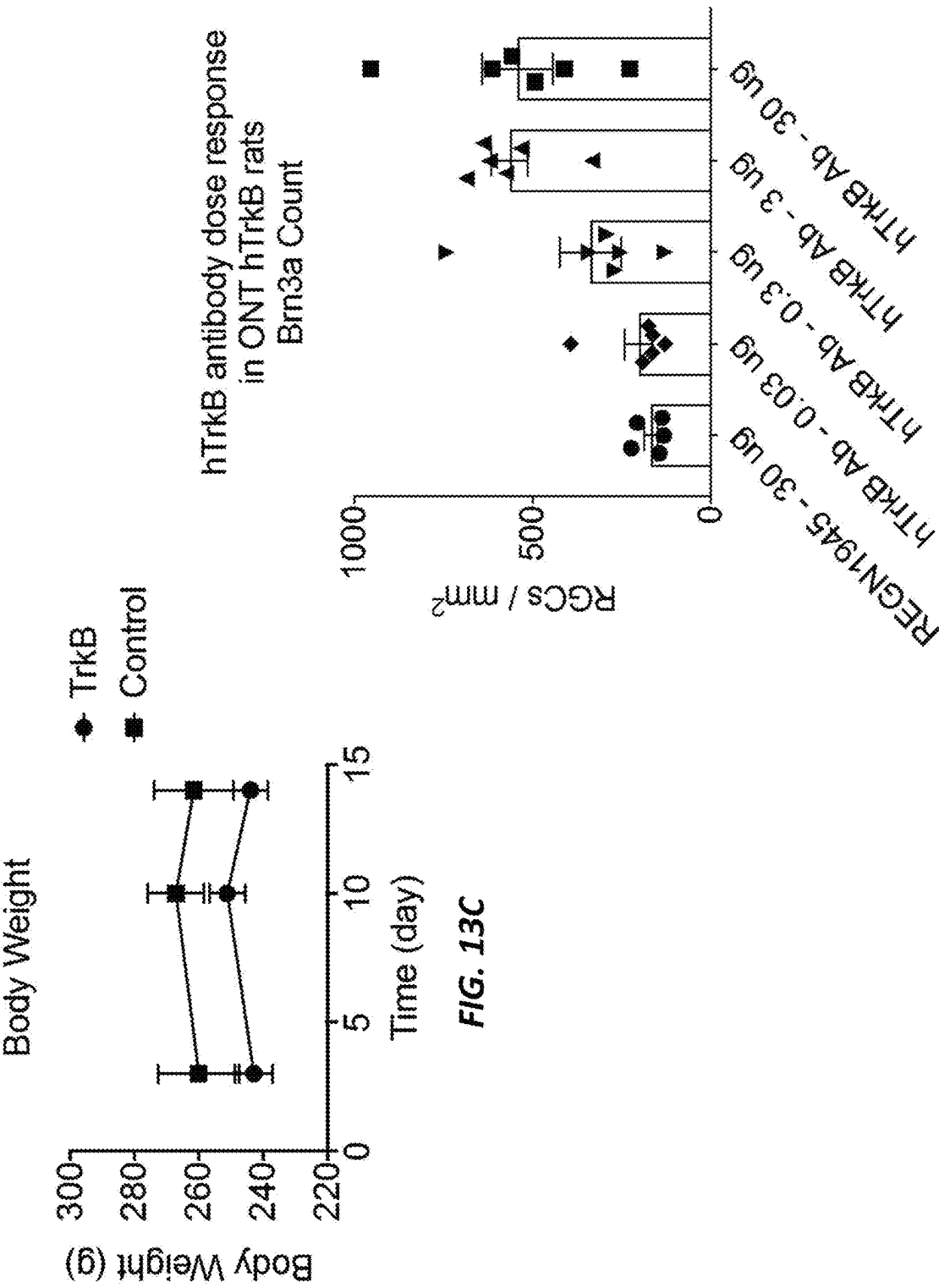


FIG. 13A



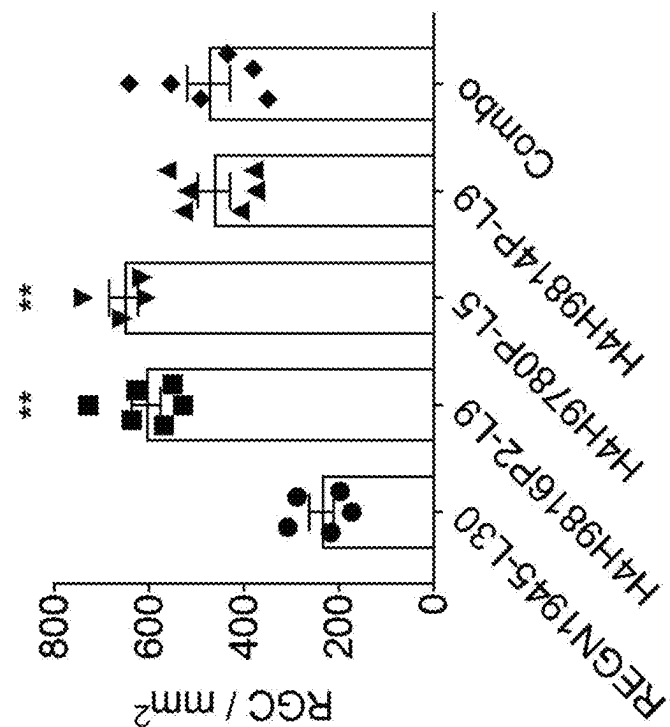


FIG. 15B

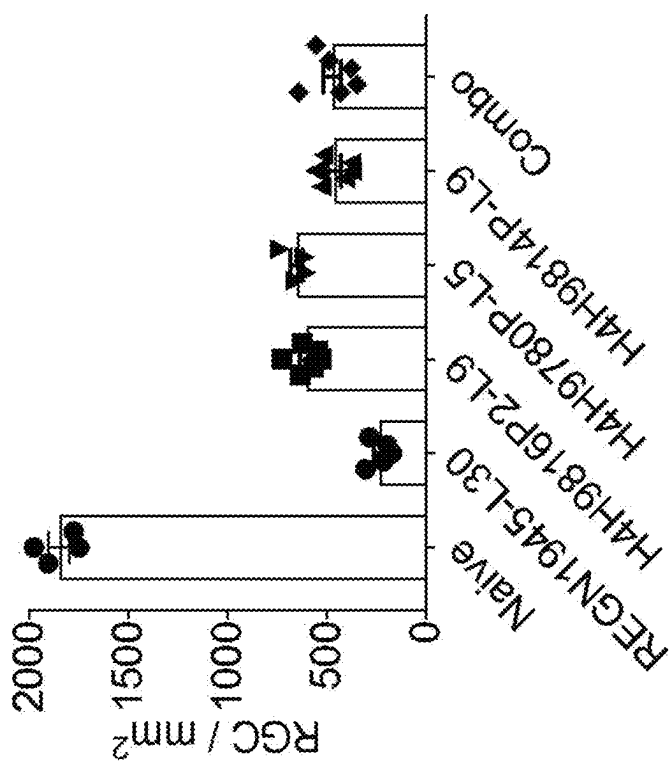


FIG. 15A

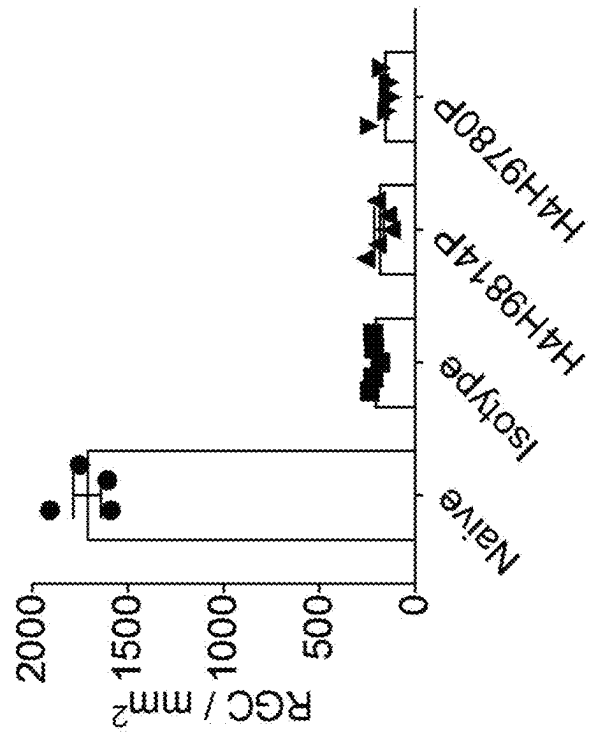


FIG. 16

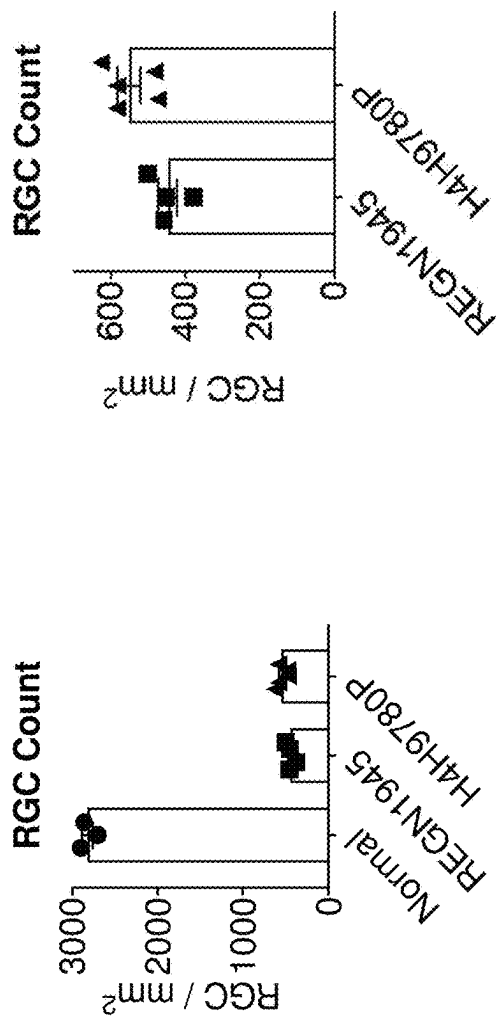


FIG. 17A

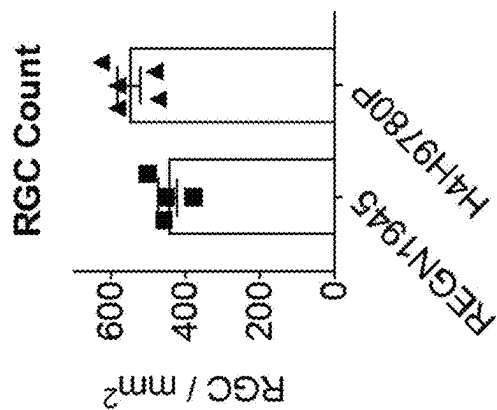


FIG. 17B

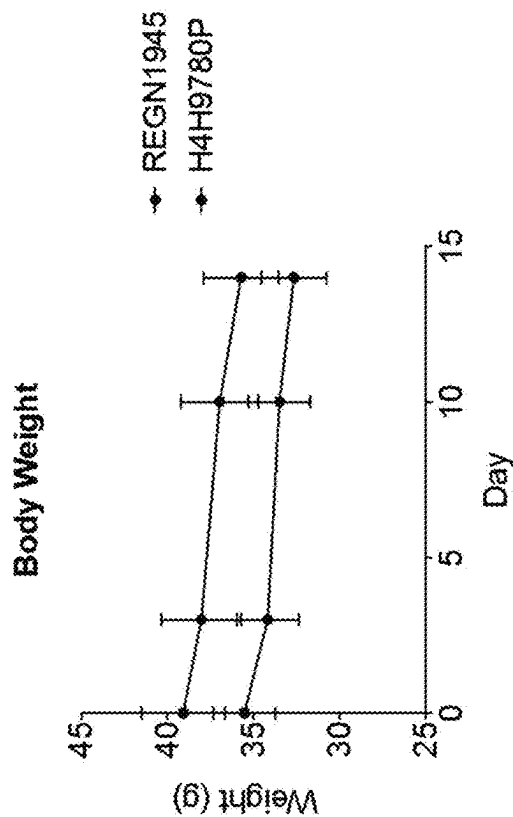


FIG. 17C

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**MOUSE COMPRISING A HUMANIZED
TRKB LOCUS****CROSS-REFERENCE TO RELATED
APPLICATIONS**

This application is a continuation of U.S. application Ser. No. 16/206,330, filed Nov. 30, 2018, which claims the benefit of U.S. Application No. 62/592,905, filed Nov. 30, 2017, and U.S. Application No. 62/661,373, filed Apr. 23, 2018, each of which is herein incorporated by reference in its entirety for all purposes.

**REFERENCE TO A SEQUENCE LISTING
SUBMITTED AS A TEXT FILE VIA EFS WEB**

The Sequence Listing written in file 580552SEQLIST.txt is 154 kilobytes, was created on Jun. 29, 2022, and is hereby incorporated by reference.

BACKGROUND

Tropomyosin receptor kinase B (TRKB) is a promising target for neuroprotection in neurodegenerative diseases such as glaucoma. TRKB is one of the most widely distributed neurotrophic receptors (NTRs) in the brain, which is highly enriched in the neocortex, hippocampus, striatum, and brainstem. Binding of brain-derived neurotrophic factor (BDNF) to TRKB receptor triggers its dimerization through conformational changes and autophosphorylation of tyrosine residues in the intracellular domain, resulting in activation of signaling pathways involving mitogen-activated protein kinase (MAPK), phosphatidylinositol 3-kinase (PI3K), and phospholipase C- γ (PLC- γ).

TRKB is important for neuronal survival, differentiation, and function, and TRKB agonists could have therapeutic potential in numerous neurological, mental, and metabolic disorders. However, there remains a need for suitable non-human animals providing the true human target or a close approximation of the true human target of human-TRKB-targeting reagents, thereby enabling testing of the efficacy and mode of action of such agents in live animals as well as pharmacokinetic and pharmacodynamics studies.

SUMMARY

Non-human animals comprising a humanized TRKB locus are provided, as well as methods of using such non-human animals. Non-human animal genomes or cells comprising a humanized TRKB locus are also provided.

In one aspect, provided are non-human animal genomes, non-human animal cells, or non-human animals comprising a humanized TRKB locus. Such non-human animal genomes, non-human animal cells, or non-human animals can comprise a genetically modified endogenous TrkB locus encoding a TRKB protein, wherein the TRKB protein comprises a cytoplasmic domain, a transmembrane domain, and an extracellular domain, and all or part of the extracellular domain is encoded by a segment of the endogenous TrkB locus that has been deleted and replaced with an orthologous human TRKB sequence.

In one aspect, provided are non-human animals comprising a humanized TrkB locus. Such non-human animals can comprise a genetically modified endogenous TrkB locus encoding a TRKB protein, wherein the TRKB protein comprises a cytoplasmic domain, a transmembrane domain, and an extracellular domain, and all or part of the extracellular

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domain is encoded by a segment of the endogenous TrkB locus that has been deleted and replaced with an orthologous human TRKB sequence.

In another aspect, provided are non-human animal cells comprising in their genome a genetically modified endogenous TrkB locus encoding a TRKB protein, wherein the TRKB protein comprises a cytoplasmic domain, a transmembrane domain, and an extracellular domain, and all or part of the extracellular domain is encoded by a segment of the endogenous TrkB locus that has been deleted and replaced with an orthologous human TRKB sequence.

In another aspect, provided are non-human animal genomes comprising a genetically modified endogenous TrkB locus encoding a TRKB protein, wherein the TRKB protein comprises a cytoplasmic domain, a transmembrane domain, and an extracellular domain, and all or part of the extracellular domain is encoded by a segment of the endogenous TrkB locus that has been deleted and replaced with an orthologous human TRKB sequence.

In some such non-human animal genomes, non-human animal cells, or non-human animals, the TRKB protein comprises a human TRKB extracellular domain. Optionally, the extracellular domain comprises the sequence set forth in SEQ ID NO: 60. Optionally, all of the extracellular domain is encoded by the segment of the endogenous TrkB locus that has been deleted and replaced with the orthologous human TRKB sequence, optionally wherein the coding sequence for the extracellular domain comprises the sequence set forth in SEQ ID NO: 72.

In some such non-human animal genomes, non-human animal cells, or non-human animals, the TRKB protein comprises an endogenous signal peptide. Optionally, the signal peptide comprises the sequence set forth in SEQ ID NO: 51 or 55. Optionally, all of the signal peptide is encoded by an endogenous TrkB sequence, optionally wherein the coding sequence for the signal peptide comprises the sequence set forth in SEQ ID NO: 63 or 67.

In some such non-human animal genomes, non-human animal cells, or non-human animals, the TRKB protein comprises an endogenous TRKB transmembrane domain. Optionally, the transmembrane domain comprises the sequence set forth in SEQ ID NO: 53 or 57. Optionally, all of the transmembrane domain is encoded by an endogenous TrkB sequence, optionally wherein the coding sequence for the transmembrane domain comprises the sequence set forth in SEQ ID NO: 65 or 69.

In some such non-human animal genomes, non-human animal cells, or non-human animals, the TRKB protein comprises an endogenous TRKB cytoplasmic domain. Optionally, the cytoplasmic domain comprises the sequence set forth in SEQ ID NO: 54 or 58. Optionally, all of the cytoplasmic domain is encoded by an endogenous TrkB sequence, optionally wherein the coding sequence for the cytoplasmic domain comprises the sequence set forth in SEQ ID NO: 66 or 70.

In some such non-human animal genomes, non-human animal cells, or non-human animals, the TRKB protein comprises an endogenous TRKB signal peptide, an endogenous TRKB transmembrane domain, and an endogenous TRKB cytoplasmic domain. Optionally, the signal peptide comprises the sequence set forth in SEQ ID NO: 51, the transmembrane domain comprises the sequence set forth in SEQ ID NO: 53, and the cytoplasmic domain comprises the sequence set forth in SEQ ID NO: 54. Optionally, the signal peptide comprises the sequence set forth in SEQ ID NO: 55, the transmembrane domain comprises the sequence set forth in SEQ ID NO: 57, and the cytoplasmic domain comprises

the sequence set forth in SEQ ID NO: 58. Optionally, all of the signal peptide, all of the transmembrane domain, and all of the cytoplasmic domain are encoded by an endogenous TrkB sequence. Optionally, the coding sequence for the signal peptide comprises the sequence set forth in SEQ ID NO: 63, the coding sequence for the transmembrane domain comprises the sequence set forth in SEQ ID NO: 65, and the coding sequence for the cytoplasmic domain comprises the sequence set forth in SEQ ID NO: 66. Optionally, the coding sequence for the signal peptide comprises the sequence set forth in SEQ ID NO: 67, the coding sequence for the transmembrane domain comprises the sequence set forth in SEQ ID NO: 69, and the coding sequence for the cytoplasmic domain comprises the sequence set forth in SEQ ID NO: 70.

In some such non-human animal genomes, non-human animal cells, or non-human animals, the TRKB protein is a chimeric non-human animal/human TRKB protein. Optionally, the extracellular domain is a human TRKB extracellular domain, the transmembrane domain is an endogenous TRKB protein transmembrane domain, and the cytoplasmic domain is an endogenous TRKB protein cytoplasmic domain. Optionally, the TRKB protein comprises the sequence set forth in SEQ ID NO: 4 or 5. Optionally, the coding sequence of the genetically modified TrkB locus encoding the TRKB protein comprises the sequence set forth in SEQ ID NO: 12 or 13.

Some such non-human animal genomes, non-human animal cells, or non-human animals are heterozygous for the genetically modified endogenous TrkB locus. Some such non-human animal genomes, non-human animal cells, or non-human animals are homozygous for the genetically modified endogenous TrkB locus.

Some such non-human animals or mammals. Optionally, the non-human animal is a rodent. Optionally, the rodent is a rat or mouse.

Some such non-human animals are rats. Optionally, the TRKB protein comprises the sequence set forth in SEQ ID NO: 5. Optionally, the coding sequence of the genetically modified TrkB locus encoding the TRKB protein comprises the sequence set forth in SEQ ID NO: 13.

Some such non-human animals are mice. Optionally, the TRKB protein comprises the sequence set forth in SEQ ID NO: 4. Optionally, the coding sequence of the genetically modified TrkB locus encoding the TRKB protein comprises the sequence set forth in SEQ ID NO: 12.

In another aspect, provided are methods of assessing the activity of a human-TRKB-targeting reagent in vivo using the above non-human animals. Some such methods comprise: (a) administering the human-TRKB-targeting reagent to the non-human animal; and (b) assessing the activity of the human-TRKB-targeting reagent in the non-human animal.

In some such methods, the assessed activity is neuroprotective activity.

In some such methods, step (a) comprises injecting the human-TRKB-targeting reagent to the non-human animal.

In some such methods, step (b) comprises assessing changes in one or more or all of body weight, body composition, metabolism, and locomotion relative to a control non-human animal. Optionally, the assessing changes in body composition comprises assessing lean mass and/or fat mass relative to a control non-human animal. Optionally, the assessing changes in metabolism comprises assessing changes in food consumption and/or water consumption.

In some such methods, step (b) comprises assessing TRKB phosphorylation and/or activation of the MAPK/ERK and PI3K/Akt pathways relative to a control non-human animal.

In some such methods, step (b) comprises assessing neuroprotective activity. In some such methods, step (b) comprises assessing neuroprotective activity, and the non-human animal is a rat. In some such methods, step (b) comprises assessing retinal ganglion cell viability. Optionally, assessing retinal ganglion cell viability comprises assessing retinal ganglion cell density. Optionally, retinal ganglion cell density is measured in dissected retinas stained for retinal ganglion cells. Optionally, retinal ganglion cell viability is assessed in a complete optic nerve transection model after optic nerve injury. Optionally, retinal ganglion cell viability is assessed in an optic nerve crush model.

In some such methods, the human-TRKB-targeting reagent is an antigen-binding protein. Optionally, the antigen-binding protein is a human TRKB agonist antibody. In some such methods, the human-TRKB-targeting reagent is a small molecule. Optionally, the small molecule is a human TRKB agonist.

In another aspect, provided are targeting vectors for generating a genetically modified endogenous TrkB locus encoding a TRKB protein, wherein the TRKB protein comprises a cytoplasmic domain, a transmembrane domain, and an extracellular domain, and all or part of the extracellular domain is encoded by a segment of the endogenous TrkB locus that has been deleted and replaced with an orthologous human TRKB sequence, and wherein the targeting vector comprises an insert nucleic acid comprising the orthologous human TRKB sequence flanked by a 5' homology arm targeting a 5' target sequence at the endogenous TrkB locus and a 3' homology arm targeting a 3' target sequence at the endogenous TrkB locus.

In another aspect, provided are methods of making any of the non-human animals described above. Some such methods can comprise (a) introducing into a non-human animal pluripotent cell that is not a one-cell stage embryo: (i) an exogenous repair template comprising an insert nucleic acid flanked by a 5' homology arm that hybridizes to a 5' target sequence at the endogenous TrkB locus and a 3' homology arm that hybridizes to a 3' target sequence at the endogenous TrkB locus, wherein the insert nucleic acid comprises the orthologous human TRKB sequence; and (ii) a nuclease agent targeting a target sequence within the endogenous TrkB locus, wherein the genome is modified to comprise the genetically modified endogenous TrkB locus; (b) introducing the modified non-human animal pluripotent cell into a host embryo; and (c) implanting the host embryo into a surrogate mother to produce a genetically modified F0 generation non-human animal comprising the genetically modified endogenous TrkB locus. Optionally, the pluripotent cell is an embryonic stem (ES) cell. Optionally, the nuclease agent is a Cas9 protein and a guide RNA that targets a guide RNA target sequence within the endogenous TrkB locus. Optionally, step (a) further comprises introducing into the non-human animal pluripotent cell a second guide RNA that targets a second guide RNA target sequence within the endogenous TrkB locus. Optionally, the exogenous repair template is a large targeting vector that is at least 10 kb in length, or wherein the exogenous repair template is a large targeting vector in which the sum total of the 5' homology arm and the 3' homology arm is at least 10 kb in length.

Some such methods comprise: (a) introducing into a non-human animal one-cell stage embryo: (i) an exogenous

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repair template comprising an insert nucleic acid flanked by a 5' homology arm that hybridizes to a 5' target sequence at the endogenous TrkB locus and a 3' homology arm that hybridizes to a 3' target sequence at the endogenous TrkB locus, wherein the insert nucleic acid comprises the orthologous human TRKB sequence; and (ii) a nuclease agent targeting a target sequence within the endogenous TrkB locus, wherein the genome is modified to comprise the genetically modified endogenous TrkB locus; and (b) implanting the modified non-human animal one-cell stage embryo into a surrogate mother to produce a genetically modified F0 generation non-human animal comprising the genetically modified endogenous TrkB locus. Optionally, the nuclease agent is a Cas9 protein and a guide RNA that targets a guide RNA target sequence within the endogenous TrkB locus. Optionally, step (a) further comprises introducing into the non-human one-cell stage embryo a second guide RNA that targets a second guide RNA target sequence within the endogenous TrkB locus.

BRIEF DESCRIPTION OF THE FIGURES

FIG. 1 (not to scale) shows a schematic of the targeting scheme for humanization of the region of the mouse TrkB (mouse Ntrk2) locus encoding the extracellular domain of TRKB. The top portion of the figure shows the endogenous mouse TrkB (mouse Ntrk2) locus, and the bottom portion of the figure shows the large targeting vector.

FIG. 2 (not to scale) shows a schematic of the TAQMAN® assays for screening humanization of the mouse TrkB (mouse Ntrk2) locus. Gain-of-allele (GOA) assays include 7138hU and 7138hD. Loss-of-allele (LOA) assays include 7138U and 7138D.

FIG. 3 shows western blots assessing total TRKB levels and phospho-TRKB levels in homozygous humanized TRKB mice at 1 hour, 4 hours, and 18 hours following direct hippocampal injection of TRKB agonist antibody H4H9816P2 or isotype control antibody.

FIG. 4 (not to scale) shows a schematic of the targeting scheme for humanization of the region of the rat TrkB (rat Ntrk2) locus encoding the extracellular domain of TRKB. The top portion of the figure shows the endogenous rat TrkB (rat Ntrk2) locus, and the bottom portion of the figure shows the large targeting vector.

FIG. 5 (not to scale) shows a schematic of the TAQMAN® assays for screening humanization of the rat TrkB (rat Ntrk2) locus and the guide RNA positions (guide target sequences set forth in SEQ ID NOS: 41-44) for targeting the rat TrkB (rat Ntrk2) locus. Gain-of-allele (GOA) assays include 7138hU and 7138hD. Loss-of-allele (LOA) assays include rnoTU, rnoTM, and rnoTD. CRISPR assays designed to cover the region that is disrupted by CRISPR/Cas9 targeting include rnoTGU and rnoTGD. Retention assays include rnoTAU2 and rnoTAD2.

FIG. 6 shows an alignment of the mouse, rat, and human TRKB (NTRK2) proteins.

FIG. 7 shows western blots of phospho-TrkB, total TrkB, phospho-Akt, total AKT, phospho-ERK, and total ERK at 15 minutes and 2 hours after treatment of primary cortical neurons isolated from postnatal day 1 homozygous humanized TRKB mouse pups with various TrkB agonist antibodies or BDNF.

FIG. 8 shows pharmacokinetic profiles of anti-TrkB antibody in H4H9816P2 in homozygous TrkB^{hu/hu} and wild type mice.

FIG. 9 shows cell survival in differentiated human neuroblastoma SH-SY5Y cells treated with different doses of

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TrkB agonist antibodies or BDNF. TrkB mAb1 is H4H9816P2; TrkB mAb2 is a control TrkB agonist antibody with affinity for human TrkB, rat TrkB, and mouse TrkB. A human isotype control antibody was used as a negative control. Data were normalized to the serum-free media without antibodies.

FIG. 10 shows cell survival in primary mouse retinal ganglion cells treated with different doses of TrkB agonist antibody or BDNF. TrkB mAb2 is a control TrkB agonist antibody with affinity for human TrkB, rat TrkB, and mouse TrkB. Data were normalized to the serum-free media without antibodies.

FIGS. 11A and 11B show retinal ganglion cell density in retinas dissected and stained for retinal ganglion cells in wild type rats and mice, respectively, following optic nerve transection and treatment with BDNF, TrkB agonist antibody, isotype control antibody, or vehicle control. Rats were given BDNF (2.5 µg), TrkB agonist antibody (18 µg), isotype control antibody (18 µg), or vehicle control intravitreally at 3 days and 10 days after optic nerve transection. Mice were given BDNF (2.5 µg), TrkB agonist antibody (10 µg), isotype control antibody (10 µg), or vehicle control intravitreally at 3 days and 10 days after optic nerve transection. TrkB mAb2 is a control TrkB agonist antibody with affinity for human TrkB, rat TrkB, and mouse TrkB.

FIGS. 12A and 12B show retinal ganglion cell density in retinas dissected and stained for retinal ganglion cells in wild type mice and rats, respectively, following optic nerve transection or optic nerve crush and treatment with various doses of BDNF. FIG. 12A shows BDNF dose response in an optic nerve crush (ONC) model in WT mice. FIG. 12B shows BDNF dose response in an optic nerve transection model in WT rat from 0.13 µg to 30 µg.

FIG. 13A shows retinal ganglion cell density in retinas dissected and stained for retinal ganglion cells in homozygous, heterozygous, or wild-type TrkB rats given either TrkB agonist antibody or isotype control antibody intravitreally at 3 and 10 days after optic nerve transection (***p<0.0001; ***p<0.001; two way ANOVA). Retinas were dissected 14 days after transection. TrkB mAb1 is H4H9816P2.

FIG. 13B shows retinal ganglion cell density in non-injured eyes dissected from homozygous, heterozygous, or wild-type TrkB rats.

FIG. 13C shows the body weight of the human TRKB homozygous mice given TrkB agonist antibody (H4H9816P2; TrkB) or isotype control antibody (REGN1945; Control).

FIG. 14 shows retinal ganglion cell density in retinas dissected and stained for retinal ganglion cells in human TRKB homozygous rats given either TrkB agonist antibody (hTrkB; H4H9816P2) or isotype control antibody (REGN1945) intravitreally at 3 and 10 days after optic nerve transection. Retinas were dissected 14 days after transection.

FIGS. 15A and 15B show retinal ganglion cell density in retinas dissected and stained for retinal ganglion cells in human TRKB homozygous rats given different TrkB agonist antibodies (H4H9816P2-L9, H4H9814P-L9, H4H9780P-L5, or a combination of all three) or isotype control antibody (REGN1945) intravitreally at 3 and 10 days after optic nerve transection (**p<0.01; Kruskal-Wallis test compared to isotype control antibody). Retinas were dissected 14 days after transection. FIG. 15A includes a naïve control (noninjured contralateral eye), and FIG. 15B does not.

FIG. 16 shows retinal ganglion cell density in retinas dissected and stained for retinal ganglion cells in wild type rats given different TrkB agonist antibodies (H4H9780P and

H4H9814P) or isotype control antibody (REGN1945) intravitreally at 3 and 10 days after optic nerve transection. Retinas were dissected 14 days after transection.

FIGS. 17A and 17B show retinal ganglion cell density in retinas dissected and stained for retinal ganglion cells in human TRKB homozygous mice given TrkB agonist antibody (H4H9780P) or isotype control antibody (REGN1945) intravitreally at 3 and 10 days after optic nerve transection. Retinas were dissected 14 days after transection. FIG. 17A includes a normal control (noninjured contralateral eye), and FIG. 17B does not.

FIG. 17C shows the body weight of the human TRKB homozygous mice given TrkB agonist antibody (H4H9780P) or isotype control antibody (REGN1945).

DEFINITIONS

The terms “protein,” “polypeptide,” and “peptide,” used interchangeably herein, include polymeric forms of amino acids of any length, including coded and non-coded amino acids and chemically or biochemically modified or derivatized amino acids. The terms also include polymers that have been modified, such as polypeptides having modified peptide backbones. The term “domain” refers to any part of a protein or polypeptide having a particular function or structure.

Proteins are said to have an “N-terminus” and a “C-terminus.” The term “N-terminus” relates to the start of a protein or polypeptide, terminated by an amino acid with a free amine group ($-\text{NH}_2$). The term “C-terminus” relates to the end of an amino acid chain (protein or polypeptide), terminated by a free carboxyl group ($-\text{COOH}$).

The terms “nucleic acid” and “polynucleotide,” used interchangeably herein, include polymeric forms of nucleotides of any length, including ribonucleotides, deoxyribonucleotides, or analogs or modified versions thereof. They include single-, double-, and multi-stranded DNA or RNA, genomic DNA, cDNA, DNA-RNA hybrids, and polymers comprising purine bases, pyrimidine bases, or other natural, chemically modified, biochemically modified, non-natural, or derivatized nucleotide bases.

Nucleic acids are said to have “5' ends” and “3' ends” because mononucleotides are reacted to make oligonucleotides in a manner such that the 5' phosphate of one mononucleotide pentose ring is attached to the 3' oxygen of its neighbor in one direction via a phosphodiester linkage. An end of an oligonucleotide is referred to as the “5' end” if its 5' phosphate is not linked to the 3' oxygen of a mononucleotide pentose ring. An end of an oligonucleotide is referred to as the “3' end” if its 3' oxygen is not linked to a 5' phosphate of another mononucleotide pentose ring. A nucleic acid sequence, even if internal to a larger oligonucleotide, also may be said to have 5' and 3' ends. In either a linear or circular DNA molecule, discrete elements are referred to as being “upstream” or 5' of the “downstream” or 3' elements.

The term “genomically integrated” refers to a nucleic acid that has been introduced into a cell such that the nucleotide sequence integrates into the genome of the cell and is capable of being inherited by progeny thereof. Any protocol may be used for the stable incorporation of a nucleic acid into the genome of a cell.

The term “targeting vector” refers to a recombinant nucleic acid that can be introduced by homologous recombination, non-homologous-end-joining-mediated ligation, or any other means of recombination to a target position in the genome of a cell.

The term “viral vector” refers to a recombinant nucleic acid that includes at least one element of viral origin and includes elements sufficient for or permissive of packaging into a viral vector particle. The vector and/or particle can be utilized for the purpose of transferring DNA, RNA, or other nucleic acids into cells either *ex vivo* or *in vivo*. Numerous forms of viral vectors are known.

The term “wild type” includes entities having a structure and/or activity as found in a normal (as contrasted with mutant, diseased, altered, or so forth) state or context. Wild type genes and polypeptides often exist in multiple different forms (e.g., alleles).

The term “endogenous” refers to a nucleic acid sequence that occurs naturally within a cell or non-human animal. For example, an endogenous TrkB sequence of a non-human animal refers to a native TrkB sequence that naturally occurs at the TrkB locus in the non-human animal.

“Exogenous” molecules or sequences include molecules or sequences that are not normally present in a cell in that form. Normal presence includes presence with respect to the particular developmental stage and environmental conditions of the cell. An exogenous molecule or sequence, for example, can include a mutated version of a corresponding endogenous sequence within the cell, such as a humanized version of the endogenous sequence, or can include a sequence corresponding to an endogenous sequence within the cell but in a different form (i.e., not within a chromosome). In contrast, endogenous molecules or sequences include molecules or sequences that are normally present in that form in a particular cell at a particular developmental stage under particular environmental conditions.

The term “heterologous” when used in the context of a nucleic acid or a protein indicates that the nucleic acid or protein comprises at least two portions that do not naturally occur together in the same molecule. For example, the term “heterologous,” when used with reference to portions of a nucleic acid or portions of a protein, indicates that the nucleic acid or protein comprises two or more sub-sequences that are not found in the same relationship to each other (e.g., joined together) in nature. As one example, a “heterologous” region of a nucleic acid vector is a segment of nucleic acid within or attached to another nucleic acid molecule that is not found in association with the other molecule in nature. For example, a heterologous region of a nucleic acid vector could include a coding sequence flanked by sequences not found in association with the coding sequence in nature. Likewise, a “heterologous” region of a protein is a segment of amino acids within or attached to another peptide molecule that is not found in association with the other peptide molecule in nature (e.g., a fusion protein, or a protein with a tag). Similarly, a nucleic acid or protein can comprise a heterologous label or a heterologous secretion or localization sequence.

“Codon optimization” takes advantage of the degeneracy of codons, as exhibited by the multiplicity of three-base pair codon combinations that specify an amino acid, and generally includes a process of modifying a nucleic acid sequence for enhanced expression in particular host cells by replacing at least one codon of the native sequence with a codon that is more frequently or most frequently used in the genes of the host cell while maintaining the native amino acid sequence. For example, a nucleic acid encoding a Cas9 protein can be modified to substitute codons having a higher frequency of usage in a given prokaryotic or eukaryotic cell, including a bacterial cell, a yeast cell, a human cell, a non-human cell, a mammalian cell, a rodent cell, a mouse cell, a rat cell, a hamster cell, or any other host cell, as

compared to the naturally occurring nucleic acid sequence. Codon usage tables are readily available, for example, at the “Codon Usage Database.” These tables can be adapted in a number of ways. See Nakamura et al. (2000) *Nucleic Acids Research* 28:292, herein incorporated by reference in its entirety for all purposes. Computer algorithms for codon optimization of a particular sequence for expression in a particular host are also available (see, e.g., Gene Forge).

The term “locus” refers to a specific location of a gene (or significant sequence), DNA sequence, polypeptide-encoding sequence, or position on a chromosome of the genome of an organism. For example, a “TrkB locus” may refer to the specific location of a TrkB gene, TrkB DNA sequence, TrkB-encoding sequence, or TrkB position on a chromosome of the genome of an organism that has been identified as to where such a sequence resides. A “TrkB locus” may comprise a regulatory element of a TrkB gene, including, for example, an enhancer, a promoter, 5' and/or 3' untranslated region (UTR), or a combination thereof.

The term “gene” refers to a DNA sequence in a chromosome that codes for a product (e.g., an RNA product and/or a polypeptide product) and includes the coding region interrupted with non-coding introns and sequence located adjacent to the coding region on both the 5' and 3' ends such that the gene corresponds to the full-length mRNA (including the 5' and 3' untranslated sequences). The term “gene” also includes other non-coding sequences including regulatory sequences (e.g., promoters, enhancers, and transcription factor binding sites), polyadenylation signals, internal ribosome entry sites, silencers, insulating sequence, and matrix attachment regions. These sequences may be close to the coding region of the gene (e.g., within 10 kb) or at distant sites, and they influence the level or rate of transcription and translation of the gene.

The term “allele” refers to a variant form of a gene. Some genes have a variety of different forms, which are located at the same position, or genetic locus, on a chromosome. A diploid organism has two alleles at each genetic locus. Each pair of alleles represents the genotype of a specific genetic locus. Genotypes are described as homozygous if there are two identical alleles at a particular locus and as heterozygous if the two alleles differ.

A “promoter” is a regulatory region of DNA usually comprising a TATA box capable of directing RNA polymerase II to initiate RNA synthesis at the appropriate transcription initiation site for a particular polynucleotide sequence. A promoter may additionally comprise other regions which influence the transcription initiation rate. The promoter sequences disclosed herein modulate transcription of an operably linked polynucleotide. A promoter can be active in one or more of the cell types disclosed herein (e.g., a eukaryotic cell, a non-human mammalian cell, a human cell, a rodent cell, a pluripotent cell, a one-cell stage embryo, a differentiated cell, or a combination thereof). A promoter can be, for example, a constitutively active promoter, a conditional promoter, an inducible promoter, a temporally restricted promoter (e.g., a developmentally regulated promoter), or a spatially restricted promoter (e.g., a cell-specific or tissue-specific promoter). Examples of promoters can be found, for example, in WO 2013/176772, herein incorporated by reference in its entirety for all purposes.

“Operable linkage” or being “operably linked” includes juxtaposition of two or more components (e.g., a promoter and another sequence element) such that both components function normally and allow the possibility that at least one of the components can mediate a function that is exerted upon at least one of the other components. For example, a

promoter can be operably linked to a coding sequence if the promoter controls the level of transcription of the coding sequence in response to the presence or absence of one or more transcriptional regulatory factors. Operable linkage can include such sequences being contiguous with each other or acting in trans (e.g., a regulatory sequence can act at a distance to control transcription of the coding sequence).

The term “variant” refers to a nucleotide sequence differing from the sequence most prevalent in a population (e.g., by one nucleotide) or a protein sequence different from the sequence most prevalent in a population (e.g., by one amino acid).

The term “fragment” when referring to a protein means a protein that is shorter or has fewer amino acids than the full-length protein. The term “fragment” when referring to a nucleic acid means a nucleic acid that is shorter or has fewer nucleotides than the full-length nucleic acid. A fragment can be, for example, an N-terminal fragment (i.e., removal of a portion of the C-terminal end of the protein), a C-terminal fragment (i.e., removal of a portion of the N-terminal end of the protein), or an internal fragment.

“Sequence identity” or “identity” in the context of two polynucleotides or polypeptide sequences makes reference to the residues in the two sequences that are the same when aligned for maximum correspondence over a specified comparison window. When percentage of sequence identity is used in reference to proteins, residue positions which are not identical often differ by conservative amino acid substitutions, where amino acid residues are substituted for other amino acid residues with similar chemical properties (e.g., charge or hydrophobicity) and therefore do not change the functional properties of the molecule. When sequences differ in conservative substitutions, the percent sequence identity may be adjusted upwards to correct for the conservative nature of the substitution. Sequences that differ by such conservative substitutions are said to have “sequence similarity” or “similarity.” Means for making this adjustment are well known. Typically, this involves scoring a conservative substitution as a partial rather than a full mismatch, thereby increasing the percentage sequence identity. Thus, for example, where an identical amino acid is given a score of 1 and a non-conservative substitution is given a score of zero, a conservative substitution is given a score between zero and 1. The scoring of conservative substitutions is calculated, e.g., as implemented in the program PC/GENE (Intelligenetics, Mountain View, California).

“Percentage of sequence identity” includes the value determined by comparing two optimally aligned sequences (greatest number of perfectly matched residues) over a comparison window, wherein the portion of the polynucleotide sequence in the comparison window may comprise additions or deletions (i.e., gaps) as compared to the reference sequence (which does not comprise additions or deletions) for optimal alignment of the two sequences. The percentage is calculated by determining the number of positions at which the identical nucleic acid base or amino acid residue occurs in both sequences to yield the number of matched positions, dividing the number of matched positions by the total number of positions in the window of comparison, and multiplying the result by 100 to yield the percentage of sequence identity. Unless otherwise specified (e.g., the shorter sequence includes a linked heterologous sequence), the comparison window is the full length of the shorter of the two sequences being compared.

Unless otherwise stated, sequence identity/similarity values include the value obtained using GAP Version 10 using the following parameters: % identity and % similarity for a

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nucleotide sequence using GAP Weight of 50 and Length Weight of 3, and the nws gapdna.cmp scoring matrix; % identity and % similarity for an amino acid sequence using GAP Weight of 8 and Length Weight of 2, and the BLOSUM62 scoring matrix; or any equivalent program thereof. “Equivalent program” includes any sequence comparison program that, for any two sequences in question, generates an alignment having identical nucleotide or amino acid residue matches and an identical percent sequence identity when compared to the corresponding alignment generated by GAP Version 10.

The term “conservative amino acid substitution” refers to the substitution of an amino acid that is normally present in the sequence with a different amino acid of similar size, charge, or polarity. Examples of conservative substitutions include the substitution of a non-polar (hydrophobic) residue such as isoleucine, valine, or leucine for another non-polar residue. Likewise, examples of conservative substitutions include the substitution of one polar (hydrophilic) residue for another such as between arginine and lysine, between glutamine and asparagine, or between glycine and serine. Additionally, the substitution of a basic residue such as lysine, arginine, or histidine for another, or the substitution of one acidic residue such as aspartic acid or glutamic acid for another acidic residue are additional examples of conservative substitutions. Examples of non-conservative substitutions include the substitution of a non-polar (hydrophobic) amino acid residue such as isoleucine, valine, leucine, alanine, or methionine for a polar (hydrophilic) residue such as cysteine, glutamine, glutamic acid or lysine and/or a polar residue for a non-polar residue. Typical amino acid categorizations are summarized below.

Alanine	Ala	A	Nonpolar	Neutral	1.8
Arginine	Arg	R	Polar	Positive	-4.5
Asparagine	Asn	N	Polar	Neutral	-3.5
Aspartic acid	Asp	D	Polar	Negative	-3.5
Cysteine	Cys	C	Nonpolar	Neutral	2.5
Glutamic acid	Glu	E	Polar	Negative	-3.5
Glutamine	Gln	Q	Polar	Neutral	-3.5
Glycine	Gly	G	Nonpolar	Neutral	-0.4
Histidine	His	H	Polar	Positive	-3.2
Isoleucine	Ile	I	Nonpolar	Neutral	4.5
Leucine	Leu	L	Nonpolar	Neutral	3.8
Lysine	Lys	K	Polar	Positive	-3.9
Methionine	Met	M	Nonpolar	Neutral	1.9
Phenylalanine	Phe	F	Nonpolar	Neutral	2.8
Proline	Pro	P	Nonpolar	Neutral	-1.6
Serine	Ser	S	Polar	Neutral	-0.8
Threonine	Thr	T	Polar	Neutral	-0.7
Tryptophan	Trp	W	Nonpolar	Neutral	-0.9
Tyrosine	Tyr	Y	Polar	Neutral	-1.3
Valine	Val	V	Nonpolar	Neutral	4.2

A “homologous” sequence (e.g., nucleic acid sequence) includes a sequence that is either identical or substantially similar to a known reference sequence, such that it is, for example, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical to the known reference sequence. Homologous sequences can include, for example, orthologous sequence and paralogous sequences. Homologous genes, for example, typically descend from a common ancestral DNA sequence, either through a speciation event (orthologous genes) or a genetic duplication event (paralogous genes). “Orthologous” genes include genes in different species that evolved from a common ancestral gene by speciation. Orthologs typically retain the same function in

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the course of evolution. “Paralogous” genes include genes related by duplication within a genome. Paralogs can evolve new functions in the course of evolution.

The term “in vitro” includes artificial environments and to processes or reactions that occur within an artificial environment (e.g., a test tube). The term “in vivo” includes natural environments (e.g., a cell or organism or body) and to processes or reactions that occur within a natural environment. The term “ex vivo” includes cells that have been removed from the body of an individual and to processes or reactions that occur within such cells.

The term “reporter gene” refers to a nucleic acid having a sequence encoding a gene product (typically an enzyme) that is easily and quantifiably assayed when a construct comprising the reporter gene sequence operably linked to a heterologous promoter and/or enhancer element is introduced into cells containing (or which can be made to contain) the factors necessary for the activation of the promoter and/or enhancer elements. Examples of reporter genes include, but are not limited, to genes encoding beta-galactosidase (lacZ), the bacterial chloramphenicol acetyltransferase (cat) genes, firefly luciferase genes, genes encoding beta-glucuronidase (GUS), and genes encoding fluorescent proteins. A “reporter protein” refers to a protein encoded by a reporter gene.

The term “fluorescent reporter protein” as used herein means a reporter protein that is detectable based on fluorescence wherein the fluorescence may be either from the reporter protein directly, activity of the reporter protein on a fluorogenic substrate, or a protein with affinity for binding to a fluorescent tagged compound. Examples of fluorescent proteins include green fluorescent proteins (e.g., GFP, GFP-2, tagGFP, turboGFP, eGFP, Emerald, Azami Green, Monomeric Azami Green, CopGFP, AceGFP, and ZsGreen1), yellow fluorescent proteins (e.g., YFP, eYFP, Citrine, Venus, YPet, PhiYFP, and ZsYellow1), blue fluorescent proteins (e.g., BFP, eBFP, eBFP2, Azurite, mKalama1, GFPuv, Sapphire, and T-sapphire), cyan fluorescent proteins (e.g., CFP, eCFP, Cerulean, CyPet, AmCyan1, and Midoriishi-Cyan), red fluorescent proteins (e.g., RFP, mKate, mKate2, mPlum, DsRed monomer, mCherry, mRFP1, DsRed-Express, DsRed2, DsRed-Monomer, HcRed-Tandem, HcRed1, AsRed2, eqFP611, mRaspberry, mStrawberry, and Jred), orange fluorescent proteins (e.g., mOrange, mKO, Kusabira-Orange, Monomeric Kusabira-Orange, mTangerine, and tdTomato), and any other suitable fluorescent protein whose presence in cells can be detected by flow cytometry methods.

The term “recombination” includes any process of exchange of genetic information between two polynucleotides and can occur by any mechanism. Recombination in response to double-strand breaks (DSBs) occurs principally through two conserved DNA repair pathways: non-homologous end joining (NHEJ) and homologous recombination (HR). See Kasperek & Humphrey (2011) *Seminars in Cell & Dev. Biol.* 22:886-897, herein incorporated by reference in its entirety for all purposes. Likewise, repair of a target nucleic acid mediated by an exogenous donor nucleic acid can include any process of exchange of genetic information between the two polynucleotides.

NHEJ includes the repair of double-strand breaks in a nucleic acid by direct ligation of the break ends to one another or to an exogenous sequence without the need for a homologous template. Ligation of non-contiguous sequences by NHEJ can often result in deletions, insertions, or translocations near the site of the double-strand break. For example, NHEJ can also result in the targeted integration of

an exogenous donor nucleic acid through direct ligation of the break ends with the ends of the exogenous donor nucleic acid (i.e., NHEJ-based capture). Such NHEJ-mediated targeted integration can be preferred for insertion of an exogenous donor nucleic acid when homology directed repair (HDR) pathways are not readily usable (e.g., in non-dividing cells, primary cells, and cells which perform homology-based DNA repair poorly). In addition, in contrast to homology-directed repair, knowledge concerning large regions of sequence identity flanking the cleavage site is not needed, which can be beneficial when attempting targeted insertion into organisms that have genomes for which there is limited knowledge of the genomic sequence. The integration can proceed via ligation of blunt ends between the exogenous donor nucleic acid and the cleaved genomic sequence, or via ligation of sticky ends (i.e., having 5' or 3' overhangs) using an exogenous donor nucleic acid that is flanked by overhangs that are compatible with those generated by a nuclease agent in the cleaved genomic sequence. See, e.g., US 2011/020722, WO 2014/033644, WO 2014/089290, and Maresca et al. (2013) *Genome Res.* 23(3):539-546, each of which is herein incorporated by reference in its entirety for all purposes. If blunt ends are ligated, target and/or donor resection may be needed to generation regions of microhomology needed for fragment joining, which may create unwanted alterations in the target sequence.

Recombination can also occur via homology directed repair (HDR) or homologous recombination (HR). HDR or HR includes a form of nucleic acid repair that can require nucleotide sequence homology, uses a "donor" molecule as a template for repair of a "target" molecule (i.e., the one that experienced the double-strand break), and leads to transfer of genetic information from the donor to target. Without wishing to be bound by any particular theory, such transfer can involve mismatch correction of heteroduplex DNA that forms between the broken target and the donor, and/or synthesis-dependent strand annealing, in which the donor is used to resynthesize genetic information that will become part of the target, and/or related processes. In some cases, the donor polynucleotide, a portion of the donor polynucleotide, a copy of the donor polynucleotide, or a portion of a copy of the donor polynucleotide integrates into the target DNA. See Wang et al. (2013) *Cell* 153:910-918; Mandalos et al. (2012) *PLOS ONE* 7:e45768:1-9; and Wang et al. (2013) *Nat Biotechnol.* 31:530-532, each of which is herein incorporated by reference in its entirety for all purposes.

The term "antigen-binding protein" includes any protein that binds to an antigen. Examples of antigen-binding proteins include an antibody, an antigen-binding fragment of an antibody, a multispecific antibody (e.g., a bi-specific antibody), an scFv, a bis-scFv, a diabody, a triabody, a tetrabody, a V-NAR, a VHH, a VL, a F(ab), a F(ab)₂, a DVD (dual variable domain antigen-binding protein), an SVD (single variable domain antigen-binding protein), a bispecific T-cell engager (BiTE), or a Davisbody (U.S. Pat. No. 8,586,713, herein incorporated by reference herein in its entirety for all purposes).

As used herein, the expression "anti-TRKB antibody" includes both monovalent antibodies with a single specificity, as well as bispecific antibodies comprising a first arm that binds TRKB and a second arm that binds a second (target) antigen, wherein the anti-TRKB arm comprises, for example, any of the HCVR/LCVR or CDR sequences as set forth in Table 22 herein. The expression "anti-TrkB antibody" also includes antibody-drug conjugates (ADCs) comprising an anti-TRKB antibody or antigen-binding portion thereof conjugated to a drug or toxin (i.e., cytotoxic agent).

The expression "anti-TRKB antibody" also includes antibody-radionuclide conjugates (ARCs) comprising an anti-TRKB antibody or antigen-binding portion thereof conjugated to a radionuclide.

The term "anti-TRKB antibody" as used herein means any antigen-binding molecule or molecular complex comprising at least one complementarity determining region (CDR) that specifically binds to or interacts with TRKB or a portion of TRKB. The term "antibody" includes immunoglobulin molecules comprising four polypeptide chains, two heavy (H) chains and two light (L) chains inter-connected by disulfide bonds, as well as multimers thereof (e.g., IgM). Each heavy chain comprises a heavy chain variable region (abbreviated herein as HCVR or V_H) and a heavy chain constant region. The heavy chain constant region comprises three domains, C_H1, C_H2 and C_H3. Each light chain comprises a light chain variable region (abbreviated herein as LCVR or V_L) and a light chain constant region. The light chain constant region comprises one domain (C_L1). The V_H and V_L regions can be further subdivided into regions of hypervariability, termed complementarity determining regions (CDRs), interspersed with regions that are more conserved, termed framework regions (FR). Each V_H and V_L is composed of three CDRs and four FRs, arranged from amino-terminus to carboxy-terminus in the following order: FR1, CDR1, FR2, CDR2, FR3, CDR3, FR4. In different embodiments of the invention, the FRs of the anti-TRKB antibody (or antigen-binding portion thereof) may be identical to the human germline sequences, or may be naturally or artificially modified. An amino acid consensus sequence may be defined based on a side-by-side analysis of two or more CDRs.

The term "antibody" as used herein also includes antigen-binding fragments of full length antibody molecules. The terms "antigen-binding portion" of an antibody, "antigen-binding fragment" of an antibody, and the like, as used herein, include any enzymatically obtainable, synthetic, or genetically engineered polypeptide or glycoprotein that specifically binds an antigen to form a complex. Antigen-binding fragments of an antibody may be derived, for example, from full antibody molecules using any suitable standard techniques such as proteolytic digestion or recombinant genetic engineering techniques involving the manipulation and expression of DNA encoding antibody variable and optionally constant domains. Such DNA is known and/or is readily available from, e.g., commercial sources, DNA libraries (including, e.g., phage-antibody libraries), or can be synthesized. The DNA may be sequenced and manipulated chemically or by using molecular biology techniques, for example, to arrange one or more variable and/or constant domains into a suitable configuration, or to introduce codons, create cysteine residues, modify, add or delete amino acids, and so forth.

Non-limiting examples of antigen-binding fragments include: (i) Fab fragments; (ii) F(ab')₂ fragments; (iii) Fd fragments; (iv) Fv fragments; (v) single-chain Fv (scFv) molecules; (vi) dAb fragments; and (vii) minimal recognition units consisting of the amino acid residues that mimic the hypervariable region of an antibody (e.g., an isolated complementarity determining region (CDR) such as a CDR3 peptide), or a constrained FR3-CDR3-FR4 peptide. Other engineered molecules, such as domain-specific antibodies, single domain antibodies, domain-deleted antibodies, chimeric antibodies, CDR-grafted antibodies, diabodies, triabodies, tetrabodies, minibodies, nanobodies (e.g., monovalent nanobodies, bivalent nanobodies, etc.), small modular immunopharmaceuticals (SMIPs), and shark variable

IgNAR domains, are also encompassed within the expression “antigen-binding fragment” as used herein.

An antigen-binding fragment of an antibody will typically comprise at least one variable domain. The variable domain may be of any size or amino acid composition and will generally comprise at least one CDR, which is adjacent to or in frame with one or more framework sequences. In antigen-binding fragments having a V_H domain associated with a V_L domain, the V_H and V_L domains may be situated relative to one another in any suitable arrangement. For example, the variable region may be dimeric and contain V_H - V_H , V_H - V_L or V_L - V_L dimers. Alternatively, the antigen-binding fragment of an antibody may contain a monomeric V_H or V_L domain.

In certain embodiments, an antigen-binding fragment of an antibody may contain at least one variable domain covalently linked to at least one constant domain. Non-limiting, exemplary configurations of variable and constant domains that may be found within an antigen-binding fragment of an antibody of the present invention include: (i) V_H - C_H1 ; (ii) V_H - C_H2 ; (iii) V_H - C_H3 ; (iv) V_H - C_H1 - C_H2 ; (v) V_H - C_H1 - C_H2 - C_H3 ; (vi) V_H - C_H2 - C_H3 ; (vii) V_H - C_L ; (viii) V_L - C_H1 ; (ix) V_L - C_H2 ; (x) V_L - C_H3 ; (xi) V_L - C_H1 - C_H2 ; (xii) V_L - C_H1 - C_H2 - C_H3 ; (xiii) V_L - C_H2 - C_H3 ; and (xiv) V_L - C_L . In any configuration of variable and constant domains, including any of the exemplary configurations listed above, the variable and constant domains may be either directly linked to one another or may be linked by a full or partial hinge or linker region. A hinge region may consist of at least 2 (e.g., 5, 10, 15, 20, 40, 60 or more) amino acids, which result in a flexible or semi-flexible linkage between adjacent variable and/or constant domains in a single polypeptide molecule. Moreover, an antigen-binding fragment of an antibody of the present invention may comprise a homo-dimer or hetero-dimer (or other multimer) of any of the variable and constant domain configurations listed above in non-covalent association with one another and/or with one or more monomeric V_H or V_L domain (e.g., by disulfide bond(s)).

As with full antibody molecules, antigen-binding fragments may be monospecific or multispecific (e.g., bispecific). A multispecific antigen-binding fragment of an antibody will typically comprise at least two different variable domains, wherein each variable domain is capable of specifically binding to a separate antigen or to a different epitope on the same antigen. Any multispecific antibody format, including the exemplary bispecific antibody formats disclosed herein, may be adapted for use in the context of an antigen-binding fragment of an antibody of the present invention using routine techniques.

The term “epitope” refers to an antigenic determinant that interacts with a specific antigen binding site in the variable region of an antibody molecule known as a paratope. A single antigen may have more than one epitope. Thus, different antibodies may bind to different areas on an antigen and may have different biological effects. Epitopes may be either conformational or linear. A conformational epitope is produced by spatially juxtaposed amino acids from different segments of the linear polypeptide chain. A linear epitope is one produced by adjacent amino acid residues in a polypeptide chain. In certain circumstance, an epitope may include moieties of saccharides, phosphoryl groups, or sulfonyl groups on the antigen.

The term “specifically binds,” or “binds specifically to,” or the like, means that an antibody or antigen-binding fragment thereof forms a complex with an antigen that is relatively stable under physiologic conditions. Specific binding can be characterized by an equilibrium dissociation

constant of at least about 1×10^{-6} M or less (e.g., a smaller K_D denotes a tighter binding). Methods for determining whether two molecules specifically bind are well-known and include, for example, equilibrium dialysis, surface plasmon resonance, and the like. As described herein, antibodies have been identified by surface plasmon resonance, e.g., BIA-CORE™, which bind specifically to TRKB. Moreover, multi-specific antibodies that bind to TRKB protein and one or more additional antigens or a bi-specific that binds to two different regions of TRKB are nonetheless considered antibodies that “specifically bind,” as used herein.

The anti-TRKB antibodies disclosed herein may comprise one or more amino acid substitutions, insertions and/or deletions in the framework and/or CDR regions of the heavy and light chain variable domains. Such mutations can be readily ascertained by comparing the amino acid sequences disclosed herein to sequences available from, for example, public antibody sequence databases. Once obtained, antibodies and antigen-binding fragments that contain one or more mutations can be easily tested for one or more desired property such as, improved binding specificity, increased binding affinity, improved or enhanced antagonistic or agonistic biological properties (as the case may be), reduced immunogenicity, and so forth. Antibodies and antigen-binding fragments obtained in this general manner are included.

Also included are anti-TRKB antibodies comprising variants of any of the HCVR, LCVR, and/or CDR amino acid sequences disclosed herein having one or more conservative substitutions. For example, the present invention includes anti-TRKB antibodies having HCVR, LCVR, and/or CDR amino acid sequences with, e.g., 10 or fewer, 8 or fewer, 6 or fewer, 4 or fewer, etc. conservative amino acid substitutions relative to any of the HCVR, LCVR, and/or CDR amino acid sequences set forth in Table 22.

The term “substantial identity” or “substantially identical,” when referring to a nucleic acid or fragment thereof in the context of anti-TRKB antibodies, indicates that, when optimally aligned with appropriate nucleotide insertions or deletions with another nucleic acid (or its complementary strand), there is nucleotide sequence identity in at least about 95%, and more preferably at least about 96%, 97%, 98% or 99% of the nucleotide bases, as measured by any well-known algorithm of sequence identity, such as FASTA, BLAST or Gap, as discussed below. A nucleic acid molecule having substantial identity to a reference nucleic acid molecule may, in certain instances, encode a polypeptide having the same or substantially similar amino acid sequence as the polypeptide encoded by the reference nucleic acid molecule.

As applied to polypeptides in the context of anti-TRKB antibodies, the term “substantial similarity” or “substantially similar” means that two peptide sequences, when optimally aligned, such as by the programs GAP or BESTFIT using default gap weights, share at least 95% sequence identity, even more preferably at least 98% or 99% sequence identity. Preferably, residue positions which are not identical differ by conservative amino acid substitutions. A “conservative amino acid substitution” as applied to polypeptides in the context of anti-TRKB antibodies is one in which an amino acid residue is substituted by another amino acid residue having a side chain (R group) with similar chemical properties (e.g., charge or hydrophobicity). In general, a conservative amino acid substitution will not substantially change the functional properties of a protein. In cases where two or more amino acid sequences differ from each other by conservative substitutions, the percent sequence identity or degree of similarity may be adjusted upwards to correct for the conservative nature of the substitution. Means for mak-

ing this adjustment are well-known to those of skill in the art. See, e.g., Pearson (1994) *Methods Mol. Biol.* 24: 307-331, herein incorporated by reference in its entirety for all purposes. Examples of groups of amino acids that have side chains with similar chemical properties include: (1) aliphatic side chains: glycine, alanine, valine, leucine and isoleucine; (2) aliphatic-hydroxyl side chains: serine and threonine; (3) amide-containing side chains: asparagine and glutamine; (4) aromatic side chains: phenylalanine, tyrosine, and tryptophan; (5) basic side chains: lysine, arginine, and histidine; (6) acidic side chains: aspartate and glutamate; and (7) sulfur-containing side chains are cysteine and methionine. Preferred conservative amino acids substitution groups are: valine-leucine-isoleucine, phenylalanine-tyrosine, lysine-arginine, alanine-valine, glutamate-aspartate, and asparagine-glutamine. Alternatively, a conservative replacement is any change having a positive value in the PAM250 log-likelihood matrix disclosed in Gonnet et al. (1992) *Science* 256:1443-1445, herein incorporated by reference in its entirety for all purposes. A "moderately conservative" replacement is any change having a nonnegative value in the PAM250 log-likelihood matrix.

Compositions or methods "comprising" or "including" one or more recited elements may include other elements not specifically recited. For example, a composition that "comprises" or "includes" a protein may contain the protein alone or in combination with other ingredients. The transitional phrase "consisting essentially of" means that the scope of a claim is to be interpreted to encompass the specified elements recited in the claim and those that do not materially affect the basic and novel characteristic(s) of the claimed invention. Thus, the term "consisting essentially of" when used in a claim of this invention is not intended to be interpreted to be equivalent to "comprising."

"Optional" or "optionally" means that the subsequently described event or circumstance may or may not occur and that the description includes instances in which the event or circumstance occurs and instances in which it does not.

Designation of a range of values includes all integers within or defining the range, and all subranges defined by integers within the range.

Unless otherwise apparent from the context, the term "about" encompasses values within a standard margin of error of measurement (e.g., SEM) of a stated value.

The term "and/or" refers to and encompasses any and all possible combinations of one or more of the associated listed items, as well as the lack of combinations when interpreted in the alternative ("or").

The term "or" refers to any one member of a particular list and also includes any combination of members of that list.

The singular forms of the articles "a," "an," and "the" include plural references unless the context clearly dictates otherwise. For example, the term "a protein" or "at least one protein" can include a plurality of proteins, including mixtures thereof.

Statistically significant means $p \leq 0.05$.

DETAILED DESCRIPTION

I. Overview

Disclosed herein are non-human animal genomes, non-human animal cells, and non-human animals comprising a humanized TRKB locus and methods of using such non-human animal cells and non-human animals. Non-human animal cells or non-human animals comprising a humanized TRKB locus express a human TRKB protein or a chimeric

TRKB protein comprising one or more fragments of a human TRKB protein (e.g., all or part of the human TRKB extracellular domain).

A humanized TRKB allele (e.g., resulting from replacing all or part of the non-human animal genomic DNA one-for-one with orthologous human genomic DNA) will provide the true human target or a close approximation of the true human target of human-TRKB-targeting reagents (e.g., agonist antibodies or agonist small molecules designed to target human TRKB), thereby enabling testing of the efficacy and mode of action of such agents in live animals as well as pharmacokinetic and pharmacodynamics studies. For example, as shown in the working examples disclosed herein, intravitreal administration of human-TRKB-agonist antibodies has a significant neuroprotective effect after optic nerve injury in humanized TrkB rats.

II. Non-Human Animals Comprising a Humanized TRKB Locus

The non-human animal genomes, non-human animal cells, and non-human animals disclosed herein comprise a humanized TRKB locus. Cells or non-human animals comprising a humanized TRKB locus express a human TRKB protein or a partially humanized, chimeric TRKB protein in which one or more fragments of the native TRKB protein have been replaced with corresponding fragments from human TRKB (e.g., all or part of the extracellular domain).

A. TRKB

The cells and non-human animals described herein comprise a humanized TRKB locus. TRKB (also known as BDNF-NT-3 growth factors receptor, GP145-TrkB, Trk-B, TrkB, neurotrophic tyrosine kinase receptor type 2, TrkB tyrosine kinase, tropomyosin-related kinase B, tropomyosin receptor kinase B, neurotrophic receptor tyrosine kinase 2, and NTRK2) is encoded by the TRKB gene (also known as NTRK2, OBHD, TRK-B, and GP 145-TRKB). TRKB is a receptor tyrosine kinase involved in the development and maturation of the central and the peripheral nervous systems through regulation of neuron survival, proliferation, migration, differentiation, and synapse formation and plasticity. TRKB is a receptor for BDNF/brain-derived neurotrophic factor and NTF4/neurotrophin-4. Alternatively, TRKB can also bind NTF3/neurotrophin-3, which is less efficient in activating the receptor but regulates neuron survival through TRKB. Upon ligand-binding, TRKB undergoes homodimerization, autophosphorylation, and activation. The canonical isoform of TRKB is expressed in the central and peripheral nervous system. In the central nervous system (CNS), expression is observed in the cerebral cortex, hippocampus, thalamus, choroid plexus, granular layer of the cerebellum, brain stem, and spinal cord. In the peripheral nervous system, it is expressed in many cranial ganglia, the ophthalmic nerve, the vestibular system, multiple facial structures, the submaxillary glands, and dorsal root ganglia.

Human TRKB maps to human 9q21.33 on chromosome 9 (NCBI RefSeq Gene ID 4915; Assembly GRCh38.p7; location NC_000009.12 (84668368 . . . 85027070)). The gene has been reported to have 23 exons. The wild type human TRKB protein has been assigned UniProt accession number Q16620. At least seven isoforms are known (Q16620-1 through Q16620-7). The sequence for one isoform, Q16620-4 (identical to NCBI Accession No. NP_006171.2), is set forth in SEQ ID NO: 3. An mRNA (cDNA) encoding the canonical isoform is assigned NCBI Accession No. AF410899.1 and is set forth in SEQ ID NO: 8. Another example of an mRNA (cDNA) encoding a human TRKB isoform is assigned RefSeq mRNA ID NM_006180.4. An exemplary coding sequence (CDS) is set forth in SEQ ID

NO: 11. The full-length human TRKB protein set forth in SEQ ID NO: 3 has 838 amino acids, including a signal peptide (amino acids 1-31), an extracellular domain (amino acids 32-430), a transmembrane domain (amino acids 431-454), and a cytoplasmic domain (amino acids 455-838). Delineations between these domains are as designated in UniProt. Reference to human TRKB includes the canonical (wild type) forms as well as all allelic forms and isoforms. Any other forms of human TRKB have amino acids numbered for maximal alignment with the wild type form, aligned amino acids being designated the same number. An example of another isoform of human TRKB is Q16620-1 (identical to NCBI Accession No. NP_001018074.1), set forth in SEQ ID NO: 75. An mRNA (cDNA) encoding this isoform is assigned NCBI Accession No. NM_001018064.2 and is set forth in SEQ ID NO: 76. An exemplary coding sequence (CDS) for this isoform (CCDS ID CCDS35050.1) is set forth in SEQ ID NO: 77.

Rat TrkB maps to rat 17p14 on chromosome 17 (NCBI RefSeq Gene ID 25054; Assembly Rnor_6.0; location NC_005116.4 (5934651 . . . 6245778, complement)). The gene has been reported to have 23 exons. The wild type rat TRKB protein has been assigned UniProt accession number Q63604. At least three isoforms are known (Q63604-1 through Q63604-3). The sequence for the canonical isoform, Q63604-1 (identical to NCBI Accession No. NP_036863.1), is set forth in SEQ ID NO: 2. An mRNA (cDNA) encoding the canonical isoform is assigned NCBI Accession No. NM_012731.2 and is set forth in SEQ ID NO: 7. Another example of an mRNA (cDNA) encoding a rat TRKB isoform is assigned RefSeq mRNA ID M55291. An exemplary coding sequence (CDS) is set forth in SEQ ID NO: 10. The canonical full-length rat TRKB protein set forth in SEQ ID NO: 2 has 821 amino acids, including a signal peptide (amino acids 1-31), an extracellular domain (amino acids 32-429), a transmembrane domain (amino acids 430-453), and a cytoplasmic domain (amino acids 454-821). Delineations between these domains are as designated in UniProt. Reference to rat TRKB includes the canonical (wild type) forms as well as all allelic forms and isoforms. Any other forms of rat TRKB have amino acids numbered for maximal alignment with the wild type form, aligned amino acids being designated the same number.

Mouse TrkB maps to mouse 13 B1; 13 31.2 cM on chromosome 12 (NCBI RefSeq Gene ID 18212; Assembly GRCm38.p4 (GCF_000001635.24); location NC_000079.6 (58806569 . . . 59133970)). The gene has been reported to have 23 exons. The wild type mouse TRKB protein has been assigned UniProt accession number P15209. At least four isoforms are known (P15209-1 through P15209-4). The sequence for the canonical isoform, P15209-1 (identical to NCBI Accession Nos. NP_001020245.1 and NP_001269890.1), is set forth in SEQ ID NO: 1. An exemplary mRNA (cDNA) isoform encoding the canonical isoform is assigned NCBI Accession No. NM_001025074.2 and is set forth in SEQ ID NO: 6. An exemplary coding sequence (CDS) (CCDS ID CCDS26573.1) is set forth in SEQ ID NO: 9. The canonical full-length mouse TRKB protein set forth in SEQ ID NO: 1 has 821 amino acids, including a signal peptide (amino acids 1-31), an extracellular domain (amino acids 32-429), a transmembrane domain (amino acids 430-453), and a cytoplasmic domain (amino acids 454-821). Delineations between these domains are as designated in UniProt. Reference to mouse TRKB includes the canonical (wild type) forms as well as all allelic forms and isoforms. Any other forms of mouse TRKB have

amino acids numbered for maximal alignment with the wild type form, aligned amino acids being designated the same number.

B. Humanized TRKB Loci

A humanized TRKB locus can be a TrkB locus in which the entire TrkB gene is replaced with the corresponding orthologous human TRKB sequence, or it can be a TrkB locus in which only a portion of the TrkB gene is replaced with the corresponding orthologous human TRKB sequence (i.e., humanized). Optionally, the corresponding orthologous human TRKB sequence is modified to be codon-optimized based on codon usage in the non-human animal. Replaced (i.e., humanized) regions can include coding regions such as an exon, non-coding regions such as an intron, an untranslated region, or a regulatory region (e.g., a promoter, an enhancer, or a transcriptional repressor-binding element), or any combination thereof. As one example, exons corresponding to 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, or all 23 exons of the human TRKB gene can be humanized. For example, exons corresponding to exons 3-10 of the human TRKB gene can be humanized, including the segment of exon 2 (coding exon 1) from the codon encoding amino acid 33, beginning just after the signal peptide. Alternatively, a region of TrkB encoding an epitope recognized by an anti-human-TRKB antigen-binding protein or a region targeted by human-TRKB-targeting reagent (e.g., a small molecule) can be humanized. Likewise, introns corresponding to 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, or all 22 introns of the human TRKB gene can be humanized or can remain endogenous. For example, introns corresponding to the introns between exons 2 and 10 (i.e., introns 2-9, between coding exon 1 and exon 10) of the human TRKB gene can be humanized, optionally including part of the intron following exon 10 (i.e., intron 10). Flanking untranslated regions including regulatory sequences can also be humanized or remain endogenous. For example, the 5' untranslated region (UTR), the 3' UTR, or both the 5' UTR and the 3' UTR can be humanized, or the 5' UTR, the 3' UTR, or both the 5' UTR and the 3' UTR can remain endogenous. In a specific example, both the 5' UTR and the 3' UTR remain endogenous. Depending on the extent of replacement by orthologous sequences, regulatory sequences, such as a promoter, can be endogenous or supplied by the replacing human orthologous sequence. For example, the humanized TRKB locus can include the endogenous non-human animal TrkB promoter.

One or more or all of the regions encoding the signal peptide, the cytoplasmic domain, the transmembrane domain, or the extracellular can be humanized, or one or more of such regions can remain endogenous. Exemplary coding sequences for a mouse TRKB signal peptide, extracellular domain, transmembrane domain, and cytoplasmic domain are set forth in SEQ ID NOS: 63-66, respectively. Exemplary coding sequences for a rat TRKB signal peptide, extracellular domain, transmembrane domain, and cytoplasmic domain are set forth in SEQ ID NOS: 67-70, respectively. Exemplary coding sequences for a human TRKB signal peptide, extracellular domain, transmembrane domain, and cytoplasmic domain are set forth in SEQ ID NOS: 71-74, respectively.

For example, all or part of the region of the TrkB locus encoding the signal peptide can be humanized, and/or all or part of the region of the TrkB locus encoding the extracellular domain can be humanized, and/or all or part of the region of the TrkB locus encoding the transmembrane domain can be humanized, and/or all or part of the region of

the TrkB locus encoding the cytoplasmic domain can be humanized. In one example, all or part of the region of the TrkB locus encoding the extracellular domain is humanized. Optionally, the CDS of the human TRKB extracellular domain comprises, consists essentially of, or consists of a sequence that is at least 85%, 90%, 95%, 96%, 97%, 98%, 99%, or 100% identical to SEQ ID NO: 72 (or degenerates thereof). The TRKB protein can retain the activity of the native TRKB (e.g., retains the ability to become phosphorylated, retains the ability to activate downstream signaling pathways such as the PI3K/AKT and MAPK/ERK pathways, or retains the ability to regulate neuron survival, proliferation, migration, differentiation, or synapse formation and plasticity or produce any of the phenotypes disclosed elsewhere herein). For example, the region of the TrkB locus encoding the extracellular domain can be humanized such that a chimeric TRKB protein is produced with an endogenous signal peptide, an endogenous cytoplasmic domain, an endogenous transmembrane domain, and a humanized extracellular domain.

One or more of the regions encoding the signal peptide, the cytoplasmic domain, the transmembrane domain, or the extracellular can remain endogenous. For example, the region encoding the signal peptide and/or the cytoplasmic domain and/or the transmembrane domain can remain endogenous. Optionally, the CDS of the endogenous TRKB signal peptide comprises, consists essentially of, or consists of a sequence that is at least 85%, 90%, 95%, 96%, 97%, 98%, 99%, or 100% identical to SEQ ID NO: 63 or 67 (or degenerates thereof). Optionally, the CDS of the endogenous TRKB transmembrane domain comprises, consists essentially of, or consists of a sequence that is at least 85%, 90%, 95%, 96%, 97%, 98%, 99%, or 100% identical to SEQ ID NO: 65 or 69 (or degenerates thereof). Optionally, the CDS of the endogenous TRKB cytoplasmic domain comprises, consists essentially of, or consists of a sequence that is at least 85%, 90%, 95%, 96%, 97%, 98%, 99%, or 100% identical to SEQ ID NO: 66 or 70 (or degenerates thereof). In each case, the TRKB protein can retain the activity of the native TRKB.

The TRKB protein encoded by the humanized TRKB locus can comprise one or more domains that are from a human TRKB protein and/or one or more domains that are from an endogenous (i.e., native) TRKB protein. Exemplary amino acid sequences for a mouse TRKB signal peptide, extracellular domain, transmembrane domain, and cytoplasmic domain are set forth in SEQ ID NOS: 51-54, respectively. Exemplary amino acid sequences for a rat TRKB signal peptide, extracellular domain, transmembrane domain, and cytoplasmic domain are set forth in SEQ ID NOS: 55-58, respectively. Exemplary amino acid sequences for a human TRKB signal peptide, extracellular domain, transmembrane domain, and cytoplasmic domain are set forth in SEQ ID NOS: 59-62, respectively.

The TRKB protein can comprise one or more or all of a human TRKB signal peptide, a human TRKB extracellular domain, a human TRKB transmembrane domain, and a human TRKB cytoplasmic domain. As one example, the TRKB protein can comprise a human TRKB extracellular domain.

The TRKB protein encoded by the humanized TRKB locus can also comprise one or more domains that are from the endogenous (i.e., native) non-human animal TRKB protein. As one example, the TRKB protein encoded by the humanized TRKB locus can comprise a signal peptide from the endogenous (i.e., native) non-human animal TRKB protein and/or a cytoplasmic domain from the endogenous

(i.e., native) non-human animal TRKB protein and/or a transmembrane domain from the endogenous (i.e., native) non-human animal TRKB protein.

Domains in a chimeric TRKB protein that are from a human TRKB protein can be encoded by a fully humanized sequence (i.e., the entire sequence encoding that domain is replaced with the orthologous human TRKB sequence) or can be encoded by a partially humanized sequence (i.e., some of the sequence encoding that domain is replaced with the orthologous human TRKB sequence, and the remaining endogenous (i.e., native) sequence encoding that domain encodes the same amino acids as the orthologous human TRKB sequence such that the encoded domain is identical to that domain in the human TRKB protein). Likewise, domains in a chimeric protein that are from the endogenous TRKB protein can be encoded by a fully endogenous sequence (i.e., the entire sequence encoding that domain is the endogenous TrkB sequence) or can be encoded by a partially humanized sequence (i.e., some of the sequence encoding that domain is replaced with the orthologous human TRKB sequence, but the orthologous human TRKB sequence encodes the same amino acids as the replaced endogenous TrkB sequence such that the encoded domain is identical to that domain in the endogenous TRKB protein). For example part of the region of the TrkB locus encoding the transmembrane domain (e.g., encoding the N-terminal region of the transmembrane domain) can be replaced with orthologous human TRKB sequence, wherein the amino acid sequence of the region of the transmembrane domain encoded by the orthologous human TRKB sequence is identical to the corresponding endogenous amino acid sequence.

As one example, the TRKB protein encoded by the humanized TRKB locus can comprise a human TRKB extracellular domain. Optionally, the human TRKB extracellular domain comprises, consists essentially of, or consists of a sequence that is at least 85%, 90%, 95%, 96%, 97%, 98%, 99%, or 100% identical to SEQ ID NO: 60. The TRKB protein retains the activity of the native TRKB (e.g., retains the ability to become phosphorylated, retains the ability to activate downstream signaling pathways such as the PI3K/AKT and MAPK/ERK pathways, or retains the ability to regulate neuron survival, proliferation, migration, differentiation, or synapse formation and plasticity or produce any of the phenotypes disclosed elsewhere herein). As another example, the TRKB protein encoded by the humanized TRKB locus can comprise an endogenous non-human animal TRKB cytoplasmic domain (e.g., a mouse TRKB cytoplasmic domain or a rat TRKB cytoplasmic domain). Optionally, the non-human animal TRKB cytoplasmic domain comprises, consists essentially of, or consists of a sequence that is at least 85%, 90%, 95%, 96%, 97%, 98%, 99%, or 100% identical to SEQ ID NO: 54 or 58. As another example, the TRKB protein encoded by the humanized TRKB locus can comprise an endogenous non-human animal TRKB transmembrane domain (e.g., a mouse TRKB transmembrane domain or a rat TRKB transmembrane domain). Optionally, the non-human animal TRKB transmembrane domain comprises, consists essentially of, or consists of a sequence that is at least 85%, 90%, 95%, 96%, 97%, 98%, 99%, or 100% identical to SEQ ID NO: 53 or 57. As another example, the TRKB protein encoded by the humanized TRKB locus can comprise an endogenous non-human animal TRKB signal peptide (e.g., a mouse TRKB signal peptide or a rat TRKB signal peptide). Optionally, the non-human animal TRKB signal peptide comprises, consists essentially of, or consists of a sequence that is at least 85%,

90%, 95%, 96%, 97%, 98%, 99%, or 100% identical to SEQ ID NO: 51 or 55. In each case, the TRKB protein can retain the activity of the native TRKB. For example, the TRKB protein encoded by the humanized TRKB locus can comprise, consist essentially of, or consist of a sequence that is at least 85%, 90%, 95%, 96%, 97%, 98%, 99%, or 100% identical to SEQ ID NO: 4 or 5. Optionally, the TRKB CDS encoded by the humanized TRKB locus can comprise, consist essentially of, or consist of a sequence that is at least 85%, 90%, 95%, 96%, 97%, 98%, 99%, or 100% identical to SEQ ID NO: 12 or 13 (or degenerates thereof). In each case, the TRKB protein can retain the activity of the native TRKB.

Optionally, a humanized TRKB locus can comprise other elements. Examples of such elements can include selection cassettes, reporter genes, recombinase recognition sites, or other elements. Alternatively, the humanized TRKB locus can lack other elements (e.g., can lack a selection marker or selection cassette). Examples of suitable reporter genes and reporter proteins are disclosed elsewhere herein. Examples of suitable selection markers include neomycin phosphotransferase (neo_r), hygromycin B phosphotransferase (hyg_r), puromycin-N-acetyltransferase (puro_r), blasticidin S deaminase (bsr_r), xanthine/guanine phosphoribosyl transferase (gpt), and herpes simplex virus thymidine kinase (HSV-k). Examples of recombinases include Cre, FLP, and Dre recombinases. One example of a Cre recombinase gene is Crei, in which two exons encoding the Cre recombinase are separated by an intron to prevent its expression in a prokaryotic cell. Such recombinases can further comprise a nuclear localization signal to facilitate localization to the nucleus (e.g., NLS-Crei). Recombinase recognition sites include nucleotide sequences that are recognized by a site-specific recombinase and can serve as a substrate for a recombination event. Examples of recombinase recognition sites include FRT, FRT11, FRT71, attP, att, rox, and lox sites such as loxP, lox511, lox2272, lox66, lox71, loxM2, and lox5171.

Other elements such as reporter genes or selection cassettes can be self-deleting cassettes flanked by recombinase recognition sites. See, e.g., U.S. Pat. No. 8,697,851 and US 2013/0312129, each of which is herein incorporated by reference in its entirety for all purposes. As an example, the self-deleting cassette can comprise a Crei gene (comprises two exons encoding a Cre recombinase, which are separated by an intron) operably linked to a mouse Prm1 promoter and a neomycin resistance gene operably linked to a human ubiquitin promoter. By employing the Prm1 promoter, the self-deleting cassette can be deleted specifically in male germ cells of F0 animals. The polynucleotide encoding the selection marker can be operably linked to a promoter active in a cell being targeted. Examples of promoters are described elsewhere herein. As another specific example, a self-deleting selection cassette can comprise a hygromycin resistance gene coding sequence operably linked to one or more promoters (e.g., both human ubiquitin and EM7 promoters) followed by a polyadenylation signal, followed by a Crei coding sequence operably linked to one or more promoters (e.g., an mPrm1 promoter), followed by another polyadenylation signal, wherein the entire cassette is flanked by loxP sites.

The humanized TRKB locus can also be a conditional allele. For example, the conditional allele can be a multi-functional allele, as described in US 2011/0104799, herein incorporated by reference in its entirety for all purposes. For example, the conditional allele can comprise: (a) an actuating sequence in sense orientation with respect to transcription of a target gene; (b) a drug selection cassette (DSC) in

sense or antisense orientation; (c) a nucleotide sequence of interest (NSI) in antisense orientation; and (d) a conditional by inversion module (COIN, which utilizes an exon-splitting intron and an invertible gene-trap-like module) in reverse orientation. See, e.g., US 2011/0104799. The conditional allele can further comprise recombinable units that recombine upon exposure to a first recombinase to form a conditional allele that (i) lacks the actuating sequence and the DSC; and (ii) contains the NSI in sense orientation and the COIN in antisense orientation. See, e.g., US 2011/0104799.

One exemplary humanized TRKB locus (e.g., a humanized mouse TrkB locus or a humanized rat TrkB locus) is one in which a region in exon 2/coding exon 1 from the codon encoding amino acid 33, beginning just after the signal peptide (or the codon corresponding to the codon encoding amino acid 33 in mouse TrkB, rat TrkB, or human TRKB when optimally aligned with the mouse TrkB, rat TrkB, or human TRKB CDS, respectively) through exon 10 (or the exon corresponding to mouse TrkB, rat TrkB, or human TRKB exon 10 when optimally aligned with the mouse TrkB, rat TrkB, or human TRKB CDS, respectively), optionally including a portion of intron 10, is replaced with the corresponding human sequence. The replaced region encodes the extracellular domain of TRKB. See FIGS. 1 and 4 and SEQ ID NOS: 4 and 5.

C. Non-Human Animal Genomes, Non-Human Animal Cells, and Non-Human Animals Comprising a Humanized TRKB Locus

Non-human animal genomes, non-human animal cells, and non-human animals comprising a humanized TRKB locus as described elsewhere herein are provided. The genomes, cells, or non-human animals can be male or female. The genomes, cells, or non-human animals can be heterozygous or homozygous for the humanized TRKB locus. A diploid organism has two alleles at each genetic locus. Each pair of alleles represents the genotype of a specific genetic locus. Genotypes are described as homozygous if there are two identical alleles at a particular locus and as heterozygous if the two alleles differ.

The non-human animal genomes or cells provided herein can be, for example, any non-human animal genome or cell comprising a TrkB locus or a genomic locus homologous or orthologous to the human TRKB locus. The genomes can be from or the cells can be eukaryotic cells, which include, for example, fungal cells (e.g., yeast), plant cells, animal cells, mammalian cells, non-human mammalian cells, and human cells. The term "animal" includes any member of the animal kingdom, including, for example, mammals, fishes, reptiles, amphibians, birds, and worms. A mammalian cell can be, for example, a non-human mammalian cell, a rodent cell, a rat cell, a mouse cell, or a hamster cell. Other non-human mammals include, for example, non-human primates, monkeys, apes, orangutans, cats, dogs, rabbits, horses, bulls, deer, bison, livestock (e.g., bovine species such as cows, steer, and so forth; ovine species such as sheep, goats, and so forth; and porcine species such as pigs and boars). Birds include, for example, chickens, turkeys, ostrich, geese, ducks, and so forth. Domesticated animals and agricultural animals are also included. The term "non-human" excludes humans.

The cells can also be any type of undifferentiated or differentiated state. For example, a cell can be a totipotent cell, a pluripotent cell (e.g., a human pluripotent cell or a non-human pluripotent cell such as a mouse embryonic stem (ES) cell or a rat ES cell), or a non-pluripotent cell. Totipotent cells include undifferentiated cells that can give rise to any cell type, and pluripotent cells include undifferenti-

ated cells that possess the ability to develop into more than one differentiated cell types. Such pluripotent and/or totipotent cells can be, for example, ES cells or ES-like cells, such as an induced pluripotent stem (iPS) cells. ES cells include embryo-derived totipotent or pluripotent cells that are capable of contributing to any tissue of the developing embryo upon introduction into an embryo. ES cells can be derived from the inner cell mass of a blastocyst and are capable of differentiating into cells of any of the three vertebrate germ layers (endoderm, ectoderm, and mesoderm).

The cells provided herein can also be germ cells (e.g., sperm or oocytes). The cells can be mitotically competent cells or mitotically-inactive cells, meiotically competent cells or meiotically-inactive cells. Similarly, the cells can also be primary somatic cells or cells that are not a primary somatic cell. Somatic cells include any cell that is not a gamete, germ cell, gametocyte, or undifferentiated stem cell. For example, the cells can be neurons, such as hippocampal neurons or cortical neurons.

Suitable cells provided herein also include primary cells. Primary cells include cells or cultures of cells that have been isolated directly from an organism, organ, or tissue. Primary cells include cells that are neither transformed nor immortal. They include any cell obtained from an organism, organ, or tissue which was not previously passed in tissue culture or has been previously passed in tissue culture but is incapable of being indefinitely passed in tissue culture. Such cells can be isolated by conventional techniques and include, for example, hippocampal neurons or cortical neurons.

Other suitable cells provided herein include immortalized cells. Immortalized cells include cells from a multicellular organism that would normally not proliferate indefinitely but, due to mutation or alteration, have evaded normal cellular senescence and instead can keep undergoing division. Such mutations or alterations can occur naturally or be intentionally induced. A specific example of an immortalized cell line is a neuroblastoma cell line such as N18TG2 or T48 or a cell line such as the NIH-3T3 cell line. Numerous types of immortalized cells are well known. Immortalized or primary cells include cells that are typically used for culturing or for expressing recombinant genes or proteins.

The cells provided herein also include one-cell stage embryos (i.e., fertilized oocytes or zygotes). Such one-cell stage embryos can be from any genetic background (e.g., BALB/c, C57BL/6, 129, or a combination thereof for mice), can be fresh or frozen, and can be derived from natural breeding or in vitro fertilization.

The cells provided herein can be normal, healthy cells, or can be diseased or mutant-bearing cells.

Non-human animals comprising a humanized TRKB locus as described herein can be made by the methods described elsewhere herein. The term "animal" includes any member of the animal kingdom, including, for example, mammals, fishes, reptiles, amphibians, birds, and worms. In a specific example, the non-human animal is a non-human mammal. Non-human mammals include, for example, non-human primates, monkeys, apes, orangutans, cats, dogs, horses, bulls, deer, bison, sheep, rabbits, rodents (e.g., mice, rats, hamsters, and guinea pigs), and livestock (e.g., bovine species such as cows and steer; ovine species such as sheep and goats; and porcine species such as pigs and boars). Birds include, for example, chickens, turkeys, ostrich, geese, and ducks. Domesticated animals and agricultural animals are also included. The term "non-human animal" excludes

humans. Preferred non-human animals include, for example, rodents, such as mice and rats.

The non-human animals can be from any genetic background. For example, suitable mice can be from a 129 strain, a C57BL/6 strain, a mix of 129 and C57BL/6, a BALB/c strain, or a Swiss Webster strain. Examples of 129 strains include 129P1, 129P2, 129P3, 129X1, 129S1 (e.g., 129S1/SV, 129S1/SvIm), 129S2, 129S4, 129S5, 129S9/SvEvH, 129S6 (129/SvEvTac), 129S7, 129S8, 129T1, and 129T2. See, e.g., Festing et al. (1999) *Mammalian Genome* 10:836, herein incorporated by reference in its entirety for all purposes. Examples of C57BL strains include C57BL/A, C57BL/An, C57BL/GrFa, C57BL/Kal_wN, C57BL/6, C57BL/6J, C57BL/6ByJ, C57BL/6NJ, C57BL/10, C57BL/10ScSn, C57BL/10Cr, and C57BL/Ola. Suitable mice can also be from a mix of an aforementioned 129 strain and an aforementioned C57BL/6 strain (e.g., 50% 129 and 50% C57BL/6). Likewise, suitable mice can be from a mix of aforementioned 129 strains or a mix of aforementioned BL/6 strains (e.g., the 129S6 (129/SvEvTac) strain).

Similarly, rats can be from any rat strain, including, for example, an ACI rat strain, a Dark Agouti (DA) rat strain, a Wistar rat strain, a LEA rat strain, a Sprague Dawley (SD) rat strain, or a Fischer rat strain such as Fisher F344 or Fisher F6. Rats can also be obtained from a strain derived from a mix of two or more strains recited above. For example, a suitable rat can be from a DA strain or an ACI strain. The ACI rat strain is characterized as having black agouti, with white belly and feet and an RT1^{av1} haplotype. Such strains are available from a variety of sources including Harlan Laboratories. The Dark Agouti (DA) rat strain is characterized as having an agouti coat and an RT1^{av1} haplotype. Such rats are available from a variety of sources including Charles River and Harlan Laboratories. Some suitable rats can be from an inbred rat strain. See, e.g., US 2014/0235933, herein incorporated by reference in its entirety for all purposes.

III. Methods of Using Non-Human Animals Comprising a Humanized TRKB Locus for Assessing Efficacy of Human-TRKB-Targeting Reagents In Vivo or Ex Vivo

Various methods are provided for using the non-human animals comprising a humanized TRKB locus as described elsewhere herein for assessing or optimizing delivery or efficacy of human-TRKB-targeting reagents (e.g., therapeutic agonist molecules) in vivo or ex vivo. Because the non-human animals comprise a humanized TRKB locus, the non-human animals will more accurately reflect the efficacy of a human-TRKB-targeting reagent.

A. Methods of Testing Efficacy of Human-TRKB-Targeting Reagents In Vivo or Ex Vivo

Various methods are provided for assessing delivery or efficacy of human-TRKB-targeting reagents in vivo using non-human animals comprising a humanized TRKB locus as described elsewhere herein. Such methods can comprise: (a) introducing into the non-human animal a human-TRKB-targeting reagent; and (b) assessing the activity of the human-TRKB-targeting reagent.

The human-TRKB-targeting reagent can be a human-TRKB-targeting antibody or antigen-binding protein or any other large molecule or small molecule that targets human TRKB. Alternatively, the human-TRKB-targeting reagent can be any biological or chemical agent that targets the human TRKB locus (the human TRKB gene), the human TRKB mRNA, or the human TRKB protein. Examples of human-TRKB-targeting reagents are disclosed elsewhere herein.

Such human-TRKB-targeting reagents can be administered by any delivery method (e.g., injection, AAV, LNP, or

HDD) as disclosed in more detail elsewhere herein and by any route of administration. Means of delivering therapeutic molecules and routes of administration are disclosed in more detail elsewhere herein. In particular methods, the reagents are delivered via injection (e.g., direct hippocampal injection, subcutaneous injection, or intravitreal injection).

Methods for assessing activity of the human-TRKB-targeting reagent are well-known and are provided elsewhere herein. In some methods, assessing activity of the human-TRKB-targeting reagent (e.g., agonist activity or inhibitory activity) comprises assessing TRKB activity (e.g., TRKB phosphorylation, TRKB-mediated activation of downstream signaling pathways, or TRKB-induced phenotypes) as disclosed elsewhere herein. Assessment of activity can be in any cell type, any tissue type, or any organ type as disclosed elsewhere herein. In some methods, assessment of activity is in brain tissue (e.g., hippocampus or striatum) or neurons (e.g., retinal ganglion cells, hippocampal neurons, or cortical neurons).

If the TRKB-targeting reagent is a genome editing reagent (e.g., a nuclease agent), such methods can comprise assessing modification of the humanized TRKB locus. For example, the assessing can comprise sequencing the humanized TRKB locus in one or more cells isolated from the non-human animal (e.g., next-generation sequencing). Assessment can comprise isolating a target organ (e.g., brain) or tissue from the non-human animal and assessing modification of humanized TRKB locus in the target organ or tissue. Assessment can also comprise assessing modification of humanized TRKB locus in two or more different cell types within the target organ or tissue. Similarly, assessment can comprise isolating a non-target organ or tissue (e.g., two or more non-target organs or tissues) from the non-human animal and assessing modification of humanized TRKB locus in the non-target organ or tissue.

Such methods can also comprise measuring expression levels of the mRNA produced by the humanized TRKB locus, or by measuring expression levels of the protein encoded by the humanized TRKB locus. For example, protein levels can be measured in a particular cell, tissue, or organ type (e.g., brain), or secreted levels can be measured in the serum. Methods for assessing expression of TRKB mRNA or protein expressed from the humanized TRKB locus are provided elsewhere herein and are well-known.

The various methods provided above for assessing activity *in vivo* can also be used to assess the activity of human-TRKB-targeting reagents *ex vivo* as described elsewhere herein.

B. Methods of Optimizing Delivery or Efficacy of Human-TRKB-Targeting Reagent *In Vivo* or *Ex Vivo*

Various methods are provided for optimizing delivery of human-TRKB-targeting reagents to a cell or non-human animal or optimizing the activity or efficacy of human-TRKB-targeting reagents *in vivo*. Such methods can comprise, for example: (a) performing the method of testing the efficacy of a human-TRKB-targeting reagent as described above a first time in a first non-human animal or first cell; (b) changing a variable and performing the method a second time in a second non-human animal (i.e., of the same species) or a second cell with the changed variable; and (c) comparing the activity of the human-TRKB-targeting reagent in step (a) with the activity of the human-TRKB-targeting reagent in step (b), and selecting the method resulting in the higher efficacy or activity.

Methods of measuring delivery, efficacy, or activity of human-TRKB-targeting reagents are disclosed elsewhere herein. Higher efficacy can mean different things depending

on the desired effect within the non-human animal or cell. For example, higher efficacy can mean higher activity and/or higher specificity. Higher activity can be, for example, activity in activating TRKB or activity in inhibiting TRKB. It can refer to a higher percentage of cells being targeted within a particular target cell type (e.g., neurons such as retinal ganglion cells) or within a particular target tissue or organ (e.g., brain). Higher specificity can refer to higher specificity with respect to TRKB as compared to off-target effects, higher specificity with respect to the cell type targeted, or higher specificity with respect to the tissue or organ type targeted.

The variable that is changed can be any parameter. As one example, the changed variable can be the packaging or the delivery method by which the human-TRKB-targeting reagent or reagents are introduced into the cell or non-human animal. Examples of delivery methods are disclosed elsewhere herein. As another example, the changed variable can be the route of administration for introduction of the human-TRKB-targeting reagent or reagents into the cell or non-human animal. Examples of routes of administration are disclosed elsewhere herein.

As another example, the changed variable can be the concentration or amount of the human-TRKB-targeting reagent or reagents introduced. As another example, the changed variable can be the timing of introducing the human-TRKB-targeting reagent or reagents relative to the timing of assessing the activity or efficacy of the reagents. As another example, the changed variable can be the number of times or frequency with which the human-TRKB-targeting reagent or reagents are introduced. As another example, the changed variable can be the human-TRKB-targeting reagent or reagents that are introduced (e.g., comparing one reagent to a different reagent).

C. Human-TRKB-Targeting Reagents

A human-TRKB-targeting reagent can be any reagent that targets a human TRKB protein, a human TRKB gene, or a human TRKB mRNA. A human-TRKB-targeting reagent can be, for example, an agonist (i.e., a molecule that indirectly or directly activates human TRKB) or it can be an antagonist (i.e., an inhibitor or inhibitory reagent that blocks human TRKB activity). In a specific example, the human-TRKB-targeting reagent is a TRKB agonist. Human-TRKB-targeting reagents in the methods disclosed herein can be known human-TRKB-targeting reagents, can be putative human-TRKB-targeting reagents (e.g., candidate reagents designed to target human TRKB), or can be reagents being screened for human-TRKB-targeting activity.

For example, a human-TRKB-targeting reagent can be an antigen-binding protein (e.g., agonist antibody) targeting an epitope of a human TRKB protein. An example of such a reagent is the TRKB agonist antibody H4H9816P2. Other anti-TRKB antibodies are disclosed elsewhere herein. In some cases, the anti-TRKB antibodies bind human TRKB with a K_D of less than about 200 nM as measured by surface plasmon resonance at 25° C. or at 37° C. In other cases, the anti-TRKB antibodies bind human TRKB with a K_D of less than about 600 pM, less than about 300 pM, less than about 200 pM, less than about 150 pM, less than about 100 pM, less than about 80 pM, less than about 50 pM, less than about 40 pM, less than about 30 pM, less than about 20 pM, less than about 10 pM, less than about 5 pM, less than about 3 pM, or less than about 1 pM. In some cases, the anti-TRKB antibodies bind human TRKB with a dissociative half-life ($t_{1/2}$) of greater than about 10 minutes as measured by surface plasmon resonance at 25° C. or 37° C. In other cases, the anti-TRKB antibodies bind human TRKB with a $t_{1/2}$ of

greater than about 20 minutes, greater than about 50 minutes, greater than about 100 minutes, greater than about 120 minutes, greater than about 150 minutes, greater than about 300 minutes, greater than about 350 minutes, greater than about 400 minutes, greater than about 450 minutes, greater than about 500 minutes, greater than about 550 minutes, greater than about 600 minutes, greater than about 700 minutes, greater than about 800 minutes, greater than about 900 minutes, greater than about 1000 minutes, greater than about 1100 minutes, or greater than about 1200 minutes. As a specific example, the anti-TRKB antibody can comprise a set of six CDRs (HCDR1-HCDR2-HCDR3-LCDR1-LCDR2-LCDR3) selected from the groups set forth in Table 22 or substantially similar sequences having at least 90%, at least 95%, at least 98%, or at least 99% sequence identity thereto.

Other human-TRKB-targeting reagents include small molecules (e.g., agonists) targeting a human TRKB protein. Examples of small molecule TRKB agonists include 7,8-Dihydroxyflavone (7,8-DHF), deoxygedunin, LM22A-4 (N,N',N"-tris(2-hydroxyethyl)-1,3,5-benzenetricarboxamide), and LM22B-10 (2-[[4-[[4-[Bis-(2-hydroxyethyl)-amino]-phenyl]-(4-chloro-phenyl)-methyl]-phenyl]-(2-hydroxy-ethyl)-amino]-ethanol). See, e.g., Liu et al. (2015) *Translational Neurodegeneration* 5:2; Massa et al. (2010) *J. Clin. Invest.* 120(5):1774-1785; and Yang et al. (2016) *Neuropharmacology* 110:343-361, each of which is herein incorporated by reference in its entirety for all purposes. An example of a TRKB-targeting reagent that is an inhibitor is K252a. See, e.g., Yang et al. (2016) *Neuropharmacology* 110:343-361, herein incorporated by reference in its entirety for all purposes.

Other human-TRKB-targeting reagents include peptides or peptide mimetics (e.g., agonists) targeting a human TRKB protein. Examples of peptide mimetics that serve as human TRKB agonists are disclosed, e.g., in O'Leary et al. (2003) *J. Biol. Chem.* 278(28):25738-25744, herein incorporated by reference in its entirety for all purposes.

Other human-TRKB-targeting reagents can include genome editing reagents such as a nuclease agent (e.g., a Clustered Regularly Interspersed Short Palindromic Repeats (CRISPR)/CRISPR-associated (Cas) (CRISPR/Cas) nuclease, a zinc finger nuclease (ZFN), or a Transcription Activator-Like Effector Nuclease (TALEN)) that cleaves a recognition site within the human TRKB gene. Likewise, a human-TRKB-targeting reagent can be an exogenous donor nucleic acid (e.g., a targeting vector or single-stranded oligodeoxynucleotide (ssODN)) designed to recombine with the human TRKB gene).

Other human-TRKB-targeting reagents can include antisense oligonucleotides (e.g., siRNA or shRNA) targeting a human TRKB mRNA. Antisense oligonucleotides (ASOs) or antisense RNAs are short synthetic strings of nucleotides designed to prevent the expression of a targeted protein by selectively binding to the RNA that encodes the targeted protein and thereby preventing translation. These compounds bind to RNA with high affinity and selectivity through well characterized Watson-Crick base pairing (hybridization). RNA interference (RNAi) is an endogenous cellular mechanism for controlling gene expression in which small interfering RNAs (siRNAs) that are bound to the RNA-induced silencing complex (RISC) mediate the cleavage of target messenger RNA (mRNA).

The activity of any other known or putative human-TRKB-targeting reagent can also be assessed using the non-human animals disclosed herein. Similarly, any other

molecule can be screened for human-TRKB-targeting activity using the non-human animals disclosed herein.

D. Administering Human-TRKB-Targeting Reagents to Non-Human Animals or Cells

The methods disclosed herein can comprise introducing into a non-human animal or cell various molecules (e.g., human-TRKB-targeting reagents such as antibodies or small molecules), including nucleic acids, proteins, nucleic-acid-protein complexes, peptide mimetics, antigen-binding proteins, or small molecules. "Introducing" includes presenting to the cell or non-human animal the molecule (e.g., nucleic acid or protein or small molecule) in such a manner that it gains access to the interior of the cell or to the interior of cells within the non-human animal. The introducing can be accomplished by any means. If multiple components are introduced, they can be introduced simultaneously or sequentially in any combination. In addition, two or more of the components can be introduced into the cell or non-human animal by the same delivery method or different delivery methods. Similarly, two or more of the components can be introduced into a non-human animal by the same route of administration or different routes of administration.

Molecules introduced into the non-human animal or cell can be provided in compositions comprising a carrier increasing the stability of the introduced molecules (e.g., prolonging the period under given conditions of storage (e.g., -20° C., 4° C., or ambient temperature) for which degradation products remain below a threshold, such as below 0.5% by weight of the starting nucleic acid or protein; or increasing the stability in vivo). Non-limiting examples of such carriers include poly(lactic acid) (PLA) microspheres, poly(D,L-lactic-coglycolic-acid) (PLGA) microspheres, liposomes, micelles, inverse micelles, lipid cochleates, and lipid microtubules.

Various methods and compositions are provided herein to allow for introduction of a human-TRKB-targeting reagent into a cell or non-human animal. Methods for introducing nucleic acids into various cell types are known and include, for example, stable transfection methods, transient transfection methods, and virus-mediated methods.

Transfection protocols as well as protocols for introducing nucleic acid sequences into cells may vary. Non-limiting transfection methods include chemical-based transfection methods using liposomes; nanoparticles; calcium phosphate (Graham et al. (1973) *Virology* 52 (2): 456-67, Bacchetti et al. (1977) *Proc. Natl. Acad. Sci. USA* 74 (4): 1590-4, and Kriegler, M (1991). *Transfer and Expression: A Laboratory Manual*. New York: W. H. Freeman and Company. pp. 96-97); dendrimers; or cationic polymers such as DEAE-dextran or polyethylenimine. Non-chemical methods include electroporation, Sono-poration, and optical transfection. Particle-based transfection includes the use of a gene gun, or magnet-assisted transfection (Bertram (2006) *Current Pharmaceutical Biotechnology* 7, 277-28). Viral methods can also be used for transfection.

Introduction of human-TRKB-targeting reagents into a cell can also be mediated by electroporation, by intracytoplasmic injection, by viral infection, by adenovirus, by adeno-associated virus, by lentivirus, by retrovirus, by transfection, by lipid-mediated transfection, or by nucleofection. Nucleofection is an improved electroporation technology that enables nucleic acid substrates to be delivered not only to the cytoplasm but also through the nuclear membrane and into the nucleus. In addition, use of nucleofection in the methods disclosed herein typically requires much fewer cells than regular electroporation (e.g., only about 2 million compared with 7 million by regular electroporation). In one

example, nucleofection is performed using the LONZA® NUCLEOFECTOR™ system.

Introduction of human-TRKB-targeting reagents into a cell (e.g., a zygote) can also be accomplished by microinjection. In zygotes (i.e., one-cell stage embryos), microinjection can be into the maternal and/or paternal pronucleus or into the cytoplasm. If the microinjection is into only one pronucleus, the paternal pronucleus is preferable due to its larger size. Microinjection of an mRNA is preferably into the cytoplasm (e.g., to deliver mRNA directly to the translation machinery), while microinjection of a protein or a polynucleotide encoding a protein or encoding an RNA is preferable into the nucleus/pronucleus. Alternatively, microinjection can be carried out by injection into both the nucleus/pronucleus and the cytoplasm: a needle can first be introduced into the nucleus/pronucleus and a first amount can be injected, and while removing the needle from the one-cell stage embryo a second amount can be injected into the cytoplasm. If a protein is injected into the cytoplasm and needs to be targeted to the nucleus, it can comprise a nuclear localization signal to ensure delivery to the nucleus/pronucleus. Methods for carrying out microinjection are well known. See, e.g., Nagy et al. (Nagy A, Gertsenstein M, Vintersten K, Behringer R., 2003, *Manipulating the Mouse Embryo*. Cold Spring Harbor, New York Cold Spring Harbor Laboratory Press); see also Meyer et al. (2010) *Proc. Natl. Acad. Sci. USA* 107:15022-15026 and Meyer et al. (2012) *Proc. Natl. Acad. Sci. USA* 109:9354-9359.

Other methods for introducing human-TRKB-targeting reagents into a cell or non-human animal can include, for example, vector delivery, particle-mediated delivery, exosome-mediated delivery, lipid-nanoparticle-mediated delivery, cell-penetrating-peptide-mediated delivery, or implantable-device-mediated delivery. As specific examples, a nucleic acid or protein can be introduced into a cell or non-human animal in a carrier such as a poly(lactic acid) (PLA) microsphere, a poly(D,L-lactic-co-glycolic-acid) (PLGA) microsphere, a liposome, a micelle, an inverse micelle, a lipid cochleate, or a lipid microtubule. Some specific examples of delivery to a non-human animal include hydrodynamic delivery, virus-mediated delivery (e.g., adeno-associated virus (AAV)-mediated delivery), and lipid-nanoparticle-mediated delivery.

Introduction of human-TRKB-targeting reagents into cells or non-human animals can be accomplished by hydrodynamic delivery (HDD). Hydrodynamic delivery has emerged as a method for intracellular DNA delivery in vivo. For gene delivery to parenchymal cells, only essential DNA sequences need to be injected via a selected blood vessel, eliminating safety concerns associated with current viral and synthetic vectors. When injected into the bloodstream, DNA is capable of reaching cells in the different tissues accessible to the blood. Hydrodynamic delivery employs the force generated by the rapid injection of a large volume of solution into the incompressible blood in the circulation to overcome the physical barriers of endothelium and cell membranes that prevent large and membrane-impermeable compounds from entering parenchymal cells. In addition to the delivery of DNA, this method is useful for the efficient intracellular delivery of RNA, proteins, and other small compounds in vivo. See, e.g., Bonamassa et al. (2011) *Pharm. Res.* 28(4): 694-701, herein incorporated by reference in its entirety for all purposes.

Introduction of human-TRKB-targeting reagents can also be accomplished by virus-mediated delivery, such as AAV-mediated delivery or lentivirus-mediated delivery. Other exemplary viruses/viral vectors include retroviruses, adeno-

viruses, vaccinia viruses, poxviruses, and herpes simplex viruses. The viruses can infect dividing cells, non-dividing cells, or both dividing and non-dividing cells. The viruses can integrate into the host genome or alternatively do not integrate into the host genome. Such viruses can also be engineered to have reduced immunity. The viruses can be replication-competent or can be replication-defective (e.g., defective in one or more genes necessary for additional rounds of virion replication and/or packaging). Viruses can cause transient expression, long-lasting expression (e.g., at least 1 week, 2 weeks, 1 month, 2 months, or 3 months), or permanent expression. Exemplary viral titers (e.g., AAV titers) include 10^{12} , 10^{13} , 10^{14} , 10^{15} , and 10^{16} vector genomes/mL.

The ssDNA AAV genome consists of two open reading frames, Rep and Cap, flanked by two inverted terminal repeats that allow for synthesis of the complementary DNA strand. When constructing an AAV transfer plasmid, the transgene is placed between the two ITRs, and Rep and Cap can be supplied in trans. In addition to Rep and Cap, AAV can require a helper plasmid containing genes from adenovirus. These genes (E4, E2a, and VA) mediated AAV replication. For example, the transfer plasmid, Rep/Cap, and the helper plasmid can be transfected into HEK293 cells containing the adenovirus gene E1+ to produce infectious AAV particles. Alternatively, the Rep, Cap, and adenovirus helper genes may be combined into a single plasmid. Similar packaging cells and methods can be used for other viruses, such as retroviruses.

Multiple serotypes of AAV have been identified. These serotypes differ in the types of cells they infect (i.e., their tropism), allowing preferential transduction of specific cell types. Serotypes for CNS tissue include AAV1, AAV2, AAV4, AAV5, AAV8, and AAV9. Serotypes for heart tissue include AAV1, AAV8, and AAV9. Serotypes for kidney tissue include AAV2. Serotypes for lung tissue include AAV4, AAV5, AAV6, and AAV9. Serotypes for pancreas tissue include AAV8. Serotypes for photoreceptor cells include AAV2, AAV5, and AAV8. Serotypes for retinal pigment epithelium tissue include AAV1, AAV2, AAV4, AAV5, and AAV8. Serotypes for skeletal muscle tissue include AAV1, AAV6, AAV7, AAV8, and AAV9. Serotypes for liver tissue include AAV7, AAV8, and AAV9, and particularly AAV8.

Tropism can be further refined through pseudotyping, which is the mixing of a capsid and a genome from different viral serotypes. For example AAV2/5 indicates a virus containing the genome of serotype 2 packaged in the capsid from serotype 5. Use of pseudotyped viruses can improve transduction efficiency, as well as alter tropism. Hybrid capsids derived from different serotypes can also be used to alter viral tropism. For example, AAV-DJ contains a hybrid capsid from eight serotypes and displays high infectivity across a broad range of cell types in vivo. AAV-DJ8 is another example that displays the properties of AAV-DJ but with enhanced brain uptake. AAV serotypes can also be modified through mutations. Examples of mutational modifications of AAV2 include Y444F, Y500F, Y730F, and S662V. Examples of mutational modifications of AAV3 include Y705F, Y731F, and T492V. Examples of mutational modifications of AAV6 include S663V and T492V. Other pseudotyped/modified AAV variants include AAV2/1, AAV2/6, AAV2/7, AAV2/8, AAV2/9, AAV2.5, AAV8.2, and AAV/SASTG.

To accelerate transgene expression, self-complementary AAV (scAAV) variants can be used. Because AAV depends on the cell's DNA replication machinery to synthesize the

complementary strand of the AAV's single-stranded DNA genome, transgene expression may be delayed. To address this delay, scAAV containing complementary sequences that are capable of spontaneously annealing upon infection can be used, eliminating the requirement for host cell DNA

synthesis. To increase packaging capacity, longer transgenes may be split between two AAV transfer plasmids, the first with a 3' splice donor and the second with a 5' splice acceptor. Upon co-infection of a cell, these viruses form concatemers, are spliced together, and the full-length transgene can be expressed. Although this allows for longer transgene expression, expression is less efficient. Similar methods for increasing capacity utilize homologous recombination. For example, a transgene can be divided between two transfer plasmids but with substantial sequence overlap such that co-expression induces homologous recombination and expression of the full-length transgene.

Introduction of human-TRKB-targeting reagents can also be accomplished by lipid nanoparticle (LNP)-mediated delivery. Lipid formulations can protect biological molecules from degradation while improving their cellular uptake. Lipid nanoparticles are particles comprising a plurality of lipid molecules physically associated with each other by intermolecular forces. These include microspheres (including unilamellar and multilamellar vesicles, e.g., liposomes), a dispersed phase in an emulsion, micelles, or an internal phase in a suspension. Such lipid nanoparticles can be used to encapsulate one or more nucleic acids or proteins for delivery. Formulations which contain cationic lipids are useful for delivering polyanions such as nucleic acids. Other lipids that can be included are neutral lipids (i.e., uncharged or zwitterionic lipids), anionic lipids, helper lipids that enhance transfection, and stealth lipids that increase the length of time for which nanoparticles can exist in vivo. Examples of suitable cationic lipids, neutral lipids, anionic lipids, helper lipids, and stealth lipids can be found in WO 2016/010840 A1, herein incorporated by reference in its entirety for all purposes. An exemplary lipid nanoparticle can comprise a cationic lipid and one or more other components. In one example, the other component can comprise a helper lipid such as cholesterol. In another example, the other components can comprise a helper lipid such as cholesterol and a neutral lipid such as DSPC. In another example, the other components can comprise a helper lipid such as cholesterol, an optional neutral lipid such as DSPC, and a stealth lipid such as S010, S024, S027, S031, or S033.

The mode of delivery can be selected to decrease immunogenicity. For example, if multiple components are delivered, they may be delivered by different modes (e.g., bimodal delivery). These different modes may confer different pharmacodynamics or pharmacokinetic properties on the subject delivered molecule. For example, the different modes can result in different tissue distribution, different half-life, or different temporal distribution. Some modes of delivery (e.g., delivery of a nucleic acid vector that persists in a cell by autonomous replication or genomic integration) result in more persistent expression and presence of the molecule, whereas other modes of delivery are transient and less persistent (e.g., delivery of an RNA or a protein).

Administration in vivo can be by any suitable route including, for example, parenteral, intravenous, oral, subcutaneous, intra-arterial, intracranial, intrathecal, intraperitoneal, topical, intranasal, or intramuscular. Systemic modes of administration include, for example, oral and parenteral routes. Examples of parenteral routes include intravenous, intraarterial, intraosseous, intramuscular, intradermal, sub-

cutaneous, intranasal, and intraperitoneal routes. A specific example is intravenous infusion. Nasal instillation and intravitreal injection are other specific examples. Local modes of administration include, for example, intrathecal, intracerebroventricular, intraparenchymal (e.g., localized intraparenchymal delivery to the striatum (e.g., into the caudate or into the putamen), cerebral cortex, precentral gyms, hippocampus (e.g., into the dentate gyms or CA3 region), temporal cortex, amygdala, frontal cortex, thalamus, cerebellum, medulla, hypothalamus, tectum, tegmentum, or substantia nigra), intraocular, intraorbital, subconjunctival, intravitreal, subretinal, and transscleral routes. Significantly smaller amounts of the components (compared with systemic approaches) may exert an effect when administered locally (for example, intraparenchymal or intravitreal) compared to when administered systemically (for example, intravenously). Local modes of administration may also reduce or eliminate the incidence of potentially toxic side effects that may occur when therapeutically effective amounts of a component are administered systemically. In a specific example, a human-TRKB-targeting reagents is administered via direct hippocampal injection, subcutaneous injection, or intravitreal injection.

Compositions comprising human-TRKB-targeting reagents can be formulated using one or more physiologically and pharmaceutically acceptable carriers, diluents, excipients or auxiliaries. The formulation can depend on the route of administration chosen. The term "pharmaceutically acceptable" means that the carrier, diluent, excipient, or auxiliary is compatible with the other ingredients of the formulation and not substantially deleterious to the recipient thereof.

The frequency of administration and the number of dosages can be depend on the half-life of the human-TRKB-targeting reagents and the route of administration among other factors. The introduction of human-TRKB-targeting reagents into the cell or non-human animal can be performed one time or multiple times over a period of time. For example, the introduction can be performed at least two times over a period of time, at least three times over a period of time, at least four times over a period of time, at least five times over a period of time, at least six times over a period of time, at least seven times over a period of time, at least eight times over a period of time, at least nine times over a period of time, at least ten times over a period of time, at least eleven times, at least twelve times over a period of time, at least thirteen times over a period of time, at least fourteen times over a period of time, at least fifteen times over a period of time, at least sixteen times over a period of time, at least seventeen times over a period of time, at least eighteen times over a period of time, at least nineteen times over a period of time, or at least twenty times over a period of time.

E. Measuring Delivery, Activity, or Efficacy of Human-TRKB-Targeting Reagents In Vivo or Ex Vivo

The methods disclosed herein can further comprise detecting or measuring activity of human-TRKB-targeting reagents. Measuring the activity of such reagents (e.g., agonist activity or inhibitor activity) can comprise measuring TRKB activity. TRKB activity can be measured by any known means. For example, TRKB phosphorylation can be assessed (e.g., in the brain or neurons), activation of downstream pathways such as PI3K/AKT and MAPK/ERK by TRKB can be assessed (e.g., in the brain or neurons, such as primary cortical neurons), or cell survival can be assessed (e.g., neuron cell survival, such as retinal ganglion cell survival). For example, phosphorylation or activation of

downstream signaling pathways can be assessed at 15 minutes, 30 minutes, 1 hour, 2 hours, 4 hours, or 18 hours post-dosing. Increases in TRKB phosphorylation, activation of downstream signaling pathways, or cell survival can be indications of TRKB activation, whereas decreases can be indications of TRKB inhibition.

In non-human animals, the assessing can comprise assessing one or more or all of body weight, body composition, metabolism, and locomotion relative to a control-non-human animal (e.g., at 12 hours, 24 hours, 48 hours, 72 hours, 96 hours, or 120 hours post-dosing). See, e.g., Lin et al. (2008) *PLoS ONE* 3(4):e1900; Rios et al. (2013) *Trends in Neurosciences* 36(2):83-90; and Zorner et al. (2003) *Biol. Psychiatry* 54:972-982, each of which is herein incorporated by reference in its entirety for all purposes. Assessing changes in body composition can comprise, for example, assessing lean mass and/or fat mass. Assessing changes in metabolism can comprise, for example, assessing changes in food consumption and/or water consumption. Decreases in body weight, fat mass, lean mass, food intake, and water intake can be indications of TRKB activation, whereas increases can be indications of TRKB inhibition. Increases in locomotion can be indications of TRKB activation, whereas decreases can be indications of TRKB inhibition.

The assessing can comprise assessing neuroprotective activity. As one example, cell survival can be assessed in non-human animals. For example, rodent retinal ganglion cells (RGCs) are often used to study neurodegenerative processes associated with axonal lesion as well as to assay neuroprotective therapies. See, e.g., Nadal-Nicolás et al. (2009) *Invest. Ophthalmol. Vis. Sci.* 50(8):3860-3868, herein incorporated by reference in its entirety for all purposes. Retinal ganglion cell survival/viability can be assessed (e.g., in a complete optic nerve transection model after optic nerve injury) following treatment with a human-TRKB targeting reagent relative to a control non-human animal. For example, retinal ganglion cell survival/viability can be assessed in a complete optic nerve transection model after optic nerve injury. See, e.g., Nadal-Nicolás et al. (2009) *Invest. Ophthalmol. Vis. Sci.* 50(8):3860-3868, herein incorporated by reference in its entirety for all purposes. As another example, retinal ganglion cell survival/viability can be assessed in an optic nerve crush model. In this model, the crush injury to the optic nerve leads to gradual retinal ganglion cells apoptosis. See, e.g., Tang et al. (2011) *J. Vis. Exp.* 50:2685, herein incorporated by reference in its entirety for all purposes. Retinal ganglion cell survival/viability can be assessed, for example, by measuring retinal ganglion cell density (e.g., in retinas dissected and stained for retinal ganglion cells). Increased survival/viability can be an indication of TRKB activation, whereas decreased survival/viability can be an indication of TRKB inhibition.

If the human-TRKB-targeting reagent is a genome editing reagent, the measuring can comprise assessing the humanized TRKB locus for modifications. Various methods can be used to identify cells having a targeted genetic modification. The screening can comprise a quantitative assay for assessing modification of allele (MOA) of a parental chromosome. For example, the quantitative assay can be carried out via a quantitative PCR, such as a real-time PCR (qPCR). The real-time PCR can utilize a first primer set that recognizes the target locus and a second primer set that recognizes a non-targeted reference locus. The primer set can comprise a fluorescent probe that recognizes the amplified sequence. Other examples of suitable quantitative assays include fluorescence-mediated in situ hybridization (FISH), comparative genomic hybridization, isothermic DNA amplification,

quantitative hybridization to an immobilized probe(s), INVADER® Probes, TAQMAN® Molecular Beacon probes, or ECLIPSE™ probe technology (see, e.g., US 2005/0144655, herein incorporated by reference in its entirety for all purposes). Next-generation sequencing (NGS) can also be used for screening. Next-generation sequencing can also be referred to as “NGS” or “massively parallel sequencing” or “high throughput sequencing.” NGS can be used as a screening tool in addition to the MOA assays to define the exact nature of the targeted genetic modification and whether it is consistent across cell types or tissue types or organ types.

The assessing in a non-human animal can be in any cell type from any tissue or organ. For example, the assessment can be in multiple cell types from the same tissue or organ (e.g., the brain) or in cells from multiple locations within the tissue or organ (e.g., hippocampus and striatum). This can provide information about which cell types within a target tissue or organ are being targeted or which sections of a tissue or organ are being reached by the human-TRKB-targeting reagent. As another example, the assessment can be in multiple types of tissue or in multiple organs. In methods in which a particular tissue, organ, or cell type is being targeted, this can provide information about how effectively that tissue or organ is being targeted and whether there are off-target effects in other tissues or organs.

If the reagent is designed to inactivate the humanized TRKB locus, affect expression of the humanized TRKB locus, or prevent translation of the humanized TRKB mRNA, the measuring can comprise assessing humanized TRKB mRNA or protein expression. This measuring can be within the brain or particular cell types (e.g., neurons such as retinal ganglion cells).

IV. Methods of Making Non-Human Animals Comprising a Humanized TRKB Locus

Various methods are provided for making a non-human animal genome, non-human animal cell, or non-human animal comprising a humanized TRKB locus as disclosed elsewhere herein. Any convenient method or protocol for producing a genetically modified organism is suitable for producing such a genetically modified non-human animal. See, e.g., Cho et al. (2009) *Current Protocols in Cell Biology* 42:19.11:19.11.1-19.11.22 and Gama Sosa et al. (2010) *Brain Strutt. Funct.* 214(2-3):91-109, each of which is herein incorporated by reference in its entirety for all purposes. Such genetically modified non-human animals can be generated, for example, through gene knock-in at a targeted TrkB locus.

For example, the method of producing a non-human animal comprising a humanized TRKB locus can comprise: (1) modifying the genome of a pluripotent cell to comprise the humanized TRKB locus; (2) identifying or selecting the genetically modified pluripotent cell comprising the humanized TRKB locus; (3) introducing the genetically modified pluripotent cell into a non-human animal host embryo; and (4) implanting and gestating the host embryo in a surrogate mother. For example, the method of producing a non-human animal comprising a humanized TRKB locus can comprise: (1) modifying the genome of a pluripotent cell to comprise the humanized TRKB locus; (2) identifying or selecting the genetically modified pluripotent cell comprising the humanized TRKB locus; (3) introducing the genetically modified pluripotent cell into a non-human animal host embryo; and (4) gestating the host embryo in a surrogate mother. Optionally, the host embryo comprising modified pluripotent cell (e.g., a non-human ES cell) can be incubated until the blastocyst stage before being implanted into and gestated in

the surrogate mother to produce an F0 non-human animal. The surrogate mother can then produce an F0 generation non-human animal comprising the humanized TRKB locus.

The methods can further comprise identifying a cell or animal having a modified target genomic locus. Various methods can be used to identify cells and animals having a targeted genetic modification.

The step of modifying the genome can, for example, utilize exogenous repair templates (e.g., targeting vectors) to modify a TrkB locus to comprise a humanized TRKB locus disclosed herein. As one example, the targeting vector can be for generating a humanized TRKB gene at an endogenous TrkB locus (e.g., endogenous non-human animal TrkB locus), wherein the targeting vector comprises a 5' homology arm targeting a 5' target sequence at the endogenous TrkB locus and a 3' homology arm targeting a 3' target sequence at the endogenous TrkB locus. Exogenous repair templates can also comprise nucleic acid inserts including segments of DNA to be integrated in the TrkB locus. Integration of a nucleic acid insert in the TrkB locus can result in addition of a nucleic acid sequence of interest in the TrkB locus, deletion of a nucleic acid sequence of interest in the TrkB locus, or replacement of a nucleic acid sequence of interest in the TrkB locus (i.e., deletion and insertion). The homology arms can flank an insert nucleic acid comprising human TRKB sequence to generate the humanized TRKB locus (e.g., for deleting a segment of the endogenous TrkB locus and replacing with an orthologous human TRKB sequence).

The exogenous repair templates can be for non-homologous-end-joining-mediated insertion or homologous recombination. Exogenous repair templates can comprise deoxyribonucleic acid (DNA) or ribonucleic acid (RNA), they can be single-stranded or double-stranded, and they can be in linear or circular form. For example, a repair template can be a single-stranded oligodeoxynucleotide (ssODN).

Exogenous repair templates can also comprise a heterologous sequence that is not present at an untargeted endogenous TrkB locus. For example, an exogenous repair template can comprise a selection cassette, such as a selection cassette flanked by recombinase recognition sites.

Some exogenous repair templates comprise homology arms. If the exogenous repair template acid also comprises a nucleic acid insert, the homology arms can flank the nucleic acid insert. For ease of reference, the homology arms are referred to herein as 5' and 3' (i.e., upstream and downstream) homology arms. This terminology relates to the relative position of the homology arms to the nucleic acid insert within the exogenous repair template. The 5' and 3' homology arms correspond to regions within the TrkB locus, which are referred to herein as "5' target sequence" and "3' target sequence," respectively.

A homology arm and a target sequence "correspond" or are "corresponding" to one another when the two regions share a sufficient level of sequence identity to one another to act as substrates for a homologous recombination reaction. The term "homology" includes DNA sequences that are either identical or share sequence identity to a corresponding sequence. The sequence identity between a given target sequence and the corresponding homology arm found in the exogenous repair template can be any degree of sequence identity that allows for homologous recombination to occur. For example, the amount of sequence identity shared by the homology arm of the exogenous repair template (or a fragment thereof) and the target sequence (or a fragment thereof) can be at least 50%, 55%, 60%, 65%, 70%, 75%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or

100% sequence identity, such that the sequences undergo homologous recombination. Moreover, a corresponding region of homology between the homology arm and the corresponding target sequence can be of any length that is sufficient to promote homologous recombination. In some targeting vectors, the intended mutation in the endogenous TrkB locus is included in an insert nucleic acid flanked by the homology arms.

In cells other than one-cell stage embryos, the exogenous repair template can be a "large targeting vector" or "LTVEC," which includes targeting vectors that comprise homology arms that correspond to and are derived from nucleic acid sequences larger than those typically used by other approaches intended to perform homologous recombination in cells. LTVECs also include targeting vectors comprising nucleic acid inserts having nucleic acid sequences larger than those typically used by other approaches intended to perform homologous recombination in cells. For example, LTVECs make possible the modification of large loci that cannot be accommodated by traditional plasmid-based targeting vectors because of their size limitations. For example, the targeted locus can be (i.e., the 5' and 3' homology arms can correspond to) a locus of the cell that is not targetable using a conventional method or that can be targeted only incorrectly or only with significantly low efficiency in the absence of a nick or double-strand break induced by a nuclease agent (e.g., a Cas protein). LTVECs can be of any length and are typically at least 10 kb in length. The sum total of the 5' homology arm and the 3' homology arm in an LTVEC is typically at least 10 kb.

The screening step can comprise, for example, a quantitative assay for assessing modification of allele (MOA) of a parental chromosome. For example, the quantitative assay can be carried out via a quantitative PCR, such as a real-time PCR (qPCR). The real-time PCR can utilize a first primer set that recognizes the target locus and a second primer set that recognizes a non-targeted reference locus. The primer set can comprise a fluorescent probe that recognizes the amplified sequence.

Other examples of suitable quantitative assays include fluorescence-mediated in situ hybridization (FISH), comparative genomic hybridization, isothermic DNA amplification, quantitative hybridization to an immobilized probe(s), INVADER® Probes, TAQMAN® Molecular Beacon probes, or ECLIPSE™ probe technology (see, e.g., US 2005/0144655, incorporated herein by reference in its entirety for all purposes).

An example of a suitable pluripotent cell is an embryonic stem (ES) cell (e.g., a mouse ES cell or a rat ES cell). The modified pluripotent cell can be generated, for example, through recombination by (a) introducing into the cell one or more exogenous donor nucleic acids (e.g., targeting vectors) comprising an insert nucleic acid flanked, for example, by 5' and 3' homology arms corresponding to 5' and 3' target sites, wherein the insert nucleic acid comprises a human TRKB sequence to generate a humanized TRKB locus; and (b) identifying at least one cell comprising in its genome the insert nucleic acid integrated at the endogenous TrkB locus (i.e., identifying at least one cell comprising the humanized TRKB locus). The modified pluripotent cell can be generated, for example, through recombination by (a) introducing into the cell one or more targeting vectors comprising an insert nucleic acid flanked by 5' and 3' homology arms corresponding to 5' and 3' target sites, wherein the insert nucleic acid comprises a humanized TRKB locus; and (b) identifying at least one cell comprising in its genome the insert nucleic acid integrated at the target genomic locus.

Alternatively, the modified pluripotent cell can be generated by (a) introducing into the cell: (i) a nuclease agent, wherein the nuclease agent induces a nick or double-strand break at a target site within the endogenous TrkB locus; and (ii) one or more exogenous donor nucleic acids (e.g., targeting vectors) optionally comprising an insert nucleic acid flanked by, for example, 5' and 3' homology arms corresponding to 5' and 3' target sites located in sufficient proximity to the nuclease target site, wherein the insert nucleic acid comprises a human TRKB sequence to generate a humanized TRKB locus; and (c) identifying at least one cell comprising in its genome the insert nucleic acid integrated at the endogenous TrkB locus (i.e., identifying at least one cell comprising the humanized TRKB locus). Alternatively, the modified pluripotent cell can be generated by (a) introducing into the cell: (i) a nuclease agent, wherein the nuclease agent induces a nick or double-strand break at a recognition site within the target genomic locus; and (ii) one or more targeting vectors comprising an insert nucleic acid flanked by 5' and 3' homology arms corresponding to 5' and 3' target sites located in sufficient proximity to the recognition site, wherein the insert nucleic acid comprises the humanized TRKB locus; and (c) identifying at least one cell comprising a modification (e.g., integration of the insert nucleic acid) at the target genomic locus. Any nuclease agent that induces a nick or double-strand break into a desired recognition site can be used. Examples of suitable nucleases include a Transcription Activator-Like Effector Nuclease (TALEN), a zinc-finger nuclease (ZFN), a meganuclease, and Clustered Regularly Interspersed Short Palindromic Repeats (CRISPR)/CRISPR-associated (Cas) systems (e.g., CRISPR/Cas9 systems) or components of such systems (e.g., CRISPR/Cas9). See, e.g., US 2013/0309670 and US 2015/0159175, each of which is herein incorporated by reference in its entirety for all purposes.

The donor cell can be introduced into a host embryo at any stage, such as the blastocyst stage or the pre-morula stage (i.e., the 4 cell stage or the 8 cell stage). Progeny that are capable of transmitting the genetic modification through the germline are generated. See, e.g., U.S. Pat. No. 7,294,754, herein incorporated by reference in its entirety for all purposes.

Alternatively, the method of producing the non-human animals described elsewhere herein can comprise: (1) modifying the genome of a one-cell stage embryo to comprise the humanized TRKB locus using the methods described above for modifying pluripotent cells; (2) selecting the genetically modified embryo; and (3) implanting and gestating the genetically modified embryo into a surrogate mother. Alternatively, the method of producing the non-human animals described elsewhere herein can comprise: (1) modifying the genome of a one-cell stage embryo to comprise the humanized TRKB locus using the methods described above for modifying pluripotent cells; (2) selecting the genetically modified embryo; and (3) gestating the genetically modified embryo in a surrogate mother. Progeny that are capable of transmitting the genetic modification through the germline are generated.

Nuclear transfer techniques can also be used to generate the non-human mammalian animals. Briefly, methods for nuclear transfer can include the steps of: (1) enucleating an oocyte or providing an enucleated oocyte; (2) isolating or providing a donor cell or nucleus to be combined with the enucleated oocyte; (3) inserting the cell or nucleus into the enucleated oocyte to form a reconstituted cell; (4) implanting the reconstituted cell into the womb of an animal to form an embryo; and (5) allowing the embryo to develop. In such

methods, oocytes are generally retrieved from deceased animals, although they may be isolated also from either oviducts and/or ovaries of live animals. Oocytes can be matured in a variety of well-known media prior to enucleation. Enucleation of the oocyte can be performed in a number of well-known manners. Insertion of the donor cell or nucleus into the enucleated oocyte to form a reconstituted cell can be by microinjection of a donor cell under the zona pellucida prior to fusion. Fusion may be induced by application of a DC electrical pulse across the contact/fusion plane (electrofusion), by exposure of the cells to fusion-promoting chemicals, such as polyethylene glycol, or by way of an inactivated virus, such as the Sendai virus. A reconstituted cell can be activated by electrical and/or non-electrical means before, during, and/or after fusion of the nuclear donor and recipient oocyte. Activation methods include electric pulses, chemically induced shock, penetration by sperm, increasing levels of divalent cations in the oocyte, and reducing phosphorylation of cellular proteins (as by way of kinase inhibitors) in the oocyte. The activated reconstituted cells, or embryos, can be cultured in well-known media and then transferred to the womb of an animal. See, e.g., US 2008/0092249, WO 1999/005266, US 2004/0177390, WO 2008/017234, and U.S. Pat. No. 7,612,250, each of which is herein incorporated by reference in its entirety for all purposes.

The various methods provided herein allow for the generation of a genetically modified non-human F0 animal wherein the cells of the genetically modified F0 animal comprise the humanized TRKB locus. It is recognized that depending on the method used to generate the F0 animal, the number of cells within the F0 animal that have the humanized TRKB locus will vary. The introduction of the donor ES cells into a pre-morula stage embryo from a corresponding organism (e.g., an 8-cell stage mouse embryo) via for example, the VELOCIMOUSE® method allows for a greater percentage of the cell population of the F0 animal to comprise cells having the nucleotide sequence of interest comprising the targeted genetic modification. For example, at least 50%, 60%, 65%, 70%, 75%, 85%, 86%, 87%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or 100% of the cellular contribution of the non-human F0 animal can comprise a cell population having the targeted modification.

The cells of the genetically modified F0 animal can be heterozygous for the humanized TRKB locus or can be homozygous for the humanized TRKB locus.

All patent filings, websites, other publications, accession numbers and the like cited above or below are incorporated by reference in their entirety for all purposes to the same extent as if each individual item were specifically and individually indicated to be so incorporated by reference. If different versions of a sequence are associated with an accession number at different times, the version associated with the accession number at the effective filing date of this application is meant. The effective filing date means the earlier of the actual filing date or filing date of a priority application referring to the accession number if applicable. Likewise, if different versions of a publication, website or the like are published at different times, the version most recently published at the effective filing date of the application is meant unless otherwise indicated. Any feature, step, element, embodiment, or aspect of the invention can be used in combination with any other unless specifically indicated otherwise. Although the present invention has been described in some detail by way of illustration and example for purposes of clarity and understanding, it will be

apparent that certain changes and modifications may be practiced within the scope of the appended claims.

BRIEF DESCRIPTION OF THE SEQUENCES

The nucleotide and amino acid sequences listed in the accompanying sequence listing are shown using standard letter abbreviations for nucleotide bases, and three-letter code for amino acids. The nucleotide sequences follow the standard convention of beginning at the 5' end of the sequence and proceeding forward (i.e., from left to right in each line) to the 3' end. Only one strand of each nucleotide sequence is shown, but the complementary strand is understood to be included by any reference to the displayed strand. When a nucleotide sequence encoding an amino acid sequence is provided, it is understood that codon degenerate variants thereof that encode the same amino acid sequence are also provided. The amino acid sequences follow the standard convention of beginning at the amino terminus of the sequence and proceeding forward (i.e., from left to right in each line) to the carboxy terminus.

TABLE 1

Description of Sequences.		
SEQ ID NO	Type	Description
1	Protein	Mouse TRKB/NTRK2 protein (P15209-1; NP_001020245.1; NP_001269890.1)
2	Protein	Rat TRKB/NTRK2 protein (Q63604-1; NP_036863.1)
3	Protein	Human TRKB/NTRK2 protein (Q16620-4; NP_006171.2)
4	Protein	Mouse/Human Hybrid TRKB/NTRK2 protein
5	Protein	Rat/Human Hybrid TRKB/NTRK2 protein
6	DNA	Mouse TrkB/Ntrk2 cDNA (NM_001025074.2)
7	DNA	Rat TrkB/Ntrk2 cDNA (NM_012731.2)
8	DNA	Human TRKB/NTRK2 cDNA (AF410899.1)
9	DNA	Mouse TrkB/Ntrk2 CDS (CCDS ID CCDS26573.1)
10	DNA	Rat TrkB/Ntrk2 CDS
11	DNA	Human TRKB/NTRK2 CDS
12	DNA	Mouse/Human TRKB/NTRK2 CDS
13	DNA	Rat/Human TRKB/NTRK2 CDS
14	DNA	7138 hU Fwd
15	DNA	7138 hU Probe(FAM)
16	DNA	7138hU Rev
17	DNA	7138 hD Fwd
18	DNA	7138 hD Probe(Cal)
19	DNA	7138 hD Rev
20	DNA	7138U Fwd
21	DNA	7138U Probe(FAM)
22	DNA	7138U Rev
23	DNA	7138D Fwd
24	DNA	7138D Probe(Cal)
25	DNA	7138D Rev
26	DNA	rnoTU Fwd
27	DNA	rnoTU Probe (FAM)
28	DNA	rnoTU Rev
29	DNA	rnoTD Fwd
30	DNA	rnoTD Probe (Cal-Orange)
31	DNA	rnoTD Rev
32	DNA	rnoTM Fwd
33	DNA	rnoTM Probe (FAM)
34	DNA	rnoTM Rev
35	DNA	rnoTAU2 Fwd
36	DNA	rnoTAU2 Probe(FAM)
37	DNA	rnoTAU2 Rev
38	DNA	rnoTAD Fwd
39	DNA	rnoTAD Probe(Cal)
40	DNA	rnoTAD Rev
41	DNA	rnoGU Guide Target
42	DNA	rnoGU2 Guide Target
43	DNA	rnoGD Guide Target

TABLE 1-continued

Description of Sequences.		
SEQ ID NO	Type	Description
44	DNA	rnoGD2 Guide Target
45	DNA	rnoTGU Fwd
46	DNA	rnoTGU Probe(FAM)
47	DNA	rnoTGU Rev
48	DNA	rnoTGD Fwd
49	DNA	rnoTGD Probe(Cal)
50	DNA	rnoTGD Rev
51	Protein	Mouse TRKB/NTRK2 Signal Peptide
52	Protein	Mouse TRKB/NTRK2 Extracellular Domain
53	Protein	Mouse TRKB/NTRK2 Transmembrane Domain
54	Protein	Mouse TRKB/NTRK2 Cytoplasmic Domain
55	Protein	Rat TRKB/NTRK2 Signal Peptide
56	Protein	Rat TRKB/NTRK2 Extracellular Domain
57	Protein	Rat TRKB/NTRK2 Transmembrane Domain
58	Protein	Rat TRKB/NTRK2 Cytoplasmic Domain
59	Protein	Human TRKB/NTRK2 Signal Peptide
60	Protein	Human TRKB/NTRK2 Extracellular Domain
61	Protein	Human TRKB/NTRK2 Transmembrane Domain
62	Protein	Human TRKB/NTRK2 Cytoplasmic Domain
63	DNA	Mouse TrkB/Ntrk2 Signal Peptide CDS
64	DNA	Mouse TrkB/Ntrk2 Extracellular Domain CDS
65	DNA	Mouse TrkB/Ntrk2 Transmembrane Domain CDS
66	DNA	Mouse TrkB/Ntrk2 Cytoplasmic Domain CDS
67	DNA	Rat TrkB/Ntrk2 Signal Peptide CDS
68	DNA	Rat TrkB/Ntrk2 Extracellular Domain CDS
69	DNA	Rat TrkB/Ntrk2 Transmembrane Domain CDS
70	DNA	Rat TrkB/Ntrk2 Cytoplasmic Domain CDS
71	DNA	Human TRKB/NTRK2 Signal Peptide CDS
72	DNA	Human TRKB/NTRK2 Extracellular Domain CDS
73	DNA	Human TRKB/NTRK2 Transmembrane Domain CDS
74	DNA	Human TRKB/NTRK2 Cytoplasmic Domain CDS
75	Protein	Human TRKB/NTRK2 protein (Q16620-1; NP_001018074.1)
76	DNA	Human TRKB/NTRK2 cDNA (NM_001018064.2)
77	DNA	Human TRKB/NTRK2 CDS (CCDS ID CCDS35050.1)
78-125	DNA/Protein	Heavy and Light Chain Variable Regions and CDRs of Selected Anti-TRKB Antibodies in Table 22 and Table 23

EXAMPLES

Example 1. Generation of Mice Comprising a Humanized TRKB Locus

A large targeting vector (LTVEC) comprising a 5' homology arm comprising 41.6 kb of the mouse TrkB locus and 3' homology arm comprising 62.4 kb of the mouse TrkB locus was generated to replace a region of 65.7 kb from the mouse TrkB gene encoding the mouse TRKB extracellular domain with 74.4 kb of the corresponding human sequence of TRKB. Information on mouse and human TRKB is provided in Table 2. A description of the generation of the large targeting vector is provided in Table 3. Generation and use of large targeting vectors (LTVECs) derived from bacterial artificial chromosome (BAC) DNA through bacterial homologous recombination (BHR) reactions using VELOCIGENE® genetic engineering technology is described, e.g., in U.S. Pat. No. 6,586,251 and Valenzuela et al. (2003) *Nat. Biotechnol.* 21(6):652-659, each of which is herein incorporated by reference in its entirety for all purposes. Generation of LTVECs through in vitro assembly methods is described, e.g., in US 2015/0376628 and WO 2015/200334, each of which is herein incorporated by reference in its entirety for all purposes.

TABLE 2

Mouse and Human TRKB/NTRK2.						
	Official Symbol	NCBI Gene ID	Primary Source	RefSeq mRNA ID	UniProt ID	Genomic Assembly Location
Mouse	Ntrk2	18212	MGI: 97384	NM_001025074	P15209	GRCm38/mm10 Chr 13: 58,806,569-59,133,970 (+)
Human	Ntrk2	4915	HGNC: 8032	AF410899	Q16620	GRCh38/hg38 Chr 9: 84,669,778-85,027,070 (+)

TABLE 3

Mouse TrkB/Ntrk2 Large Targeting Vector.				
	Genome Build	Start	End	Length (bp)
5' Mouse Arm	GRCm38/mm10	Chr13: 58,767,209	Chr13: 58,808,821	41,613
Human Insert	GRCh38/hg38	Chr9: 84,670,730	Chr9: 84,745,139	74,409
3' Mouse Arm	GRCm38/mm10	Chr13: 58,874,563	Chr13: 58,936,986	62,424

Specifically, a region starting in exon 2 (coding exon 1; 25 from amino acid 32, preserving signal peptide) through exon 10, including the first 137 base pairs of intron 10 and all introns between exons 2 and 10 (i.e., between coding exon 1 and exon 10) was deleted from the mouse TrkB locus (preserving the mouse transmembrane domain encoded by exons 10 and 11). A region including exon 2/coding exon 1 (from amino acid 32, beginning after the signal peptide) through exon 10, including the first 177 base pairs of intron 10 and all introns between exons 2 and 10 (i.e., between coding exon 1 and exon 10) was inserted in place of the deleted mouse region (preserving the mouse transmembrane domain encoded by exons 10 and 11).

Sequences for the mouse TRKB signal peptide, extracellular domain, transmembrane domain, and cytoplasmic domain are set forth in SEQ ID NOS: 51-54, respectively, with the corresponding coding sequence set forth in SEQ ID NOS: 63-66, respectively. Sequences for the human TRKB signal peptide, extracellular domain, transmembrane domain, and cytoplasmic domain are set forth in SEQ ID NOS: 59-62, respectively, with the corresponding coding sequences set forth in SEQ ID NOS: 71-74, respectively. The expected encoded chimeric TRKB protein is has mouse TRKB transmembrane and intracellular domains, a mouse TRKB signal peptide, and a human TRKB extracellular domain. See FIG. 1. An alignment of the mouse and human TRKB proteins in FIG. 6. The mouse and human TrkB/ TRKB coding sequences are set forth in SEQ ID NOS: 9 and 11, respectively. The mouse and human TRKB protein sequences are set forth in SEQ ID NOS: 1 and 3, respectively. The sequences for the expected chimeric mouse/ human TRKB coding sequence and the expected chimeric mouse/human TRKB protein are set forth in SEQ ID NOS: 12 and 4, respectively.

To generate the mutant allele, the large targeting vector was introduced into F1H4 mouse embryonic stem cells. Following antibiotic selection, colonies were picked, expanded, and screened by TAQMAN®. See FIG. 2. Loss-of-allele assays were performed to detect loss of the endogenous mouse allele, and gain-of-allele assays were performed to detect gain of the humanized allele using the primers and probes set forth in Table 4.

TABLE 4

Screening Assays.			
Assay	Description	Primer/ Probe	Sequence
7138 hU	Upstream	Fwd	AGGTGGGTAGGTCCTGGAAGTG (SEQ ID NO: 14)
	Human	Probe	AATGCTGTCCCAAGAGTGGG (SEQ ID NO: 15)
	Insertion	Rev	GTCCTGCATCCCTTGCTTTG (SEQ ID NO: 16)
7138 hD	Downstream	Fwd	ATGTGGGCGTTGTGCAGTCTC (SEQ ID NO: 17)
	Human	Probe	CGCTGCAGTGCATTGAAGTACAGCA (SEQ ID NO: 18)
	Insertion	Rev	CTGTGGAGGACGTGACCAAG (SEQ ID NO: 19)
7138U	Upstream	Fwd	TCCGCTAGGATTGGTGTACTG (SEQ ID NO: 20)
	Mouse LOA	Probe	AGCCTTCTCCAGCATCGTGGCAT (SEQ ID NO: 21)
		Rev	TCCGGGTCAACGCTGTATTG (SEQ ID NO: 22)
7138D	Downstream	Fwd	TCCTGCGAGGGTTCTGAC (SEQ ID NO: 23)
	Mouse LOA	Probe	TGGGTGCTCATATGCCAGAGAAAT (SEQ ID NO: 24)
		Rev	CGATCTGTGATGGCCTGCTTAC (SEQ ID NO: 25)

Modification-of-allele (MOA) assays including loss-of-allele (LOA) and gain-of-allele (GOA) assays are described, for example, in US 2014/0178879; US 2016/0145646; WO 2016/081923; and Friendewey et al. (2010) *Methods Enzymol.* 476:295-307, each of which is herein incorporated by reference in its entirety for all purposes. The loss-of-allele (LOA) assay inverts the conventional screening logic and quantifies the number of copies in a genomic DNA sample of the native locus to which the mutation was directed. In a correctly targeted heterozygous cell clone, the LOA assay detects one of the two native alleles (for genes not on the X or Y chromosome), the other allele being disrupted by the targeted modification. The same principle can be applied in

reverse as a gain-of-allele (GOA) assay to quantify the copy number of the inserted targeting vector in a genomic DNA sample.

F0 mice were generated using the VELOCIMOUSE® method. See, e.g., U.S. Pat. Nos. 7,576,259; 7,659,442; 7,294,754; US 2008/0078000; and Poueymirou et al. (2007) *Nat. Biotechnol.* 25(1):91-99, each of which is herein incorporated by reference in its entirety for all purposes. In the VELOCIMOUSE method, targeted mouse embryonic stem (ES) cells are injected through laser-assisted injection into pre-morula stage embryos, e.g., eight-cell-stage embryos, which efficiently yields F0 generation mice that are fully ES-cell-derived. All experiments performed in humanized TRKB mice as described below were performed in mice in which the self-deleting selection cassette was self-deleted.

Example 2. In Vivo Comparison of Effect of H4H9816 and Isotype Control REGN1945 Antibodies on Body Weight and Metabolism in TrkB^{hu/hu} Mice (MAID7139)

Experimental Procedure

To determine the effect of a TRKB agonist antibody, H4H9816P2, on body weight and composition, a metabolic study of mice homozygous for the expression of human TRKB receptor in place of the mouse TRKB receptor (TrkB^{hu/hu} mice) was conducted following a single subcutaneous antibody injection. These studies were undertaken in part based on previous studies of TrkB agonists and TrkB-knockout mice. See, e.g., Lin et al. (2008) *PLoS ONE* 3(4):e1900; Rios et al. (2013) *Trends in Neurosciences* 36(2):83-90; and Zorner et al. (2003) *Biol. Psychiatry* 54:972-982, each of which is herein incorporated by reference in its entirety for all purposes. TrkB^{hu/hu} mice (male, 20 weeks old) were first transferred from group-cage to single-cage housing for two weeks of acclimatization. After this period, mice were transferred to metabolic cages (CLAMS, Columbus Instruments) to assess changes in food and water consumption, locomotion, energy expenditure, and respiration following antibody administration. Regular powdered chow was stored in a floor chamber on a spring-loaded scale (Mettler Toledo, PL602E) to measure food consumption via changes in total chow weight. Water was accessible via a cage-top spout and intake was measured by tracking changes in pump-line volume (Oxymax®/CLAMS Liquid Unit). CLAMS metabolic cages measured each of these parameters in continuous, 16-18 minute intervals throughout the dura-

tion of the study. Metabolic data were analyzed in single measures and summarized in 24-hour intervals containing one complete dark and light cycle using OXYMAX®/CLAMS software (Columbus instruments, v5.35). After acclimating to the cages for two weeks, TrkB^{hu/hu} mice received a single 50 mg/kg sub-cutaneous dose of either a TRKB agonist antibody, H4H9816P2, or an IgG4 isotype control antibody in PBS at pH7.2. A group of naïve control TrkB^{hu/hu} mice did not receive an injection. Mice were weighed immediately prior to dosing, and at 24, 48, 72, 96, and 120 hours post-dosing. In order to measure each mouse's body composition, Nuclear Magnetic Resonance Relaxometry, also referred to as Quantitative Magnetic Resonance, was performed using an EchoMRI™-500 Analyzer (EchoMRI LLC). Prior to dosing, mice were placed in a clear plastic holder and inserted into the NMR-MRI device to measure each subject's lean mass, fat mass, and hydration status. Measurements were performed over the course of 0.5-3.2 minutes per mouse, and were taken again approximately 120 hours after dosing.

Results and Conclusions

Daily body weight monitoring was performed to determine whether a single subcutaneous injection of H4H9816P2 induces weight loss in TrkB^{hu/hu} mice. Prior to dosing, there were no significant differences in the average body weight of the three treatment groups, as each had an average pre-dose body weight of 28.39-29.85 g (Table 5). At 48 hours post-dosing, however, H4H9816P2-treated TrkB^{hu/hu} mice lost an average of 1.70 g, or 5.96% of their pre-dose body weight. At the same time point, naïve and isotype control antibody-treated TrkB^{hu/hu} mice gained between 1.79-2.37% of their pre-dose body weight. H4H9816P2-treated TrkB^{hu/hu} mice continued to lose weight throughout the full time course of the study, and by 72 and 96 hours post-dosing these mice had lost an average of 8.42% and 11.80% of their pre-dose body weight, respectively. At 120 hours post-dosing, H4H9816P2-treated TrkB^{hu/hu} mice had lost an average of 12.67% of their pre-dose body weight. Conversely, naïve and isotype control-treated TrkB^{hu/hu} mice did not exhibit any loss in pre-dose body weight throughout the study. As body weight in H4H9816P2-treated TrkB^{hu/hu} mice was significantly reduced relative to both naïve and isotype controls at 48, 72, 96, and 120 hours post-dosing, it was determined that TRKB agonist antibody H4H9816P2 induced significant body weight loss in TrkB^{hu/hu} mice.

TABLE 5

Body Weight of TrkB ^{hu/hu} Mice after Dosing with TRKB Agonist Antibody H4H9816P2.						
Experimental group	Mean pre-dose body weight (g)	Mean body weight (g) 24 hours post-dose (±SD)	Mean body weight (g) 48 hours post-dose (±SD)	Mean body weight (g) 72 hours post-dose (±SD)	Mean body weight (g) 96 hours post-dose (±SD)	Mean body weight (g) 120 hours post-dose (±SD)
	Percent change from pre-dose body weight	Percent change from pre-dose body weight	Percent change from pre-dose body weight	Percent change from pre-dose body weight	Percent change from pre-dose body weight	Percent change from pre-dose body weight
	(+/-SD)	(+/-SD)	(+/-SD)	(+/-SD)	(+/-SD)	(+/-SD)
	(+/-SD)	(+/-SD)	(+/-SD)	(+/-SD)	(+/-SD)	(+/-SD)
	(+/-SD)	(+/-SD)	(+/-SD)	(+/-SD)	(+/-SD)	(+/-SD)
	(+/-SD)	(+/-SD)	(+/-SD)	(+/-SD)	(+/-SD)	(+/-SD)
Naïve (n = 3)	28.85 (+/-0.81)	29.69 (+/-0.97)	29.36 (+/-1.10)	29.32 (+/-1.29)	29.29 (+/-1.10)	28.88 (+/-1.04)
	N/A	+2.91% (+/-0.62)	+1.79% (+/-1.62)	+1.65% (+/-2.24)	+1.54% (+/-1.22)	+0.10% (+/-1.05)

TABLE 5-continued

Body Weight of TrkB ^{hu/hu} Mice after Dosing with TRKB Agonist Antibody H4H9816P2.						
Experimental group	Mean pre-dose body weight (g) (±SD) Percent change from pre-dose body weight (+/-SD)	Mean body weight (g) 24 hours post-dose (±SD) Percent change from pre-dose body weight (+/-SD)	Mean body weight (g) 48 hours post-dose (±SD) Percent change from pre-dose body weight (+/-SD)	Mean body weight (g) 72 hours post-dose (±SD) Percent change from pre-dose body weight (+/-SD)	Mean body weight (g) 96 hours post-dose (±SD) Percent change from pre-dose body weight (+/-SD)	Mean body weight (g) 120 hours post-dose (±SD) Percent change from pre-dose body weight (+/-SD)
Isotype control (n = 4)	29.21 (+/-2.68) N/A	30.27 (+/-2.51) +3.61% (+/-1.68)	29.90 (+/-2.63) +2.37% (+/-1.50)	30.08 (+/-2.69) +2.98% (+/-1.09)	29.87 (+/-2.52) +2.25% (+/-1.56)	29.69 (+/-2.68) +1.65% (+/-0.81)
H4H9816P2 (n = 4)	28.39 (+/-1.35) N/A	27.87 (+/-1.29) -1.83% (+/-0.56)	26.69 (+/-0.87) -5.96% (+/-1.88)	26.00* (+/-0.98) -8.42% (+/-1.85)	25.04** (+/-1.03) -11.80% (+/-1.52)	24.79** (+/-1.36) -12.67% (+/-1.66)

Statistical significance determined by two-way ANOVA with Tukey's multiple comparison post-hoc test is indicated (*= p < 0.05, **= p < 0.01, ***= p < 0.001, ****= p < 0.0001, compared to isotype control group: TrkB^{hu/hu} mice dosed with 50 mg/kg isotype control antibody.

The effect of TRKB agonist antibody H4H9816P2 injection on body composition was also measured by performing NMR-MRI on each subject before and after dosing. Prior to dosing, the three treatment groups of TrkB^{hu/hu} mice did not exhibit any significant differences in fat mass or lean mass, as each group had an average of 4.19-4.75 g of fat mass and 21.32-21.70 g of lean mass (Table 6). Following antibody administration, however, TrkB^{hu/hu} mice dosed with H4H9816P2 lost an average of 48.90% of their total body fat mass over the course of the study (Table 6). Naïve and isotype control antibody-treated TrkB^{hu/hu} mice lost an aver-

age of 8.49% and 9.48% of their pre-dose fat mass, respectively, which was significantly less than H4H9816P2-treated subjects (Table 6). Furthermore, H4H9816P2-treated TrkB^{hu/hu} mice lost an average of 7.84% of their lean mass throughout the study, which was significantly greater than the 2.41% and 1.75% of average pre-dose lean mass lost by naïve and isotype control antibody-treated groups, respectively (Table 6). As such, the described body weight loss could be explained by a significant loss of fat mass and a modest loss of lean mass following injection of TRKB agonist antibody H4H9816P2 in TrkB^{hu/hu} mice.

TABLE 6

Body Composition of TrkB ^{hu/hu} Mice after Dosing with TRKB Agonist Antibody H4H9816P2.						
Experimental group	Mean pre-dose fat mass (%) (±SD)	Mean fat mass (%) 120 hours post-dose (±SD)	Mean fat mass change (%) 120 hours post-dose (±SD)	Mean pre-dose lean mass (%) (±SD)	Mean lean mass (%) 120 hours post-dose (±SD)	Mean lean mass change (%) 120 hours post-dose (±SD)
Naïve (n = 3)	4.65 (+/-0.32)	4.27 (+/-0.55)	-8.49 (+/-7.18)	21.45 (+/-0.79)	20.94 (+/-0.98)	-2.41 (+/-1.81)
Isotype control (n = 4)	4.75 (+/-2.98)	4.40 (+/-2.98)	-9.48 (+/-6.00)	21.70 (+/-0.50)	21.32 (+/-0.35)	-1.75 (+/-0.98)
H4H9816P2 (n = 4)	4.19 (+/-1.15)	2.14 (+/-0.64)	-48.90**** (+/-5.06)	21.32 (+/-1.87)	19.64 (+/-1.69)	-7.84*** (+/-0.94)

Statistical significance determined by Kruskal-Wallis One-way ANOVA with Tukey's multiple comparison post-hoc test is indicated (*= p < 0.05, **= p < 0.01, ***= p < 0.001, ****= p < 0.0001, compared to isotype control group: TrkB^{hu/hu} mice dosed with 50 mg/kg isotype control antibody.

In addition to assessing the effects of TRKB agonist antibody H4H9816P2 injection on body weight and composition in $\text{TrkB}^{hu/hu}$ mice, feeding, drinking, and locomotor activity were continuously measured by metabolic cages. Prior to dosing, $\text{TrkB}^{hu/hu}$ mice consumed an average of 3.49-3.73 g of chow per day. Within 24 hours of dosing, however, H4H9816P2-treated $\text{TrkB}^{hu/hu}$ mice significantly reduced their food intake to 2.20 g of chow per day. The average level of food intake in H4H9816P2-treated $\text{TrkB}^{hu/hu}$ mice did not exceed 2.49 g of chow per day throughout the remainder of the study, while naïve and isotype antibody-treated $\text{TrkB}^{hu/hu}$ mice consistently consumed an average of 3.62-4.07 g of chow per day (Table 7).

Similarly, there were no significant differences in daily water consumption between treatment groups prior to dosing. $\text{TrkB}^{hu/hu}$ mice consumed an average of 4.67-5.55 mL of water per day in each treatment group (Table 8). After dosing, H4H9816P2-treated $\text{TrkB}^{hu/hu}$ mice reduced their water intake to 2.05-3.24 mL of water per day. This was significantly lower than naïve and isotype control antibody-treated $\text{TrkB}^{hu/hu}$ mice, which consistently consumed 4.50-5.77 mL of water per day throughout the study (Table 8). Thus, injection of the TRKB agonist antibody, H4H9816P2, appeared to result in a significant reduction of both food and water intake in $\text{TrkB}^{hu/hu}$ mice relative to both naïve and isotype controls.

TABLE 7

Food Consumption of $\text{TrkB}^{hu/hu}$ Mice after Dosing with TRKB Agonist Antibody H4H9816P2.					
Experimental group	Mean total food intake (g) 0-24 hours pre-dose ($\pm$ SD)	Mean total food intake (g) 0-24 hours post-dose ($\pm$ SD)	Mean total food intake (g) 24-48 hours post-dose ($\pm$ SD)	Mean total food intake (g) 48-72 hours post-dose ($\pm$ SD)	Mean total food intake (g) 72-96 hours post-dose ($\pm$ SD)
Naïve (n = 3)	3.51 (+/-0.53)	3.98 (+/-0.08)	3.76 (+/-0.19)	3.62 (+/-0.35)	3.91 (+/-0.18)
Isotype control (n = 4)	3.73 (+/-0.48)	4.07 (+/-0.23)	3.99 (+/-0.17)	3.89 (+/-0.22)	3.80 (+/-0.22)
H4H9816P2 (n = 4)	3.49 (+/-1.07)	2.20*** (+/-0.16)	2.08**** (+/-0.36)	2.18**** (+/-0.37)	2.49**** (+/-0.47)

Statistical significance determined by Kruskal-Wallis One-way ANOVA with Tukey's multiple comparison post-hoc test is indicated (*= p < 0.05, **= p < 0.01, ***= p < 0.001, ****= p < 0.0001, compared to isotype control group: $\text{TrkB}^{hu/hu}$ mice dosed with 50 mg/kg isotype control antibody.

TABLE 8

Water Consumption of $\text{TrkB}^{hu/hu}$ Mice after Dosing with TrkB Agonist Antibody H4H9816P2.					
Experimental group	Mean total water intake (mL) 0-24 hours pre-dose ($\pm$ SD)	Mean total water intake (mL) 0-24 hours post-dose ($\pm$ SD)	Mean total water intake (mL) 24-48 hours post-dose ($\pm$ SD)	Mean total water intake (mL) 48-72 hours post-dose ($\pm$ SD)	Mean total water intake (mL) 72-96 hours post-dose ($\pm$ SD)
Naïve (n = 3)	4.79 (+/-0.21)	5.42 (+/-0.94)	4.96 (+/-0.91)	4.57 (+/-0.56)	4.88 (+/-0.32)
Isotype control (n = 4)	5.55 (+/-1.23)	4.50 (+/-1.08)	5.08 (+/-1.39)	5.09 (+/-1.10)	5.77 (+/-1.62)
H4H9816P2 (n = 4)	4.67 (+/-1.13)	2.25** (+/-0.55)	3.24* (+/-1.10)	2.05*** (+/-0.29)	2.25**** (+/-0.24)

Statistical significance determined by Kruskal-Wallis One-way ANOVA with Tukey's multiple comparison post-hoc test is indicated (*= p < 0.05, **= p < 0.01, ***= p < 0.001, ****= p < 0.0001, compared to isotype control group: $\text{TrkB}^{hu/hu}$ mice dosed with 50 mg/kg isotype control antibody.

To determine the effects of antibody treatment on activity, locomotion was analyzed by OXYMAX®/CLAMS software (Columbus instruments, v5.35), which continuously measured the total number of x-plane ambulations of each mouse. One mouse exhibited hyperactivity prior to dosing and was removed from post-dose statistical analysis. While naïve and isotype antibody-treated subjects consistently registered an average of 11,000-15,000 ambulations per day throughout the study, H4H9816P2-treated TrkB^{hu/hu} mice registered 28,260 ambulations between 24-48 hours post-dosing, and registered 21,193 and 27,028 ambulations from 48-72 and 72-96 hours post-dosing, respectively (Table 9). H4H9816P2-treated TrkB^{hu/hu} mice registered more total ambulation counts at each time point following antibody administration, suggesting hyperactivity to be an additional effect of H4H9816P2 injection. In combination, these effects suggest that a single subcutaneous injection of the TRKB agonist antibody, H4H9816P2, induced significant changes in body weight, body composition, metabolism, and locomotion in TrkB^{hu/hu} mice.

TABLE 9

Locomotion of TrkB ^{hu/hu} Mice after Dosing with TrkB Agonist Antibody H4H9816P2.					
Experimental group	Mean total ambulations (counts) 0-24 hours pre-dose (±SD)	Mean total ambulations (counts) 0-24 hours post-dose (±SD)	Mean total ambulations (counts) 24-48 hours post-dose (±SD)	Mean total ambulations (counts) 48-72 hours post-dose (±SD)	Mean total ambulations (counts) 72-96 hours post-dose (±SD)
Naive (n = 3)	16562 (+/-3380)	14692 (+/-2792)	14387 (+/-6126)	13279 (+/-3607)	12525 (+/-4121)
Isotype Control	18105	13380	13049	11371	11468
REGN1945 (n = 4)	(+/-4085)	(+/-2730)	(+/-3376)	(+/-2552)	(+/-2088)
H4H9816P2 (n = 4)	13292 (+/-5294)	16575 (+/-6836)	28260 (+/-19874)	21193 (+/-6668)	27028* (+/-10969)

Statistical significance determined by Kruskal-Wallis One-way ANOVA with Tukey's multiple comparison post-hoc test is indicated (*= p < 0.05, **= p < 0.01, ***= p < 0.001, ****= p < 0.0001, compared to isotype control group: TrkB^{hu/hu} mice dosed with 50 mg/kg isotype control antibody.

Example 3. In Vivo Comparison of the Effect of TRKB Agonist Antibody H4H9816P2 and IgG4 Isotype Control REGN1945 on TRKB Phosphorylation in the Brain Following Stereotaxic Injection in TrkB^{hu/hu} Mice (MAID 7139)

Experimental Procedure

Tyrosine receptor kinase B (TRKB) is activated through binding of its ligand brain-derived neurotrophic factor (BDNF) at the extracellular receptor domain, which induces the dimerization and autophosphorylation of tyrosine residues in the intracellular receptor domain and subsequent activation of cytoplasmic signaling pathways. See, e.g., Haniu et al. (1997) *J. Biol. Chem.* 272(40):25296-25303 and Rogalski et al. (2000) *J. Biol. Chem.* 275(33):25082-25088, each of which is herein incorporated by reference in its entirety for all purposes. In order to determine the effect of a TRKB agonist antibody, H4H9816P2, on TRKB activation following direct hippocampal injection was performed in mice homozygous for a chimeric mouse/human TRKB receptor in which the extracellular domain has been humanized (MAID 7139) (referred to as TrkB^{hu/hu} mice). TrkB^{hu/hu} mice (N=48) received bilateral stereotaxic injections of either with 2 µL of either vehicle (PBS), REGN1945 hereby noted as IgG4 isotype control antibody (27.5 mg/mL final concentration), or TRKB agonist antibody H4H9816P2 (27.5 mg/mL final concentration) into the hippocampus, -2 mm posterior and +1.5 mm lateral to bregma. In order to minimize tissue damage, injection and needle removal were

both performed gradually over 5-minute intervals. TrkB^{hu/hu} mice were then sacrificed by CO₂ euthanasia approximately 30 minutes, 1 hour, 4 hours, or 18 hours post-injection. A terminal bleed was performed via cardiac puncture to collect blood, and mice were then transcardially perfused with cold heparinized saline. The brain was carefully removed from the skull, and a 2 mm³ section of tissue surrounding the injection site was dissected, collected in an Eppendorf tube and stored on ice. The brain section was then lysed in 300 µL of RIPA lysis buffer (ThermoFisher Scientific, Cat #89901) containing 2× protease and phosphatase inhibitors (ThermoFisher Scientific, Cat #78444) and stored on ice. The lysed tissue was then homogenized for further processing, aliquoted and stored at -80° C.

To assess TRKB phosphorylation in the brain tissue, immuno-precipitation and western blotting was performed. Anti-human TRKB antibody H4H10108N that does not compete for binding with H4H9816P2 was coupled to NHS-activated Sepharose beads (prepared using manufacturer's protocol; GE Healthcare, Cat #17-0906) and washed with DPBS three times to remove any residual preservation solution. Homogenized brain lysates were thawed on ice and diluted to a concentration of 1 mg/mL (brain weight to buffer volume) in a buffer composed of 1% NP-40, 0.1% Tween-20, protease and phosphatase inhibitors in TBST. The protein concentration of the homogenized brain lysate was quantified by performing a standard BCA assay per manufacturer's instructions (Thermo Scientific Pierce, Cat #23225). For every 100 µg of protein, 15 µL of anti-human TRKB antibody (H4H10108N) NHS-activated Sepharose

beads were added to the brain lysate solution and the mixture was incubated overnight at 4° C. with gentle shaking 20 rpm (Thermo rotator). The next day, samples were centrifuged at 1000× g for one minute, and the supernatant was then carefully removed. Beads were subsequently washed twice with 400 µL of Tris-buffered saline (Bio-Rad, Cat #1706435) with 1% Tween-20 (Sigma Aldrich, Cat #P9416) (TBST). After carefully aspirating the wash buffer, 60 µL of 0.1% Trifluoroacetic acid (TFA; Sigma-Aldrich, T62200) in water at pH 3.0 was added to each sample. The solution was mixed and allowed to stand for two minutes before being collected and transferred into a separate tube. This process was repeated with another 60 µL of 0.1% TFA at pH 3.0. The two 0.1% TFA solutions for each sample were then combined, and 2 µL of 1M Tris-HCl (ThermoFisher Scientific, Cat #15567-027), at pH 8.5, was added.

The solution was dried using a speed vacuum and then re-suspended and reduced with a mixture of 20 µL of 1× Laemmli Buffer (Bio-Rad, Cat #1610737) plus 355 nM 2-mercaptoethanol (BME; Gibco, Cat #21985-023). Samples were boiled at 95° C. for 10 minutes and loaded onto a 10-well, Mini-Protein 4-15% Tris-Glycine gel (Bio-Rad, Cat #4561086). After electrophoresis, protein samples were transferred from the Tris-Glycine gel onto a PVDF membrane (Bio-Rad, Cat #170-4156) via the Trans-Blot Turbo Transfer System (Bio-Rad, Cat #1704156) over the course of 30 minutes at a constant rate of 1.3 A and 25 V. After the transfer, the membrane was blocked with 2.5% milk (Bio-Rad, Cat #170-6406) in TBST for one hour at room temperature, and subsequently probed overnight with either an anti-phospho-TRKB antibody (Novus, Cat #NB100-92656) diluted 1:1000 in a solution of 2.5% BSA or anti-TRKB primary antibody (Cell Signaling, Cat #4603) diluted to 1:1000 in 2.5% milk TBST at 4° C. on a shaker at 30 rpm. The next day, blots were washed with TBST and incubated with an anti-rabbit IgG antibody conjugated with horseradish peroxidase (Jackson, Cat #111-035-144) at 1:1000 in 1% milk in TBST for 1 hour at room temperature. Blots were then washed again, developed with ECL solution (PerkinElmer, Inc. Cat #RPN2106), and subsequent image exposures were taken every 30 seconds.

Results and Conclusions

Immunoprecipitation and subsequent western blotting of protein derived from TrkB^{hu/hu} mouse brain lysates demonstrated that hippocampal TRKB phosphorylation was detectable in mice injected with a TRKB agonist antibody, H4H9816P2, but not in mice treated with vehicle or isotype control antibody, as shown FIG. 3. Among the time points assessed, TRKB phosphorylation peaked at 4 hours after stereotaxic injection in mice dosed with H4H9816P2. TRKB phosphorylation was also detected by western blot at 18 hours post-dosing in some, but not all mice. Conversely, injection of vehicle and IgG4 isotype control antibody did not induce TRKB phosphorylation at any time point. Western blotting also indicated that the total TRKB receptor levels were downregulated in some, but not all TrkB^{hu/hu} mice dosed with H4H9816P2 relative to vehicle and isotype control treated mice. Total TRKB levels appeared to be slightly downregulated in H4H9816P2-treated subjects at 18 hours post-dosing. Thus, these results indicate that direct injection of the TRKB agonist antibody, H4H9816P2, induces phosphorylation of hippocampal TRKB receptors in TrkB^{hu/hu} mice.

Example 4. Activation of Downstream Signaling Pathways by TrkB Agonist Antibodies in Primary Cortical Neurons from Postnatal Day 1 TrkB^{hu/hu} Mice

Experimental Procedure

All procedures were conducted in accordance with the ARVO Statement for Use of Animals in Ophthalmic and Vision Research and the Regeneron Pharmaceuticals, Inc. IACUC. Primary mouse cortical neurons were isolated and cultured from humanized TrkB mice (MAID 7139). See, e.g., Beaudoin et al. (2012) *Nat. Protoc.* 7(9):1741-1754, herein incorporated by reference in its entirety for all purposes. Western blots were performed to determine the effects of TrkB agonist antibodies on the downstream pathways of Akt and Erk (p-Akt, p-Erk1/2). Primary cortical neurons from postnatal day 1 (P1) humanized TrkB mouse pups were cultured for 4 days (DIV-4) in NeuralQ Basal Medium (Global Stem, cat. #GSM-9420) supplemented with GS21 Neural Supplement (Global Stem, cat. #GSM-3100), Glutamax (Invitrogen, cat. #35050-061) and Penicillin/Streptomycin. Cells were treated with TrkB agonist antibody H4H9816P-L1 (10 µg/mL), TrkB agonist antibody H4H9780P-L1 (10 µg/mL), TrkB agonist antibody H4H9814P-L1 (10 µg/mL), IgG4 isotype control REGN1945 (10 µg/mL), control antibody H1M8037C-L1 (10 µg/mL), or BDNF (1 µg/mL), for 15 minutes or 2 hours. Western blots were performed to determine if the agonists have a difference in downstream signaling maintenance and strength. Treated cells were rinsed and scraped in cold PBS containing 1% protease and phosphatase inhibitors (Sigma). Protein concentration was determined by Bradford protein assay (Pierce). Samples (50 µg) were separated by SDS-PAGE in 3-8% Tris-Acetate reduced gels (Novex) and transferred to a nitrocellulose membrane (Bio-Rad).

The membrane was incubated for 1 hour in blocking solution containing 5% milk and 0.1% Tween-20, pH 7.6. This was followed by overnight incubation at 4° C. in the blocking buffer containing 5% BSA, 0.1% Tween-20, and rabbit anti-phospho-Trk (Cell Signaling, cat. #9141, 1:500), rabbit anti-phospho-Akt (Cell Signaling, cat. #9271, 1:1000), or rabbit anti-phospho-ERK1/2 antibody (Sigma, cat. #E7028, 1:5000). Subsequently, the labeled proteins were visualized by incubation with a horseradish peroxidase (HRP) conjugated anti-goat, mouse or rabbit IgG followed by development with a chemiluminescence substrate for HRP (Pierce). To determine the amounts of total TrkB, MAPK or Akt present in each lane, the nitrocellulose membranes were stripped of the antibodies in stripping buffer (Pierce) for 20 min and incubated with rabbit anti-TrkB (Cell Signaling, cat. #4603, 1:1000), rabbit anti-Erk1/2 (Cell Signaling, cat. #06-182, 1:1000), or rabbit anti-Akt antibody (Cell Signaling, cat. #9272, 1:1000) and then visualized as described above. Beta-Actin (Sigma, cat. #A5316, 1:20000 and GAPDH (Sigma, cat. #G9295) were probed as sample loading control.

Materials

TABLE 10

mAB Clone IDs.		
REGN	AbPID	Lot
REGN1945	H4H9816P	L1
	H4H9780P	L1
	H4H9814P	L1
	H1M8037C	L1
	Comparator, Control antibody	

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TABLE 11

Reagents.			
Reagent/Equipment	Source	Identifier	Lot #
Penicillin/Streptomycin	Invitrogen	15140	
Fetal Bovine Serum	Invitrogen	10082-147	
GS21 Neural Supplement (50X)	GlobalStem	GSM-3100	18130001
NeuralQ Basal Medium	GlobalStem	GSM-9420	18190001
Glutamax	Invitrogen	35050-061	
Protease Inhibitor Cocktail	Sigma	P8340	
Phosphatase Inhibitor Cocktail 3	Sigma	P0044	034M4010V
RIPA lysis buffer 1x	Rockland	MB-030-0250	24805
BSA	Sigma	A8806	
Tris-Acetate 4-8% reduced gels	Invitrogen	WG1602BX10	14022684
BCA Protein Assay Kit	Pierce	23227	
ECL	Pierce	32209	
Restore Western Blot Stripping Buffer	Pierce	21059	
Nitrocellulose membrane	Bio-Rad Laboratories	1620112	

TABLE 12

Neurobasal Medium.
NeuralQ Basal Medium (Global Stem, GSM-9420) 50 mL
GS21 Neural Supplement (50X) (Global Stem, GSM-3100) 10 mL
Glutamax (Invitrogen, 35050-061) 0.5 mL
Penicillin/Streptomycin 5 mL

TABLE 13

Antibodies.
H4H9816P lot1 (10 µg/mL)
H4H9780P lot1 (10 µg/mL)
H4H9814P lot1 (10 µg/mL)
REGN1945 human IgG4 lot1 (10 µg/mL)
C2 H1M8037C lot1 (10 µg/mL)

TABLE 14

Western Blots.
p-Trk (Cell Signaling, 9141) Rb, 1:500
total TrkB (Cell Signaling, 4603) Rb 1:1000
p-Akt (Cell Signaling, 9271) Rb 1:1000
total-Akt (Cell Signaling, 9272) Rb 1:1000
p-Erk1/2 (Sigma, E7028) 1:5000
total Erk1/2 (Cell Signaling, 06-182) Rb 1:1000
b-Actin (Sigma, A5316) Ms 1:20000
GAPDH (Sigma, G9295) HRP conjugated 1:20000

Results and Conclusions

As shown in FIG. 7, while all the TrkB agonist antibodies showed activation of the MAPK/ERK and PI3K/Akt pathways at 15 minutes after the incubation, only BDNF and H4H9814P showed TrkB phosphorylation. At 2 hours after incubation, all the TrkB agonist antibodies showed activation of TrkB.

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Example 5. Pharmacokinetic Assessment of an Anti-TrkB Antibody in Humanized TrkB and Wild Type Mice

Experimental Procedure

Evaluation of the pharmacokinetics of an anti-TrkB antibody, H4H9816P2 (Lot H4H9816P2-L7), was conducted in humanized TrkB (mice homozygous for chimeric mouse/human TrkB expression, TrkB^{hu/hu}) (MAID7139) and wild type (WT) mice. Cohorts contained 5 mice per mouse strain. All mice received a single sub-cutaneous (SC) 10 mg/kg dose. Blood samples were collected at 6 hours and 1, 2, 3, 6, 9, 16, 21, and 30 days post-dosing. Blood was processed into serum and frozen at -80° C. until analyzed.

Circulating antibody concentrations were determined by total human IgG4/hlgG1 antibody analysis using the Gyro-Lab xPlore™ (Gyros, Uppsala, Sweden). Briefly, biotinylated mouse anti-human IgG4/IgG1-specific monoclonal antibody (REGN2567; Lot RSCH15088) diluted to 100 µg/mL in antibody dilution buffer (0.05% Tween-20+PBS) was captured on a Gyrolab Bioaffy 200 CD, which contained affinity columns preloaded with streptavidin-coated beads (Dynospheres™). The standard used for calibration in this assay was H4H9816P at concentrations ranging from 0.488 to 2000 ng/mL in dilution buffer (0.5% BSA+PBS) containing 0.1% normal mouse serum (NMS). Serum samples were diluted 1:100 in the antibody dilution buffer. Human IgG captured on the anti-REGN2567-coated affinity columns on the CD, run at room temperature, was detected by addition of 0.5 µg/mL Alexa-647-conjugated mouse anti-human kappa monoclonal antibody (REGN654; Lot RSCH13067) diluted in detection buffer (Rexxip F buffer); and the resultant fluorescent signal was recorded in response units (RU) by the GyroLab xPlore instrument. Sample concentrations were determined by interpolation from a standard curve that was fit using a 5-parameter logistic curve fit using the Gyrolab Evaluator Software. Average concentrations from 2 replicate experiments were used for subsequent PK analysis.

PK parameters were determined by non-compartmental analysis (NCA) using Phoenix® WinNonlin® software Version 6.3 (Certara, L. P., Princeton, NJ) and an extravascular dosing model. Using the respective mean concentration values for each antibody, all PK parameters including observed maximum concentration in serum (C_{max}), estimated half-life observed ($t_{1/2}$), and area under the concentration curve versus time up to the last measurable concentration (AUC_{last}) were determined using a linear trapezoidal rule with linear interpolation and uniform weighting.

Results and Conclusions

Following 10 mg/kg s.c. administration of anti-TrkB antibody, H4H9816P2, similar maximum concentrations (C_{max}) of antibody were observed by day 1 or 2 in both TrkB^{hu/hu} and WT mice (135 and 131 µg/mL, respectively). By day 9, H4H9816P2 exhibited steeper drug elimination in TrkB^{hu/hu} mice than in WT mice, indicating a target-mediated effect. Day 30 antibody concentrations were about 35-fold less in TrkB^{hu/hu} mice. Antibody exposure (AUC_{last}) for H4H9816P2 in WT mice was ~1.7-fold higher than seen in TrkB^{hu/hu} mice (1730 and 1020 d*µg/mL, respectively). WT mice also exhibited about a 3-fold increase in half-life ($T_{1/2}$) over TrkB^{hu/hu} mice (8.4 and 2.9 days, respectively).

A summary of the data for total anti-TrkB antibody concentrations are summarized in Table 15, mean PK parameters are described in Table 16 and mean total antibody concentrations versus time are shown in FIG. 8. In FIG. 8, mice were administered a single 10 mg/kg sub-cutaneous dose on day 0. Concentrations of total

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H4H9816P2 in serum were measured using a Gyros immunoassay. Data points on post-dose 6 hours, 1, 2, 3, 6, 9, 16, 21, and 30 days indicate the mean concentration of antibody. Total antibody concentrations of H4H9816P2 are represented as solid circles for TrkB^{hu/hu} mice and solid squares for wild type mice. Data are plotted as mean±SD.

TABLE 15

Mean Concentrations (±SD) of Total IgG in Serum Following a Single 10 mg/kg Sub-Cutaneous Injection of H4H9816P2 in TrkB ^{hu/hu} and Wild Type Mice over Time.			
Total mAb Concentration in Mouse Serum 10 mg/kg			
Antibody	Time (d)	Mean (µg/mL)	+/-SD
TrkB ^{hu/hu} Mice	0.25	72.42	4.06
	1	132.0	18.0
	2	124.9	15.9
	3	113.4	11.8
	6	78.72	9.98
	9	37.74	14.0
	16	5.592	4.97
	21	2.060	2.11
	30	0.447	0.506
WT Mice	0.25	56.73	14.5
	1	120.8	6.26
	2	131.2	7.54
	3	125.7	7.46
	6	101.9	11.4
	9	75.94	7.06
	16	42.61	16.1
	21	27.75	16.9
	30	15.52	13.0

Abbreviations: Time = time in days post single-dose injection; d = day of study; SD = standard deviation.

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TABLE 16

Summary of Pharmacokinetic Parameters.			
H4H9816P2			
Parameter	Units	TrkB ^{hu/hu} Mice	WT Mice
C _{max}	µg/mL	135 ± 15	131 ± 7.5
T _{1/2}	d	2.94 ± 1.1	8.36 ± 3.9
AUC _{last}	d□µg/mL	1020 ± 150	1730 ± 310

PK parameters were derived from mean concentration versus time profiles. T_{1/2} and AUC_{last} are based on concentrations out to day 30. Abbreviations: C_{max} = peak concentration; AUC = area under the concentration-time curve; AUC_{last} = AUC computed from time zero to the time of the last positive concentration; T_{1/2} = terminal half-life of elimination.

Example 6. Generation of Rats Comprising a Humanized TRKB Locus

A large targeting vector comprising a 5' homology arm comprising 7 kb of the rat TrkB locus and 3' homology arm comprising 47 kb of the rat TrkB locus was generated to replace a region of 68.5 kb from the rat TrkB gene encoding the rat TRKB extracellular domain with 74.4 kb of the corresponding human sequence of TRKB. Generation and use of large targeting vectors (LTVECs) derived from bacterial artificial chromosome (BAC) DNA through bacterial homologous recombination (BHR) reactions using VELOCIGENE® genetic engineering technology is described, e.g., in U.S. Pat. No. 6,586,251 and Valenzuela et al. (2003) *Nat. Biotechnol.* 21(6):652-659, each of which is herein incorporated by reference in its entirety for all purposes. Generation of LTVECs through in vitro assembly methods is described, e.g., in US 2015/0376628 and WO 2015/200334, each of which is herein incorporated by reference in its entirety for all purposes. Information on rat and human TRKB is provided in Table 17. A description of the generation of the large targeting vector is provided in Table 18.

TABLE 17

Rat and Human TRKB/NTRK2.							
	Official Symbol	NCBI GeneID	Primary Source	RefSeq mRNA ID	UniProt ID	Genomic Assembly	Location
Rat	Ntrk2	25054	RGD: 3213	NM_012731.2	Q63604	RGSC 5.0/rn5	Chr 17: 8,156,432-8,464,507 (-)
Human	Ntrk2	4915	HGNC: 8032	AF410899	Q16620	GRCh38/hg38	Chr 9: 84,669,778-85,027,070 (+)

TABLE 18

Rat TrkB/Ntrk2 Large Targeting Vector.				
	Genome Build	Start	End	Length (bp)
5' Rat Arm	RGSC 5.0/rn5	Chr17: 8,470,615	Chr17: 8,463,379	7,236
Human Insert	GRCh38/hg38	Chr9: 84,670,730	Chr9: 84,745,139	74,409
3' Rat Arm	RGSC 5.0/rn5	Chr17: 8,394,967	Chr17: 8,347,889	47,078

Specifically, a region starting in exon 2 (coding exon 1; from amino acid 32, preserving signal peptide) through exon 10, including the first 50 base pairs of intron 10 and all introns between exons 2 and 10 (i.e., between coding exon 1 and exon 10) was deleted from the rat TrkB locus (preserving the rat transmembrane domain encoded by exons 10 and 11). A region including exon 2/coding exon 1 (from amino acid 32, beginning after the signal peptide) through exon 10, including the first 66 base pairs of intron 10 and all introns between exons 2 and 10 (i.e., between coding exon 1 and exon 10) was inserted in place of the deleted rat region (preserving the rat transmembrane domain encoded by exons 10 and 11).

Sequences for the rat TRKB signal peptide, extracellular domain, transmembrane domain, and cytoplasmic domain are set forth in SEQ ID NOS: 55-58, respectively, with the corresponding coding sequence set forth in SEQ ID NOS: 67-70, respectively. Sequences for the human TRKB signal peptide, extracellular domain, transmembrane domain, and cytoplasmic domain are set forth in SEQ ID NOS: 59-62, respectively, with the corresponding coding sequence set forth in SEQ ID NOS: 71-74, respectively. The expected encoded chimeric TRKB protein is has rat TRKB transmembrane and intracellular domains, a rat TRKB signal peptide, and a human TRKB extracellular domain. See FIG. 4. An

alignment of the rat and human TRKB proteins in FIG. 6. The rat and human TrkB/TRKB coding sequences are set forth in SEQ ID NOS: 10 and 11, respectively. The rat and human TRKB protein sequences are set forth in SEQ ID NOS: 2 and 3, respectively. The sequences for the expected chimeric rat/human TRKB coding sequence and the expected chimeric rat/human TRKB protein are set forth in SEQ ID NOS: 13 and 5, respectively.

To generate the mutant allele, CRISPR/Cas9 components including four guide RNAs (guide RNA target sequences set forth in SEQ ID NOS: 41-44) were introduced into rat embryonic stem cells together with the large targeting vector. Specifically, 4×10^6 rat ES cells (Dark Agouti line DA2B) were electroporated with the following: 2 mg TrkB LTVEC; 5 mg Cas9 expression plasmid; and 5 mg each of the gRNAs: gU, gU2, gD and gD2. The electroporation conditions were: 400 V voltage; 100 mF capacitance; and 0 W resistance. Antibiotic selection was performed using G418 at a concentration of 75 mg/mL. Colonies were picked, expanded, and screened by TAQMAN®. See FIG. 5. Loss-of-allele assays were performed to detect loss of the endogenous rat allele, gain-of-allele assays were performed to detect gain of the humanized allele, and CRISPR and retention assays were performed using the primers and probes set forth in Table 19.

TABLE 19

Screening Assays.			
Assay	Description	Primer/ Probe	Sequence
moTU	Upstream LOA	Fwd	GGGCTCAGGCAGGTATATGTTG (SEQ ID NO: 26)
		Probe (FAM)	ACAGATGCTGTCCCAACATAGCAAGA (SEQ ID NO: 27)
		Rev	CCAACCCTAAGCCAGTGAAACAG (SEQ ID NO: 28)
moTM	Middle LOA	Fwd	GCAGACACTGGATGGGTCA (SEQ ID NO: 32)
		Probe (FAM)	CCATTCGCGAGTTATGAGAAGCTGCA (SEQ ID NO: 33)
		Rev	ACAGGGTTAGCTGGTGAATGGA (SEQ ID NO: 34)
moTD	Downstream LOA	Fwd	GTGCTGGAGACCAGGAGACT (SEQ ID NO: 29)
		Probe (Cal-Orange)	TGCCATACTCAGTTTATACGGTGCTGAC (SEQ ID NO: 30)
		Rev	GCCTGGTGGCTCAGTCAATG (SEQ ID NO: 31)
7138 hU	Upstream Human Insertion	Fwd	AGGTGGGTAGGTCTGGAAGTG (SEQ ID NO: 14)
		Probe (FAM)	AATGCTGTCCCAAGAGTGGG (SEQ ID NO: 15)
		Rev	GTCCTGCATCCCTTGTCTTTG (SEQ ID NO: 16)
7138 hD	Downstream Human Insertion	Fwd	ATGTGGGCGTTGTGAGTCTC (SEQ ID NO: 17)
		Probe (Cal)	CGCTGCAGTGCAATTGAAGTCAAGCA (SEQ ID NO: 18)
		Rev	CTGTGGAGGGACGTGACCAG (SEQ ID NO: 19)
rnoTAU2	Upstream Retention	Fwd	TCGGAGCACAGGACTACAG (SEQ ID NO: 35)
		Probe (FAM)	CAAGAGGAAGTGTGTCAGGAAAGC (SEQ ID NO: 36)
		Rev	AGCGTGCCTCACCTAACCTCTA (SEQ ID NO: 37)
rnoTAD2	Downstream Retention	Fwd	GCACAGCACTGTAAAGGCA (SEQ ID NO: 38)
		Probe (Cal)	ACGGAAGTCAAGGAATTGGTATTGTTGT (SEQ ID NO: 39)
		Rev	ACACAGCTATGGGAGAAAGACTG (SEQ ID NO: 40)
rnoTGU	Upstream CRISPR Assay	Fwd	CTGGGTGATTGGGACTGAGAAAG (SEQ ID NO: 45)
		Probe (FAM)	CAGCCTTGAAAGTATGGCTTGGGC (SEQ ID NO: 46)
		Rev	GCACTCGCCACCGGAAG (SEQ ID NO: 47)

TABLE 19-continued

Screening Assays.			
Assay	Description	Primer/ Probe	Sequence
rnoTGD	Downstream	Fwd	GACCAGCTCACCCCTTACTTATGG (SEQ ID NO: 48)
	CRISPR	Probe	ACTGAATGCCAAGGGTGC GTTGA (SEQ ID NO: 49)
	Assay	(Cal)	
		Rev	TCTTGGAATCCGCTGAAGAGTT (SEQ ID NO: 50)

Modification-of-allele (MOA) assays including loss-of-allele (LOA) and gain-of-allele (GOA) assays are described, for example, in US 2014/0178879; US 2016/0145646; WO 2016/081923; and Frendewey et al. (2010) *Methods Enzymol.* 476:295-307, each of which is herein incorporated by reference in its entirety for all purposes. The loss-of-allele (LOA) assay inverts the conventional screening logic and quantifies the number of copies in a genomic DNA sample of the native locus to which the mutation was directed. In a correctly targeted heterozygous cell clone, the LOA assay detects one of the two native alleles (for genes not on the X or Y chromosome), the other allele being disrupted by the targeted modification. The same principle can be applied in reverse as a gain-of-allele (GOA) assay to quantify the copy number of the inserted targeting vector in a genomic DNA sample.

Retention assays are described in US 2016/0145646 and WO 2016/081923, each of which is herein incorporated by reference in its entirety for all purposes. Retention assays distinguish between correct targeted insertions of a nucleic acid insert into a target genomic locus from random transgenic insertions of the nucleic acid insert into genomic locations outside of the target genomic locus by assessing copy numbers of DNA templates from 5' and 3' target sequences corresponding to the 5' and 3' homology arms of the targeting vector, respectively. Specifically, retention assays determine copy numbers in a genomic DNA sample of a 5' target sequence DNA template intended to be retained in the modified target genomic locus and/or the 3' target sequence DNA template intended to be retained in the modified target genomic locus. In diploid cells, correctly targeted clones will retain a copy number of two. Copy numbers greater than two generally indicate transgenic integration of the targeting vector randomly outside of the target genomic locus rather than at the target genomic locus. Copy numbers of less than two generally indicate large deletions extending beyond the region targeted for deletion.

CRISPR assays are TAQMAN® assays designed to cover the region that is disrupted by the CRISPR gRNAs. When a CRISPR gRNA cuts and creates an indel (insertion or deletion), the TAQMAN® assay will fail to amplify and thus reports CRISPR cleavage.

The positive clone CB1 was thawed, expanded, and reconfirmed by TAQMAN®. CB1 was also confirmed by successful PCR from the 5' end of the human replacement sequence to the flanking rat genomic sequence, beyond the end of the 5' homology arm. The PCR amplicon was confirmed as correct by sequencing of the ends.

F0 and F1 rats were generated using methods as described in US 2014/0235933, US 2014/0310828, WO 2014/130706, and WO 2014/172489, each of which is herein incorporated by reference in its entirety for all purposes. In such methods, confirmed targeted rat ES cell clones (e.g., Dark Agouti ES cell clones) are microinjected into blastocysts (e.g., Sprague

Dawley (SD) blastocysts), which are then transferred to pseudopregnant recipient females (e.g., SD recipient females) for gestation using standard techniques. Chimeras are identified (e.g., by coat color), and male F0 chimeras are bred to female wild-type rats of the same strain (e.g., SD females). Germline (e.g., agouti) F1 pups are then genotyped for the presence of the targeted allele. All experiments performed in humanized TRKB rats as described below were performed in rats in which the self-deleting selection cassette was self-deleted.

Example 7. In Vivo Comparison of the Effect of TRKB Agonist Antibody H4H9816P2 and IgG4 Isotype Control REGN1945 on Retinal Ganglion Cell (RGC) Survival TrkBhuhihu Rats

Experimental Procedure

All procedures were conducted in accordance with the ARVO Statement for Use of Animals in Ophthalmic and Vision Research and the Regeneron Pharmaceutical Inc. IACUC. Adult female TrkB humanized rats (MAID100010), 8-10 weeks old, each weighing 200-250 g, were used. All surgical procedures on rats were performed under general anesthesia using an intraperitoneal injection of ketamine (63 mg/kg) and xylazine (6.0 mg/kg). Eye ointment containing erythromycin (0.5%, Bausch & Lomb) was applied to protect the cornea.

Intraorbital Optic Nerve Axotomy and Intravitreal Injection. The left optic nerve (ON) was exposed intraorbitally, its dura was opened. ON was transected about 1.5 mm behind the globe. Care was taken to avoid damaging the blood supply to the retina. Intravitreal injections were performed just posterior to the pars plana with a pulled glass pipette connected to a 50 µL Hamilton syringe. Care was taken not to damage the lens. Rats with any significant postoperative complications (e.g., retinal ischemia, cataract) were excluded from further analysis. Animals were allocated to different experimental groups. One control group received intravitreal injections of 3 µL isotype control REGN1945 (46.6 µg/µL); the other group received injection of 3 µL anti-human TRKB antibody H4H9816P2 (45.7 µg/µL) at 3 and 10 days after ON axotomy.

Immunohistochemical Staining and Counting of Viable Retinal Ganglion Cells (RGCs). BRN3A (brain-specific homeobox/POU domain protein 3A) was used as a marker for surviving retinal ganglion cells (RGCs), because it has been shown to be an efficient and reliable method for selective labelling of viable RGCs in retinal whole mounts after ON injury. See, e.g., Nadal-Nicolás et al. (2009) *Invest. Ophthalmol. Vis. Sci.* 50(8):3860-3868, herein incorporated by reference in its entirety for all purposes. To immunostain for BRN3A, retinas were blocked in 10% normal donkey serum and 0.5% Triton X-100 for 1 hr, then incubated in the same medium with BRN3A antibody (1:400; Cat #:

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sc-31984, Santa Cruz) 2 hr at room temperature. After further washes retinas were incubated with Alexa594-conjugated donkey anti-goat secondary antibody (1:400; Cat #: A-11058, Invitrogen) overnight at 4° C.

Results and Conclusions

To assess the effect of the TRKB agonist antibody on RGC survival in vivo, we used a complete optic nerve transection model. TRKB agonist antibody (H4H9816P2) or isotype control antibody was applied at 3 and 10 days after surgery. Animals were euthanized 14 days after axotomy. The RGC density in the uninjured contralateral eye is similar in the three TRKB genotypes (homozygous humanized, heterozygous humanized, and wild type), averaging around 1600 per mm² as shown in Table 20. The density of surviving RGCs was assessed in retinal whole mounts using BRN3A staining. We found that in homozygous TrkB^{hu/hu} humanized rats, TRKB agonist antibody (H4H9816P2) significantly (p<0.01, Mann-Whitney test) increased RGC survival compared with controls (685±106 vs. 255±66 RGCs per mm²). In heterozygous TrkB^{hu/+} humanized rats, there is also significant (p<0.05, Mann-Whitney test) survival effect

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of TrkB agonist Ab (444±90 vs. 208±50 RGCs per mm²). In wild type rats, there was a slight but not significant increase of RGC number in rats treated with TRKB agonist antibody compared to isotype control (Table 21). In conclusion, the TRKB agonist antibody (H4H9816P2) significantly increased RGC survival in TrkB^{hu/hu} humanized rats.

TABLE 20

	RGC Quantification (RGCs/mm ²) in Uninjured Control Eye.		
	hu/hu	hu/+	+/+
10	1637.3	1720.4	1636.3
	1551.5	2064.6	1670.2
	1651.4	1738.8	1873.4
15	1628.2	2029.8	1725.4
	1804.7	1929.6	1973.4
	1741.3	1645.9	
	1739.7	1761.5	
	1698.8	1787.5	
20	1862.5	1914.0	
	1779.4		

TABLE 21

	RGC Quantification (RGCs/mm ²) after Optic Nerve Injury.									
	H4H9816P2					Isotype control Ab				
	A: Y1	A: Y2	A: Y3	A: Y4	A: Y5	B: Y1	B: Y2	B: Y3	B: Y4	B: Y5
Hu/Hu	790.1	737.1	756.3	587.8	555.7	322.8	295.0	286.9	171.3	197.9
Hu/+	530.4	457.5	522.9	390.6	319.2	231.0	184.6	265.1	151.3	
+/+	320.9	355.5	256.9	342.7		112.3				

Example 8. Heavy and Light Chain Variable Region Amino Acid Sequences of Anti-TRKB Antibodies Used in Examples

Several fully human anti-TRKB antibodies (i.e., antibodies possessing human variable domains and human constant domains) were tested in the examples, including those designated as H4H9780P, H4H9814P, and H4H9816P2. Table 22 sets forth the amino acid sequence identifiers of the heavy and light chain variable regions and CDRs of selected anti-TRKB antibodies used in the examples. Table 23 sets forth the nucleic acid sequence identifiers of the heavy and light chain variable regions and CDRs of selected anti-TRKB antibodies used in the examples. These antibodies are described in more detail in U.S. patent application Ser. No. 16/202,881, filed Nov. 28, 2018, which is herein incorporated by reference in its entirety for all purposes.

TABLE 22

Ab Name	Amino Acid SEQ ID NOS for Anti-TRKB Antibodies.							
	VH	HCDR1	HCDR2	HCDR3	VK	LCDR1	LCDR2	LCDR3
H4H9780P	79	81	83	85	87	89	91	93
H4H9814P	95	97	99	101	103	105	107	109
H4H9816P2	111	113	115	117	119	121	123	125

TABLE 23

Nucleic Acid SEQ ID NOS for Anti-TRKB Antibodies.								
Ab Name	VH	HCDR1	HCDR2	HCDR3	VK	LCDR1	LCDR2	LCDR3
H4H9780P	78	80	82	84	86	88	90	92
H4H9814P	94	96	98	100	102	104	106	108
H4H9816P2	110	112	114	116	118	120	122	124

Antibodies are typically referred to herein according to the following nomenclature: Fc prefix (e.g., “H4H”), followed by a numerical identifier (e.g., “9780,” “9816,” etc., as shown in Table 22), followed by a “P” or “P2” suffix. The H4H prefix in the antibody designations indicates the particular Fc region isotype of the antibody. Thus, according to this nomenclature, an antibody may be referred to herein as, e.g., “H4H9780P,” which indicates a human IgG4 Fc region. Variable regions are fully human if denoted by the first “H” in the antibody designation. As will be appreciated by a person of ordinary skill in the art, an antibody having a particular Fc isotype can be converted to an antibody with a different Fc isotype (e.g., an antibody with a mouse IgG1 Fc can be converted to an antibody with a human IgG4, etc.), but in any event, the variable domains (including the CDRs)—which are indicated by the numerical identifiers shown in Table 22—will remain the same, and the binding properties to antigen are expected to be identical or substantially similar regardless of the nature of the Fc domain.

Example 9. Neuroprotective Effect of Anti-Human TrkB Agonist Antibodies in Humanized TrkB Rats

The experiments below were undertaken to evaluate the neuroprotective effect of the endogenous TRKB agonist, brain-derived neurotrophic factor (BDNF), and a TRKB agonist monoclonal antibody (mAb) in wild-type (WT) mice and rats and in humanized TrkB mice and rats.

The in vitro effects of BDNF and TRKB Ab were quantified by cell survival assays using differentiated human neuroblastoma cell line SH-SY5Y. In vitro, BDNF or TRKB Ab significantly increased cell survival in retinoic-acid-differentiated SH-SY5Y cells. The effects showed bell shaped dose responses with the optimal dose of 1 µg/mL for BDNF or 10 µg/mL for TRKB Ab. Neuroblastoma cell line SH-SY5Y was cultured in differentiation media containing all-trans 10 µM retinoic acid for 4 days. The culture was changed to serum-free differentiation media containing different dose of antibodies (0.01-100 µg/mL). Two days later, CCK8 reagent was added, plates were incubated for 3-4 hours, and OD450 was measured to determine percentage of surviving cells. Data were normalized to the serum-free media without antibodies. As shown in FIG. 9, TRKB mAbs (TrkB mAb1 is H4H9816P2; TrkB mAb2 is a control TrkB agonist antibody) dose-dependently increased the survival of SH-SY5Y cells. Human isotype control had no effect on SH-SY5Y cell survival. Serum-free media without antibodies resulted in 100% survival.

Retinas from P2 C57BL/6J mice were then dissected and dissociated. Retinal ganglion cells were purified by immuno-panning and cultured in a 96-well plate with treatment or no treatment. After 24 hours, MTT (3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide) was added to each well to calculate cell survival for each group. As shown in FIG. 10, BDNF had a bell-shaped response curve and optimal dose at 1 µg/mL. TrkB mAb2 (a control TrkB agonist antibody with affinity for human TrkB, mouse TrkB,

and rat TrkB) may have a bell-shaped curve as well at higher doses but shows neuroprotective effect.

To test the in vivo neuroprotective effect, WT and humanized TrkB mice and rats were used. Animals received intravitreal (IVT) injections of BDNF or TRKB mAb day 3 and 10 post-optic-nerve transection (ONT). Retinal ganglion cell (RGC) number was quantified using HALO software (Indica Labs) at 1 week for mouse or 2 weeks for rat after optic nerve transection by Brn3a IHC on retinal flat mounts.

RGC death in TrkB^{hu/hu} mice was similar to WT mice at 1 or 2 weeks after optic nerve transection. BDNF or TRKB Ab had small or no significant neuroprotective effect in WT or TrkB^{hu/hu} mice. In contrast, there was significant RGC neuroprotection in TrkB^{hu/hu} rats with IVT TRKB Ab. A decrease in body weight was observed in TrkB^{hu/hu} mice but not rats after IVT TRKB Ab treatment. BDNF had no effect on body weight in either mouse or rat.

FIGS. 11A and 11B show the results of an experiment assessing neuroprotection in an optic nerve transection model in WT mice and rats. In FIG. 11A, 8-9 week old Dark Agouti rats were given BDNF (5 µg), TrkB mAb2 (18 µg), isotype control antibody (18 µg), or vehicle control intravitreally at 3 days and 10 days after transection. TrkB mAb2 is a control TrkB agonist antibody with affinity for human TrkB, mouse TrkB, and rat TrkB. Retinas were dissected and stained for retinal ganglion cells 14 days after transection. BDNF and TRKB mAb showed significant neuroprotection as measured by retinal ganglion cell (RGC) density. In FIG. 11B, 8-week-old C57BL/6J WT mice were given BDNF (2.5 µg), TrkB mAb2 (10 µg), isotype control antibody (10 µg), or vehicle control intravitreally at 3 days and 10 days after transection. TrkB mAb2 is a control TrkB agonist antibody with affinity for human TrkB, mouse TrkB, and rat TrkB. There was no significant neuroprotection. Thus, BDNF and TRKB mAb treatment resulted in significant increases in RGC density in dissected retinas in wild type rats after optic nerve transection, whereas no significant effect on RGC density was observed in the same model in wild type mice.

FIGS. 12A and 12B show BDNF dose response in WT mice and rats. In FIG. 12A, BDNF dose response in an optic nerve crush (ONC) model in WT mice shows a small window of neuroprotection. FIG. 12B shows a BDNF dose response in an optic nerve transection model in WT rat from 0.13 µg to 30 µg. There is bell-shaped response similar to the in vitro data, with the optimal dose at 0.8 µg. Retinas were dissected and stained for retinal ganglion cells 14 days after transection. Thus, BDNF treatment resulted in much more pronounced dose response curve as measured by RGC density in dissected retinas in wild type rats after optic nerve transection compared to the much less pronounced BDNF dose response curve in as measured by RGC density in dissected retinas in wild type mice after optic nerve crush.

Neuroprotective effect of TRKB Abs was next tested in humanized TrkB rats. The results in FIGS. 13A and 13B show that intravitreal injection of TRKB mAb in optic-nerve-transected humanized TrkB rats show significant neuroprotection of retinal ganglion cells. Human TRKB homozygous, human TRKB heterozygous, or wild-type TrkB rats were given either TrkB mAb1 or isotype control antibody intravitreally (3 μ L) at 3 and 10 days after optic nerve transection. Fourteen days after transection, retinas were dissected and stained for RGCs. The rats were females that were 17-19 weeks old. As shown in FIG. 13A, rats treated with TrkB mAb1 (H4H9816P2) showed neuroprotection in all three genotypes compared to corresponding rats treated with isotype control antibody. Isotype-control-treated homozygous and heterozygous rats for human TRKB have higher RGC density than isotype-control-treated wild-type rats. FIG. 13B shows no RGC number difference in the naïve eyes between genotypes. FIG. 13C shows body weight of human TRKB homozygous mice given either TrkB agonist antibody (H4H9816P2) or isotype control antibody (REGN1945) at 14 days after transection.

Rat retinal whole-mount RGC isodensity maps were then created showing Brn3a labeled cells of non-injured and treated injured eyes in the three genotypes (data not shown). Whole mount reconstruction was prepared with the aid of motorized stage on fluorescence microscope (Nikon Eclipse Ti). RGCs were counted using an image analysis software (HALO®; Indica Labs, Corrales, NM, USA). Isodensity maps were generated through Matlab. Higher RGC density was observed with the humanized TrkB rats treated with TrkB mAb1 (H4H9816P2) compared to the isotype-control-treated rats (data not shown).

Taken together, the data shown in FIGS. 11A, 11B, 12A, 12B, and 13A-13C demonstrate that intravitreal administration of TRKB agonist mAb has a significant neuroprotective effect after optic nerve injury in humanized TrkB rats, in contrast to the small or no significant neuroprotective effect observed after optic nerve injury in humanized TrkB mice.

To further evaluate the effect of TRKB agonist antibodies on RGC survival in rats in the optic-nerve transection (ONT) model, a dose-response study was undertaken. Human TRKB homozygous rats (MAID100010; 75% SD, 25% DA) that were 1-9 months old were used. Six rats were used in each group. Human TRKB homozygous rats were given different doses of either TrkB mAb1 or isotype control antibody (REGN1945) intravitreally (3 μ L) at 3 and 10 days after optic nerve transection. Fourteen days after transection,

retinas were dissected and stained for RGCs. As shown in FIG. 14, TrkB mAb1 dose-dependently increased RGC survival in the TrkB humanized rats.

Next, the neuroprotective effect of different TrkB agonist antibodies was compared in human TRKB homozygous rats in the optic-nerve transection (ONT) model. Humanized TrkB rats (MAID100010; 75% SD, 25% DA) that were 8-10 weeks old were used. Five to six rats were used in each group. Human TRKB homozygous rats were given either H4H9816P2-L9 (10 μ g), H4H9814P-L9 (10 μ g), H4H9780P-L5 (10 μ g), a combination of all three (3.3 μ g each), or isotype control antibody (REGN1945; 10 μ g) intravitreally (3 μ L) at 3 and 10 days after optic nerve transection. Fourteen days after transection, retinas were dissected and stained for RGCs. The results are shown in FIGS. 15A and 15B. Each TrkB agonist antibody had a neuroprotective effect compared to the isotype control antibody. Body weight in each group was similar (data not shown).

In contrast, the TrkB agonist antibodies H4H9780P and H4H9814P did not have any neuroprotective effect in wild type rats. Neuroprotective effect was assessed in wild type rats using the optic-nerve transection (ONT) model. Female wild type rats that were 8-10 weeks old were used. Five to six rats were used in each group. Wild type rats were given either H4H9780P (120 μ g), H4H9814P (120 μ g), or isotype control antibody (REGN1945; 120 μ g) intravitreally (3 μ L) at 3 and 10 days after optic nerve transection. Fourteen days after transection, retinas were dissected and stained for RGCs. As shown in FIG. 16, neither TrkB agonist antibody had a significant neuroprotective effect in wild type rats.

In addition, TrkB agonist antibody (H4H9780P) did not have a neuroprotective effect in human TRKB homozygous mice. Male human TRKB homozygous mice (MAID7139; 75% C57BL/6, 25% 129) that were 5 months old were used. Five to six mice were used in each group. Human TRKB homozygous mice were given either H4H9780P (40 μ g per eye) or isotype control antibody (REGN1945; 40 μ g per eye) intravitreally (1 μ L) at 3 and 10 days after optic nerve transection. Fourteen days after transection, retinas were dissected and stained for RGCs. As shown in FIGS. 17A and 17B, the TrkB agonist antibody did not have a neuroprotective effect in human TRKB homozygous mice in contrast to the neuroprotective effect seen in human TRKB homozygous rats. FIG. 17C shows body weight of human TRKB homozygous mice given either H4H9780P or isotype control antibody at 14 days after transection.

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 20           25           30
Pro Thr Ser Cys Lys Cys Ser Ser Ala Arg Ile Trp Cys Thr Glu Pro
 35           40           45
Ser Pro Gly Ile Val Ala Phe Pro Arg Leu Glu Pro Asn Ser Val Asp
 50           55           60
Pro Glu Asn Ile Thr Glu Ile Leu Ile Ala Asn Gln Lys Arg Leu Glu
 65           70           75           80
Ile Ile Asn Glu Asp Asp Val Glu Ala Tyr Val Gly Leu Arg Asn Leu
 85           90           95
Thr Ile Val Asp Ser Gly Leu Lys Phe Val Ala Tyr Lys Ala Phe Leu
100           105           110
Lys Asn Ser Asn Leu Arg His Ile Asn Phe Thr Arg Asn Lys Leu Thr
115           120           125
Ser Leu Ser Arg Arg His Phe Arg His Leu Asp Leu Ser Asp Leu Ile
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145           150           155           160
Thr Leu Gln Glu Thr Lys Ser Ser Pro Asp Thr Gln Asp Leu Tyr Cys
165           170           175
Leu Asn Glu Ser Ser Lys Asn Met Pro Leu Ala Asn Leu Gln Ile Pro
180           185           190
Asn Cys Gly Leu Pro Ser Ala Arg Leu Ala Ala Pro Asn Leu Thr Val
195           200           205
Glu Glu Gly Lys Ser Val Thr Leu Ser Cys Ser Val Gly Gly Asp Pro
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Leu Pro Thr Leu Tyr Trp Asp Val Gly Asn Leu Val Ser Lys His Met
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-continued

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				405					410					415	
Pro	Ser	Thr	Asp	Val	Ala	Asp	Gln	Ser	Asn	Arg	Glu	His	Leu	Ser	Val
			420				425						430		
Tyr	Ala	Val	Val	Val	Ile	Ala	Ser	Val	Val	Gly	Phe	Cys	Leu	Leu	Val
	435					440					445				
Met	Leu	Leu	Leu	Leu	Lys	Leu	Ala	Arg	His	Ser	Lys	Phe	Gly	Met	Lys
450					455						460				
Gly	Pro	Ala	Ser	Val	Ile	Ser	Asn	Asp	Asp	Asp	Ser	Ala	Ser	Pro	Leu
465					470				475						480
His	His	Ile	Ser	Asn	Gly	Ser	Asn	Thr	Pro	Ser	Ser	Ser	Glu	Gly	Gly
			485					490						495	
Pro	Asp	Ala	Val	Ile	Ile	Gly	Met	Thr	Lys	Ile	Pro	Val	Ile	Glu	Asn
			500					505					510		
Pro	Gln	Tyr	Phe	Gly	Ile	Thr	Asn	Ser	Gln	Leu	Lys	Pro	Asp	Thr	Phe
		515					520					525			
Val	Gln	His	Ile	Lys	Arg	His	Asn	Ile	Val	Leu	Lys	Arg	Glu	Leu	Gly
	530					535					540				
Glu	Gly	Ala	Phe	Gly	Lys	Val	Phe	Leu	Ala	Glu	Cys	Tyr	Asn	Leu	Cys
545					550					555					560
Pro	Glu	Gln	Asp	Lys	Ile	Leu	Val	Ala	Val	Lys	Thr	Leu	Lys	Asp	Ala
			565					570						575	
Ser	Asp	Asn	Ala	Arg	Lys	Asp	Phe	His	Arg	Glu	Ala	Glu	Leu	Leu	Thr
			580					585					590		
Asn	Leu	Gln	His	Glu	His	Ile	Val	Lys	Phe	Tyr	Gly	Val	Cys	Val	Glu
	595						600					605			
Gly	Asp	Pro	Leu	Ile	Met	Val	Phe	Glu	Tyr	Met	Lys	His	Gly	Asp	Leu
610					615						620				
Asn	Lys	Phe	Leu	Arg	Ala	His	Gly	Pro	Asp	Ala	Val	Leu	Met	Ala	Glu
625					630					635					640
Gly	Asn	Pro	Pro	Thr	Glu	Leu	Thr	Gln	Ser	Gln	Met	Leu	His	Ile	Ala
			645					650						655	
Gln	Gln	Ile	Ala	Ala	Gly	Met	Val	Tyr	Leu	Ala	Ser	Gln	His	Phe	Val
			660					665					670		
His	Arg	Asp	Leu	Ala	Thr	Arg	Asn	Cys	Leu	Val	Gly	Glu	Asn	Leu	Leu
		675					680					685			
Val	Lys	Ile	Gly	Asp	Phe	Gly	Met	Ser	Arg	Asp	Val	Tyr	Ser	Thr	Asp
	690					695					700				
Tyr	Tyr	Arg	Val	Gly	Gly	His	Thr	Met	Leu	Pro	Ile	Arg	Trp	Met	Pro
705					710					715					720
Pro	Glu	Ser	Ile	Met	Tyr	Arg	Lys	Phe	Thr	Thr	Glu	Ser	Asp	Val	Trp
			725					730						735	
Ser	Leu	Gly	Val	Val	Leu	Trp	Glu	Ile	Phe	Thr	Tyr	Gly	Lys	Gln	Pro
			740				745						750		
Trp	Tyr	Gln	Leu	Ser	Asn	Asn	Glu	Val	Ile	Glu	Cys	Ile	Thr	Gln	Gly
		755				760						765			
Arg	Val	Leu	Gln	Arg	Pro	Arg	Thr	Cys	Pro	Gln	Glu	Val	Tyr	Glu	Leu
	770				775						780				
Met	Leu	Gly	Cys	Trp	Gln	Arg	Glu	Pro	His	Thr	Arg	Lys	Asn	Ile	Lys
785					790					795					800
Ser	Ile	His	Thr	Leu	Leu	Gln	Asn	Leu	Ala	Lys	Ala	Ser	Pro	Val	Tyr

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805	810	815
Leu Asp Ile Leu Gly		
820		
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<223> OTHER INFORMATION: Extracellular Domain		
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<223> OTHER INFORMATION: Transmembrane Domain		
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<223> OTHER INFORMATION: Cytoplasmic Domain		
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Gly Leu Cys Leu Leu Val Leu Gly Phe Trp Arg Ala Ser Leu Ala Cys		
20 25 30		
Pro Met Ser Cys Lys Cys Ser Thr Thr Arg Ile Trp Cys Thr Glu Pro		
35 40 45		
Ser Pro Gly Ile Val Ala Phe Pro Arg Leu Glu Pro Asn Ser Ile Asp		
50 55 60		
Pro Glu Asn Ile Thr Glu Ile Leu Ile Ala Asn Gln Lys Arg Leu Glu		
65 70 75 80		
Ile Ile Asn Glu Asp Asp Val Glu Ala Tyr Val Gly Leu Lys Asn Leu		
85 90 95		
Thr Ile Val Asp Ser Gly Leu Lys Phe Val Ala Tyr Lys Ala Phe Leu		
100 105 110		
Lys Asn Gly Asn Leu Arg His Ile Asn Phe Thr Arg Asn Lys Leu Thr		
115 120 125		
Ser Leu Ser Arg Arg His Phe Arg His Leu Asp Leu Ser Asp Leu Ile		
130 135 140		
Leu Thr Gly Asn Pro Phe Thr Cys Ser Cys Asp Ile Met Trp Leu Lys		
145 150 155 160		
Thr Leu Gln Glu Thr Lys Ser Ser Pro Asp Thr Gln Asp Leu Tyr Cys		
165 170 175		
Leu Asn Glu Ser Ser Lys Asn Thr Pro Leu Ala Asn Leu Gln Ile Pro		
180 185 190		
Asn Cys Gly Leu Pro Ser Ala Arg Leu Ala Ala Pro Asn Leu Thr Val		
195 200 205		
Glu Glu Gly Lys Ser Val Thr Ile Ser Cys Ser Val Gly Gly Asp Pro		
210 215 220		
Leu Pro Thr Leu Tyr Trp Asp Val Gly Asn Leu Val Ser Lys His Met		
225 230 235 240		
Asn Glu Thr Ser His Thr Gln Gly Ser Leu Arg Ile Thr Asn Ile Ser		
245 250 255		
Ser Asp Asp Ser Gly Lys Gln Ile Ser Cys Val Ala Glu Asn Leu Val		
260 265 270		

Gly	Glu	Asp	Gln	Asp	Ser	Val	Asn	Leu	Thr	Val	His	Phe	Ala	Pro	Thr
	275						280					285			
Ile	Thr	Phe	Leu	Glu	Ser	Pro	Thr	Ser	Asp	His	His	Trp	Cys	Ile	Pro
	290					295					300				
Phe	Thr	Val	Arg	Gly	Asn	Pro	Lys	Pro	Ala	Leu	Gln	Trp	Phe	Tyr	Asn
305					310					315					320
Gly	Ala	Ile	Leu	Asn	Glu	Ser	Lys	Tyr	Ile	Cys	Thr	Lys	Ile	His	Val
				325					330					335	
Thr	Asn	His	Thr	Glu	Tyr	His	Gly	Cys	Leu	Gln	Leu	Asp	Asn	Pro	Thr
			340					345					350		
His	Met	Asn	Asn	Gly	Asp	Tyr	Thr	Leu	Met	Ala	Lys	Asn	Glu	Tyr	Gly
	355						360					365			
Lys	Asp	Glu	Arg	Gln	Ile	Ser	Ala	His	Phe	Met	Gly	Arg	Pro	Gly	Val
	370					375					380				
Asp	Tyr	Glu	Thr	Asn	Pro	Asn	Tyr	Pro	Glu	Val	Leu	Tyr	Glu	Asp	Trp
385					390					395					400
Thr	Thr	Pro	Thr	Asp	Ile	Gly	Asp	Thr	Thr	Asn	Lys	Ser	Asn	Glu	Ile
				405					410					415	
Pro	Ser	Thr	Asp	Val	Ala	Asp	Gln	Thr	Asn	Arg	Glu	His	Leu	Ser	Val
			420					425					430		
Tyr	Ala	Val	Val	Val	Ile	Ala	Ser	Val	Val	Gly	Phe	Cys	Leu	Leu	Val
	435						440					445			
Met	Leu	Leu	Leu	Leu	Lys	Leu	Ala	Arg	His	Ser	Lys	Phe	Gly	Met	Lys
	450					455					460				
Gly	Pro	Ala	Ser	Val	Ile	Ser	Asn	Asp	Asp	Asp	Ser	Ala	Ser	Pro	Leu
465					470				475						480
His	His	Ile	Ser	Asn	Gly	Ser	Asn	Thr	Pro	Ser	Ser	Ser	Glu	Gly	Gly
				485					490					495	
Pro	Asp	Ala	Val	Ile	Ile	Gly	Met	Thr	Lys	Ile	Pro	Val	Ile	Glu	Asn
			500					505					510		
Pro	Gln	Tyr	Phe	Gly	Ile	Thr	Asn	Ser	Gln	Leu	Lys	Pro	Asp	Thr	Phe
	515						520					525			
Val	Gln	His	Ile	Lys	Arg	His	Asn	Ile	Val	Leu	Lys	Arg	Glu	Leu	Gly
	530					535					540				
Glu	Gly	Ala	Phe	Gly	Lys	Val	Phe	Leu	Ala	Glu	Cys	Tyr	Asn	Leu	Cys
545					550					555					560
Pro	Glu	Gln	Asp	Lys	Ile	Leu	Val	Ala	Val	Lys	Thr	Leu	Lys	Asp	Ala
				565					570					575	
Ser	Asp	Asn	Ala	Arg	Lys	Asp	Phe	His	Arg	Glu	Ala	Glu	Leu	Leu	Thr
			580					585					590		
Asn	Leu	Gln	His	Glu	His	Ile	Val	Lys	Phe	Tyr	Gly	Val	Cys	Val	Glu
	595						600					605			
Gly	Asp	Pro	Leu	Ile	Met	Val	Phe	Glu	Tyr	Met	Lys	His	Gly	Asp	Leu
	610					615					620				
Asn	Lys	Phe													

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Val Lys Ile Gly Asp Phe Gly Met Ser Arg Asp Val Tyr Ser Thr Asp
 690                695                700

Tyr Tyr Arg Val Gly Gly His Thr Met Leu Pro Ile Arg Trp Met Pro
 705                710                715                720

Pro Glu Ser Ile Met Tyr Arg Lys Phe Thr Thr Glu Ser Asp Val Trp
                725                730                735

Ser Leu Gly Val Val Leu Trp Glu Ile Phe Thr Tyr Gly Lys Gln Pro
                740                745                750

Trp Tyr Gln Leu Ser Asn Asn Glu Val Ile Glu Cys Ile Thr Gln Gly
                755                760                765

Arg Val Leu Gln Arg Pro Arg Thr Cys Pro Gln Glu Val Tyr Glu Leu
 770                775                780

Met Leu Gly Cys Trp Gln Arg Glu Pro His Thr Arg Lys Asn Ile Lys
 785                790                795                800

Asn Ile His Thr Leu Leu Gln Asn Leu Ala Lys Ala Ser Pro Val Tyr
                805                810                815

Leu Asp Ile Leu Gly
                820

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<210> SEQ ID NO 3
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<213> ORGANISM: Homo sapiens
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<223> OTHER INFORMATION: Signal Peptide
<220> FEATURE:
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<222> LOCATION: (32)..(430)
<223> OTHER INFORMATION: Extracellular Domain
<220> FEATURE:
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<222> LOCATION: (431)..(454)
<223> OTHER INFORMATION: Transmembrane Domain
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<222> LOCATION: (455)..(838)
<223> OTHER INFORMATION: Cytoplasmic Domain

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<400> SEQUENCE: 3

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Met Ser Ser Trp Ile Arg Trp His Gly Pro Ala Met Ala Arg Leu Trp
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Gly Phe Cys Trp Leu Val Val Gly Phe Trp Arg Ala Ala Phe Ala Cys
 20                25                30

Pro Thr Ser Cys Lys Cys Ser Ala Ser Arg Ile Trp Cys Ser Asp Pro
 35                40                45

Ser Pro Gly Ile Val Ala Phe Pro Arg Leu Glu Pro Asn Ser Val Asp
 50                55                60

Pro Glu Asn Ile Thr Glu Ile Phe Ile Ala Asn Gln Lys Arg Leu Glu
 65                70                75                80

Ile Ile Asn Glu Asp Asp Val Glu Ala Tyr Val Gly Leu Arg Asn Leu
 85                90                95

Thr Ile Val Asp Ser Gly Leu Lys Phe Val Ala His Lys Ala Phe Leu
 100               105               110

Lys Asn Ser Asn Leu Gln His Ile Asn Phe Thr Arg Asn Lys Leu Thr
 115               120               125

Ser Leu Ser Arg Lys His Phe Arg His Leu Asp Leu Ser Glu Leu Ile
 130               135               140

Leu Val Gly Asn Pro Phe Thr Cys Ser Cys Asp Ile Met Trp Ile Lys

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145					150					155					160
Thr	Leu	Gln	Glu	Ala	Lys	Ser	Ser	Pro	Asp	Thr	Gln	Asp	Leu	Tyr	Cys
				165					170					175	
Leu	Asn	Glu	Ser	Ser	Lys	Asn	Ile	Pro	Leu	Ala	Asn	Leu	Gln	Ile	Pro
			180					185					190		
Asn	Cys	Gly	Leu	Pro	Ser	Ala	Asn	Leu	Ala	Ala	Pro	Asn	Leu	Thr	Val
		195					200					205			
Glu	Glu	Gly	Lys	Ser	Ile	Thr	Leu	Ser	Cys	Ser	Val	Ala	Gly	Asp	Pro
	210					215					220				
Val	Pro	Asn	Met	Tyr	Trp	Asp	Val	Gly	Asn	Leu	Val	Ser	Lys	His	Met
225					230					235					240
Asn	Glu	Thr	Ser	His	Thr	Gln	Gly	Ser	Leu	Arg	Ile	Thr	Asn	Ile	Ser
			245						250					255	
Ser	Asp	Asp	Ser	Gly	Lys	Gln	Ile	Ser	Cys	Val	Ala	Glu	Asn	Leu	Val
			260					265					270		
Gly	Glu	Asp	Gln	Asp	Ser	Val	Asn	Leu	Thr	Val	His	Phe	Ala	Pro	Thr
		275					280					285			
Ile	Thr	Phe	Leu	Glu	Ser	Pro	Thr	Ser	Asp	His	His	Trp	Cys	Ile	Pro
	290					295					300				
Phe	Thr	Val	Lys	Gly	Asn	Pro	Lys	Pro	Ala	Leu	Gln	Trp	Phe	Tyr	Asn
305					310					315					320
Gly	Ala	Ile	Leu	Asn	Glu	Ser	Lys	Tyr	Ile	Cys	Thr	Lys	Ile	His	Val
			325						330					335	
Thr	Asn	His	Thr	Glu	Tyr	His	Gly	Cys	Leu	Gln	Leu	Asp	Asn	Pro	Thr
			340					345					350		
His	Met	Asn	Asn	Gly	Asp	Tyr	Thr	Leu	Ile	Ala	Lys	Asn	Glu	Tyr	Gly
		355					360					365			
Lys	Asp	Glu	Lys	Gln	Ile	Ser	Ala	His	Phe	Met	Gly	Trp	Pro	Gly	Ile
	370					375					380				
Asp	Asp	Gly	Ala	Asn	Pro	Asn	Tyr	Pro	Asp	Val	Ile	Tyr	Glu	Asp	Tyr
385					390					395					400
Gly	Thr	Ala	Ala	Asn	Asp	Ile	Gly	Asp	Thr	Thr	Asn	Arg	Ser	Asn	Glu
			405						410					415	
Ile	Pro	Ser	Thr	Asp	Val	Thr	Asp	Lys	Thr	Gly	Arg	Glu	His	Leu	Ser
			420					425					430		
Val	Tyr	Ala	Val	Val	Val	Ile	Ala	Ser	Val	Val	Gly	Phe	Cys	Leu	Leu
		435				440						445			
Val	Met	Leu	Phe	Leu	Leu	Lys	Leu	Ala	Arg	His	Ser	Lys	Phe	Gly	Met
	450					455					460				
Lys	Asp	Phe	Ser	Trp	Phe	Gly	Phe	Gly	Lys	Val	Lys	Ser	Arg	Gln	Gly
465					470					475					480
Val	Gly	Pro	Ala	Ser	Val	Ile	Ser	Asn	Asp	Asp	Asp	Ser	Ala	Ser	Pro
			485						490					495	
Leu	His	His	Ile	Ser	Asn	Gly	Ser	Asn	Thr	Pro	Ser	Ser	Ser	Glu	Gly
			500					505					510		
Gly	Pro	Asp	Ala	Val	Ile	Ile	Gly	Met	Thr	Lys	Ile	Pro	Val	Ile	Glu
		515					520					525			
Asn	Pro	Gln	Tyr	Phe	Gly	Ile	Thr	Asn	Ser	Gln	Leu	Lys	Pro	Asp	Thr
	530					535					540				
Phe	Val	Gln	His	Ile	Lys	Arg	His	Asn	Ile	Val	Leu	Lys	Arg	Glu	Leu
545					550					555					560
Gly	Glu	Gly	Ala	Phe	Gly	Lys	Val	Phe	Leu	Ala	Glu	Cys	Tyr	Asn	Leu
			565					570						575	

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Cys Pro Glu Gln Asp Lys Ile Leu Val Ala Val Lys Thr Leu Lys Asp
 580 585 590
 Ala Ser Asp Asn Ala Arg Lys Asp Phe His Arg Glu Ala Glu Leu Leu
 595 600 605
 Thr Asn Leu Gln His Glu His Ile Val Lys Phe Tyr Gly Val Cys Val
 610 615 620
 Glu Gly Asp Pro Leu Ile Met Val Phe Glu Tyr Met Lys His Gly Asp
 625 630 635 640
 Leu Asn Lys Phe Leu Arg Ala His Gly Pro Asp Ala Val Leu Met Ala
 645 650 655
 Glu Gly Asn Pro Pro Thr Glu Leu Thr Gln Ser Gln Met Leu His Ile
 660 665 670
 Ala Gln Gln Ile Ala Ala Gly Met Val Tyr Leu Ala Ser Gln His Phe
 675 680 685
 Val His Arg Asp Leu Ala Thr Arg Asn Cys Leu Val Gly Glu Asn Leu
 690 695 700
 Leu Val Lys Ile Gly Asp Phe Gly Met Ser Arg Asp Val Tyr Ser Thr
 705 710 715 720
 Asp Tyr Tyr Arg Val Gly Gly His Thr Met Leu Pro Ile Arg Trp Met
 725 730 735
 Pro Pro Glu Ser Ile Met Tyr Arg Lys Phe Thr Thr Glu Ser Asp Val
 740 745 750
 Trp Ser Leu Gly Val Val Leu Trp Glu Ile Phe Thr Tyr Gly Lys Gln
 755 760 765
 Pro Trp Tyr Gln Leu Ser Asn Asn Glu Val Ile Glu Cys Ile Thr Gln
 770 775 780
 Gly Arg Val Leu Gln Arg Pro Arg Thr Cys Pro Gln Glu Val Tyr Glu
 785 790 795 800
 Leu Met Leu Gly Cys Trp Gln Arg Glu Pro His Met Arg Lys Asn Ile
 805 810 815
 Lys Gly Ile His Thr Leu Leu Gln Asn Leu Ala Lys Ala Ser Pro Val
 820 825 830
 Tyr Leu Asp Ile Leu Gly
 835

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 <213> ORGANISM: Artificial Sequence
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 <223> OTHER INFORMATION: Signal Peptide
 <220> FEATURE:
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 <223> OTHER INFORMATION: Mouse Sequence
 <220> FEATURE:
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 <222> LOCATION: (32)..(430)
 <223> OTHER INFORMATION: Extracellular Domain
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 <223> OTHER INFORMATION: Transmembrane Domain

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<223> OTHER INFORMATION: Mouse Sequence
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<222> LOCATION: (455)..(822)
<223> OTHER INFORMATION: Cytoplasmic Domain

<400> SEQUENCE: 4

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20          25          30
Pro Thr Ser Cys Lys Cys Ser Ala Ser Arg Ile Trp Cys Ser Asp Pro
35          40          45
Ser Pro Gly Ile Val Ala Phe Pro Arg Leu Glu Pro Asn Ser Val Asp
50          55          60
Pro Glu Asn Ile Thr Glu Ile Phe Ile Ala Asn Gln Lys Arg Leu Glu
65          70          75          80
Ile Ile Asn Glu Asp Asp Val Glu Ala Tyr Val Gly Leu Arg Asn Leu
85          90          95
Thr Ile Val Asp Ser Gly Leu Lys Phe Val Ala His Lys Ala Phe Leu
100         105         110
Lys Asn Ser Asn Leu Gln His Ile Asn Phe Thr Arg Asn Lys Leu Thr
115         120         125
Ser Leu Ser Arg Lys His Phe Arg His Leu Asp Leu Ser Glu Leu Ile
130         135         140
Leu Val Gly Asn Pro Phe Thr Cys Ser Cys Asp Ile Met Trp Ile Lys
145         150         155         160
Thr Leu Gln Glu Ala Lys Ser Ser Pro Asp Thr Gln Asp Leu Tyr Cys
165         170         175
Leu Asn Glu Ser Ser Lys Asn Ile Pro Leu Ala Asn Leu Gln Ile Pro
180         185         190
Asn Cys Gly Leu Pro Ser Ala Asn Leu Ala Ala Pro Asn Leu Thr Val
195         200         205
Glu Glu Gly Lys Ser Ile Thr Leu Ser Cys Ser Val Ala Gly Asp Pro
210         215         220
Val Pro Asn Met Tyr Trp Asp Val Gly Asn Leu Val Ser Lys His Met
225         230         235         240
Asn Glu Thr Ser His Thr Gln Gly Ser Leu Arg Ile Thr Asn Ile Ser
245         250         255
Ser Asp Asp Ser Gly Lys Gln Ile Ser Cys Val Ala Glu Asn Leu Val
260         265         270
Gly Glu Asp Gln Asp Ser Val Asn Leu Thr Val His Phe Ala Pro Thr
275         280         285
Ile Thr Phe Leu Glu Ser Pro Thr Ser Asp His His Trp Cys Ile Pro
290         295         300
Phe Thr Val Lys Gly Asn Pro Lys Pro Ala Leu Gln Trp Phe Tyr Asn
305         310         315         320
Gly Ala Ile Leu Asn Glu Ser Lys Tyr Ile Cys Thr Lys Ile His Val
325         330         335
Thr Asn His Thr Glu Tyr His Gly Cys Leu Gln Leu Asp Asn Pro Thr
340         345         350
His Met Asn Asn Gly Asp Tyr Thr Leu Ile Ala Lys Asn Glu Tyr Gly
355         360         365

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Lys 370	Asp	Glu	Lys	Gln	Ile	Ser 375	Ala	His	Phe	Met 380	Gly	Trp	Pro	Gly	Ile
Asp 385	Asp	Gly	Ala	Asn	Pro 390	Asn	Tyr	Pro	Asp	Val 395	Ile	Tyr	Glu	Asp	Tyr 400
Gly	Thr	Ala	Ala	Asn 405	Asp	Ile	Gly	Asp	Thr 410	Thr	Asn	Arg	Ser	Asn 415	Glu
Ile	Pro	Ser	Thr 420	Asp	Val	Thr	Asp	Lys 425	Thr	Gly	Arg	Glu	His 430	Leu	Ser
Val	Tyr 435	Ala	Val	Val	Val	Ile	Ala 440	Ser	Val	Val	Gly	Phe 445	Cys	Leu	Leu
Val 450	Met	Leu	Leu	Leu	Leu	Lys 455	Leu	Ala	Arg	His	Ser 460	Lys	Phe	Gly	Met
Lys 465	Gly	Pro	Ala	Ser	Val 470	Ile	Ser	Asn	Asp	Asp 475	Asp	Ser	Ala	Ser	Pro 480
Leu	His	His	Ile 485	Ser	Asn	Gly	Ser	Asn	Thr 490	Pro	Ser	Ser	Ser	Glu 495	Gly
Gly	Pro	Asp	Ala 500	Val	Ile	Ile	Gly	Met 505	Thr	Lys	Ile	Pro	Val 510	Ile	Glu
Asn	Pro	Gln	Tyr 515	Phe	Gly	Ile	Thr 520	Asn	Ser	Gln	Leu	Lys 525	Pro	Asp	Thr
Phe 530	Val	Gln	His	Ile	Lys	Arg 535	His	Asn	Ile	Val	Leu 540	Lys	Arg	Glu	Leu
Gly 545	Glu	Gly	Ala	Phe	Gly 550	Lys	Val	Phe	Leu	Ala 555	Glu	Cys	Tyr	Asn	Leu 560
Cys	Pro	Glu	Gln	Asp 565	Lys	Ile	Leu	Val	Ala 570	Val	Lys	Thr	Leu	Lys 575	Asp
Ala	Ser	Asp	Asn 580	Ala	Arg	Lys	Asp	Phe 585	His	Arg	Glu	Ala 590	Glu	Leu	Leu
Thr	Asn 595	Leu	Gln	His	Glu	His 600	Ile	Val	Lys	Phe	Tyr	Gly 605	Val	Cys	Val
Glu 610	Gly	Asp	Pro	Leu	Ile	Met 615	Val	Phe	Glu	Tyr	Met 620	Lys	His	Gly	Asp
Leu 625	Asn	Lys	Phe	Leu	Arg 630	Ala	His	Gly	Pro	Asp 635	Ala	Val	Leu	Met	Ala 640
Glu	Gly	Asn	Pro	Pro 645	Thr	Glu	Leu	Thr	Gln 650	Ser	Gln	Met	Leu	His 655	Ile
Ala	Gln	Gln	Ile 660	Ala	Ala	Gly	Met	Val 665	Tyr	Leu	Ala	Ser 670	Gln	His	Phe
Val	His 675	Arg	Asp	Leu	Ala	Thr 680	Arg	Asn	Cys	Leu	Val	Gly 685	Glu	Asn	Leu
Leu 690	Val	Lys	Ile	Gly	Asp	Phe 695	Gly	Met	Ser	Arg	Asp 700	Val	Tyr	Ser	Thr
Asp 705	Tyr	Tyr	Arg	Val	Gly 710	Gly	His	Thr	Met	Leu 715	Pro	Ile	Arg	Trp	Met 720
Pro	Pro	Glu	Ser	Ile 725	Met	Tyr	Arg	Lys	Phe 730	Thr	Thr	Glu	Ser	Asp	Val 735
Trp	Ser	Leu	Gly 740	Val	Val	Leu	Trp	Glu 745	Ile	Phe	Thr	Tyr 750	Gly	Lys	Gln
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<210> SEQ ID NO 7

<211> LENGTH: 4750

<212> TYPE: DNA

<213> ORGANISM: Rattus norvegicus

<400> SEQUENCE: 7

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<211> LENGTH: 4057
<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

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<400> SEQUENCE: 8

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<210> SEQ ID NO 9
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<212> TYPE: DNA
<213> ORGANISM: Mus musculus
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<222> LOCATION: (1)..(93)
<223> OTHER INFORMATION: Signal Peptide
<220> FEATURE:
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<222> LOCATION: (94)..(1287)
<223> OTHER INFORMATION: Extracellular Domain
<220> FEATURE:
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<223> OTHER INFORMATION: Transmembrane Domain
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<400> SEQUENCE: 9

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<210> SEQ ID NO 10
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<221> NAME/KEY: misc_feature
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<223> OTHER INFORMATION: Signal Peptide
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<222> LOCATION: (94)..(1287)
<223> OTHER INFORMATION: Extracellular Domain
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<223> OTHER INFORMATION: Transmembrane Domain
<220> FEATURE:
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<223> OTHER INFORMATION: Cytoplasmic Domain

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<220> FEATURE:
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<223> OTHER INFORMATION: Transmembrane Domain
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<223> OTHER INFORMATION: Human Sequence
<220> FEATURE:
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<211> LENGTH: 2469

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<212> TYPE: DNA
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<220> FEATURE:
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<223> OTHER INFORMATION: Signal Peptide
<220> FEATURE:
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<220> FEATURE:
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<222> LOCATION: (94)..(1290)
<223> OTHER INFORMATION: Extracellular Domain
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (94)..(1296)
<223> OTHER INFORMATION: Human Sequence
<220> FEATURE:
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<223> OTHER INFORMATION: Transmembrane Domain
<220> FEATURE:
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<222> LOCATION: (1297)..(2466)
<223> OTHER INFORMATION: Rat Sequence
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<223> OTHER INFORMATION: Cytoplasmic Domain

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<400> SEQUENCE: 14

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<210> SEQ ID NO 15
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<400> SEQUENCE: 15

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<400> SEQUENCE: 16

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<210> SEQ ID NO 17
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<212> TYPE: DNA
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<220> FEATURE:
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<400> SEQUENCE: 17

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<212> TYPE: DNA
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<400> SEQUENCE: 18

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<210> SEQ ID NO 19
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<400> SEQUENCE: 19

ctgtggaggg acgtgaccag 20

<210> SEQ ID NO 20
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<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 20

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<210> SEQ ID NO 21
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<212> TYPE: DNA
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<400> SEQUENCE: 21

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<210> SEQ ID NO 22
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<400> SEQUENCE: 22

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<210> SEQ ID NO 23
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<400> SEQUENCE: 23

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<400> SEQUENCE: 24

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<210> SEQ ID NO 25
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 25

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<210> SEQ ID NO 26
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<400> SEQUENCE: 26

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<210> SEQ ID NO 27
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<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 27

acagatgctg tcccaaacad agcaaga 27

<210> SEQ ID NO 28
<211> LENGTH: 23
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 28

ccaaccctaa gccagtgaag cag 23

<210> SEQ ID NO 29
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 29

gtgctggaga ccaggagact 20

<210> SEQ ID NO 30
<211> LENGTH: 28
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence

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<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 30

tgccatactc agtttatacg gtgctgac 28

<210> SEQ ID NO 31
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 31

gcctggtggc tcagtcaatg 20

<210> SEQ ID NO 32
<211> LENGTH: 19
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 32

gcagacactg gatgggtca 19

<210> SEQ ID NO 33
<211> LENGTH: 26
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 33

ccattcgca gttatgagaa gctgca 26

<210> SEQ ID NO 34
<211> LENGTH: 22
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 34

acagggttag ctggtgaatg ga 22

<210> SEQ ID NO 35
<211> LENGTH: 19
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 35

tcggagcaca ggactacag 19

<210> SEQ ID NO 36
<211> LENGTH: 25
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 36

caagaggaac tgtgtccagg aaage 25

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<210> SEQ ID NO 37
<211> LENGTH: 22
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 37

agcgtgcctc acctaacctc ta

22

<210> SEQ ID NO 38
<211> LENGTH: 19
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 38

gcacagcact gtaaaggca

19

<210> SEQ ID NO 39
<211> LENGTH: 29
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 39

acggaactcg aaggaattgg tattgttgt

29

<210> SEQ ID NO 40
<211> LENGTH: 23
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 40

acacagctat gggagaaaga ctg

23

<210> SEQ ID NO 41
<211> LENGTH: 23
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 41

ccggttggcg agtgcgcatg cac

23

<210> SEQ ID NO 42
<211> LENGTH: 23
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 42

ccttcgggtt ggcgagtgcg cat

23

<210> SEQ ID NO 43
<211> LENGTH: 23
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

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<400> SEQUENCE: 43
ccaagggtgc gttgatggat cta 23

<210> SEQ ID NO 44
<211> LENGTH: 23
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 44
gaatgccaaag ggtgcgttga tgg 23

<210> SEQ ID NO 45
<211> LENGTH: 23
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 45
ctgggtgatt gggactgaga aag 23

<210> SEQ ID NO 46
<211> LENGTH: 24
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 46
cagccttgaa agtatggctt gggc 24

<210> SEQ ID NO 47
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 47
gcactcgcca accggaag 18

<210> SEQ ID NO 48
<211> LENGTH: 23
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 48
gaccagctca cccttactta tgg 23

<210> SEQ ID NO 49
<211> LENGTH: 23
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 49
actgaatgcc aagggtgcgt tga 23

<210> SEQ ID NO 50

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<211> LENGTH: 23
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 50

tcttggaat ccgctgaaga gtt 23

<210> SEQ ID NO 51
 <211> LENGTH: 31
 <212> TYPE: PRT
 <213> ORGANISM: Mus musculus

<400> SEQUENCE: 51

Met Ser Pro Trp Leu Lys Trp His Gly Pro Ala Met Ala Arg Leu Trp
 1 5 10 15

Gly Leu Cys Leu Leu Val Leu Gly Phe Trp Arg Ala Ser Leu Ala
 20 25 30

<210> SEQ ID NO 52
 <211> LENGTH: 398
 <212> TYPE: PRT
 <213> ORGANISM: Mus musculus

<400> SEQUENCE: 52

Cys Pro Thr Ser Cys Lys Cys Ser Ser Ala Arg Ile Trp Cys Thr Glu
 1 5 10 15

Pro Ser Pro Gly Ile Val Ala Phe Pro Arg Leu Glu Pro Asn Ser Val
 20 25 30

Asp Pro Glu Asn Ile Thr Glu Ile Leu Ile Ala Asn Gln Lys Arg Leu
 35 40 45

Glu Ile Ile Asn Glu Asp Asp Val Glu Ala Tyr Val Gly Leu Arg Asn
 50 55 60

Leu Thr Ile Val Asp Ser Gly Leu Lys Phe Val Ala Tyr Lys Ala Phe
 65 70 75 80

Leu Lys Asn Ser Asn Leu Arg His Ile Asn Phe Thr Arg Asn Lys Leu
 85 90 95

Thr Ser Leu Ser Arg Arg His Phe Arg His Leu Asp Leu Ser Asp Leu
 100 105 110

Ile Leu Thr Gly Asn Pro Phe Thr Cys Ser Cys Asp Ile Met Trp Leu
 115 120 125

Lys Thr Leu Gln Glu Thr Lys Ser Ser Pro Asp Thr Gln Asp Leu Tyr
 130 135 140

Cys Leu Asn Glu Ser Ser Lys Asn Met Pro Leu Ala Asn Leu Gln Ile
 145 150 155 160

Pro Asn Cys Gly Leu Pro Ser Ala Arg Leu Ala Ala Pro Asn Leu Thr
 165 170 175

Val Glu Glu Gly Lys Ser Val Thr Leu Ser Cys Ser Val Gly Gly Asp
 180 185 190

Pro Leu Pro Thr Leu Tyr Trp Asp Val Gly Asn Leu Val Ser Lys His
 195 200 205

Met Asn Glu Thr Ser His Thr Gln Gly Ser Leu Arg Ile Thr Asn Ile
 210 215 220

Ser Ser Asp Asp Ser Gly Lys Gln Ile Ser Cys Val Ala Glu Asn Leu
 225 230 235 240

Val Gly Glu Asp Gln Asp Ser Val Asn Leu Thr Val His Phe Ala Pro
 245 250 255

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Thr Ile Thr Phe Leu Glu Ser Pro Thr Ser Asp His His Trp Cys Ile
 260 265 270
 Pro Phe Thr Val Arg Gly Asn Pro Lys Pro Ala Leu Gln Trp Phe Tyr
 275 280 285
 Asn Gly Ala Ile Leu Asn Glu Ser Lys Tyr Ile Cys Thr Lys Ile His
 290 295 300
 Val Thr Asn His Thr Glu Tyr His Gly Cys Leu Gln Leu Asp Asn Pro
 305 310 315 320
 Thr His Met Asn Asn Gly Asp Tyr Thr Leu Met Ala Lys Asn Glu Tyr
 325 330 335
 Gly Lys Asp Glu Arg Gln Ile Ser Ala His Phe Met Gly Arg Pro Gly
 340 345 350
 Val Asp Tyr Glu Thr Asn Pro Asn Tyr Pro Glu Val Leu Tyr Glu Asp
 355 360 365
 Trp Thr Thr Pro Thr Asp Ile Gly Asp Thr Thr Asn Lys Ser Asn Glu
 370 375 380
 Ile Pro Ser Thr Asp Val Ala Asp Gln Ser Asn Arg Glu His
 385 390 395

<210> SEQ ID NO 53
 <211> LENGTH: 24
 <212> TYPE: PRT
 <213> ORGANISM: Mus musculus

<400> SEQUENCE: 53

Leu Ser Val Tyr Ala Val Val Val Ile Ala Ser Val Val Gly Phe Cys
 1 5 10 15
 Leu Leu Val Met Leu Leu Leu Leu
 20

<210> SEQ ID NO 54
 <211> LENGTH: 368
 <212> TYPE: PRT
 <213> ORGANISM: Mus musculus

<400> SEQUENCE: 54

Lys Leu Ala Arg His Ser Lys Phe Gly Met Lys Gly Pro Ala Ser Val
 1 5 10 15
 Ile Ser Asn Asp Asp Asp Ser Ala Ser Pro Leu His His Ile Ser Asn
 20 25 30
 Gly Ser Asn Thr Pro Ser Ser Ser Glu Gly Gly Pro Asp Ala Val Ile
 35 40 45
 Ile Gly Met Thr Lys Ile Pro Val Ile Glu Asn Pro Gln Tyr Phe Gly
 50 55 60
 Ile Thr Asn Ser Gln Leu Lys Pro Asp Thr Phe Val Gln His Ile Lys
 65 70 75 80
 Arg His Asn Ile Val Leu Lys Arg Glu Leu Gly Glu Gly Ala Phe Gly
 85 90 95
 Lys Val Phe Leu Ala Glu Cys Tyr Asn Leu Cys Pro Glu Gln Asp Lys
 100 105 110
 Ile Leu Val Ala Val Lys Thr Leu Lys Asp Ala Ser Asp Asn Ala Arg
 115 120 125
 Lys Asp Phe His Arg Glu Ala Glu Leu Leu Thr Asn Leu Gln His Glu
 130 135 140
 His Ile Val Lys Phe Tyr Gly Val Cys Val Glu Gly Asp Pro Leu Ile
 145 150 155 160

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Met Val Phe Glu Tyr Met Lys His Gly Asp Leu Asn Lys Phe Leu Arg
      165                      170                      175

Ala His Gly Pro Asp Ala Val Leu Met Ala Glu Gly Asn Pro Pro Thr
      180                      185                      190

Glu Leu Thr Gln Ser Gln Met Leu His Ile Ala Gln Gln Ile Ala Ala
      195                      200                      205

Gly Met Val Tyr Leu Ala Ser Gln His Phe Val His Arg Asp Leu Ala
      210                      215                      220

Thr Arg Asn Cys Leu Val Gly Glu Asn Leu Leu Val Lys Ile Gly Asp
      225                      230                      235                      240

Phe Gly Met Ser Arg Asp Val Tyr Ser Thr Asp Tyr Tyr Arg Val Gly
      245                      250                      255

Gly His Thr Met Leu Pro Ile Arg Trp Met Pro Pro Glu Ser Ile Met
      260                      265                      270

Tyr Arg Lys Phe Thr Thr Glu Ser Asp Val Trp Ser Leu Gly Val Val
      275                      280                      285

Leu Trp Glu Ile Phe Thr Tyr Gly Lys Gln Pro Trp Tyr Gln Leu Ser
      290                      295                      300

Asn Asn Glu Val Ile Glu Cys Ile Thr Gln Gly Arg Val Leu Gln Arg
      305                      310                      315                      320

Pro Arg Thr Cys Pro Gln Glu Val Tyr Glu Leu Met Leu Gly Cys Trp
      325                      330                      335

Gln Arg Glu Pro His Thr Arg Lys Asn Ile Lys Ser Ile His Thr Leu
      340                      345                      350

Leu Gln Asn Leu Ala Lys Ala Ser Pro Val Tyr Leu Asp Ile Leu Gly
      355                      360                      365

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<210> SEQ ID NO 55
<211> LENGTH: 31
<212> TYPE: PRT
<213> ORGANISM: Rattus norvegicus

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<400> SEQUENCE: 55

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Met Ser Pro Trp Pro Arg Trp His Gly Pro Ala Met Ala Arg Leu Trp
1      5      10      15

Gly Leu Cys Leu Leu Val Leu Gly Phe Trp Arg Ala Ser Leu Ala
      20      25      30

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<210> SEQ ID NO 56
<211> LENGTH: 398
<212> TYPE: PRT
<213> ORGANISM: Rattus norvegicus

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<400> SEQUENCE: 56

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Cys Pro Met Ser Cys Lys Cys Ser Thr Thr Arg Ile Trp Cys Thr Glu
1      5      10      15

Pro Ser Pro Gly Ile Val Ala Phe Pro Arg Leu Glu Pro Asn Ser Ile
      20      25      30

Asp Pro Glu Asn Ile Thr Glu Ile Leu Ile Ala Asn Gln Lys Arg Leu
      35      40      45

Glu Ile Ile Asn Glu Asp Asp Val Glu Ala Tyr Val Gly Leu Lys Asn
      50      55      60

Leu Thr Ile Val Asp Ser Gly Leu Lys Phe Val Ala Tyr Lys Ala Phe
      65      70      75      80

Leu Lys Asn Gly Asn Leu Arg His Ile Asn Phe Thr Arg Asn Lys Leu
      85      90      95

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Thr Ser Leu Ser Arg Arg His Phe Arg His Leu Asp Leu Ser Asp Leu
 100 105 110
 Ile Leu Thr Gly Asn Pro Phe Thr Cys Ser Cys Asp Ile Met Trp Leu
 115 120 125
 Lys Thr Leu Gln Glu Thr Lys Ser Ser Pro Asp Thr Gln Asp Leu Tyr
 130 135 140
 Cys Leu Asn Glu Ser Ser Lys Asn Thr Pro Leu Ala Asn Leu Gln Ile
 145 150 155 160
 Pro Asn Cys Gly Leu Pro Ser Ala Arg Leu Ala Ala Pro Asn Leu Thr
 165 170 175
 Val Glu Glu Gly Lys Ser Val Thr Ile Ser Cys Ser Val Gly Gly Asp
 180 185 190
 Pro Leu Pro Thr Leu Tyr Trp Asp Val Gly Asn Leu Val Ser Lys His
 195 200 205
 Met Asn Glu Thr Ser His Thr Gln Gly Ser Leu Arg Ile Thr Asn Ile
 210 215 220
 Ser Ser Asp Asp Ser Gly Lys Gln Ile Ser Cys Val Ala Glu Asn Leu
 225 230 235 240
 Val Gly Glu Asp Gln Asp Ser Val Asn Leu Thr Val His Phe Ala Pro
 245 250 255
 Thr Ile Thr Phe Leu Glu Ser Pro Thr Ser Asp His His Trp Cys Ile
 260 265 270
 Pro Phe Thr Val Arg Gly Asn Pro Lys Pro Ala Leu Gln Trp Phe Tyr
 275 280 285
 Asn Gly Ala Ile Leu Asn Glu Ser Lys Tyr Ile Cys Thr Lys Ile His
 290 295 300
 Val Thr Asn His Thr Glu Tyr His Gly Cys Leu Gln Leu Asp Asn Pro
 305 310 315 320
 Thr His Met Asn Asn Gly Asp Tyr Thr Leu Met Ala Lys Asn Glu Tyr
 325 330 335
 Gly Lys Asp Glu Arg Gln Ile Ser Ala His Phe Met Gly Arg Pro Gly
 340 345 350
 Val Asp Tyr Glu Thr Asn Pro Asn Tyr Pro Glu Val Leu Tyr Glu Asp
 355 360 365
 Trp Thr Thr Pro Thr Asp Ile Gly Asp Thr Thr Asn Lys Ser Asn Glu
 370 375 380
 Ile Pro Ser Thr Asp Val Ala Asp Gln Thr Asn Arg Glu His
 385 390 395

<210> SEQ ID NO 57
 <211> LENGTH: 24
 <212> TYPE: PRT
 <213> ORGANISM: Rattus norvegicus

<400> SEQUENCE: 57

Leu Ser Val Tyr Ala Val Val Val Ile Ala Ser Val Val Gly Phe Cys
 1 5 10 15
 Leu Leu Val Met Leu Leu Leu Leu
 20

<210> SEQ ID NO 58
 <211> LENGTH: 368
 <212> TYPE: PRT
 <213> ORGANISM: Rattus norvegicus

<400> SEQUENCE: 58

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Lys Leu Ala Arg His Ser Lys Phe Gly Met Lys Gly Pro Ala Ser Val
1           5           10           15
Ile Ser Asn Asp Asp Asp Ser Ala Ser Pro Leu His His Ile Ser Asn
20           25           30
Gly Ser Asn Thr Pro Ser Ser Ser Glu Gly Gly Pro Asp Ala Val Ile
35           40           45
Ile Gly Met Thr Lys Ile Pro Val Ile Glu Asn Pro Gln Tyr Phe Gly
50           55           60
Ile Thr Asn Ser Gln Leu Lys Pro Asp Thr Phe Val Gln His Ile Lys
65           70           75           80
Arg His Asn Ile Val Leu Lys Arg Glu Leu Gly Glu Gly Ala Phe Gly
85           90           95
Lys Val Phe Leu Ala Glu Cys Tyr Asn Leu Cys Pro Glu Gln Asp Lys
100          105          110
Ile Leu Val Ala Val Lys Thr Leu Lys Asp Ala Ser Asp Asn Ala Arg
115          120          125
Lys Asp Phe His Arg Glu Ala Glu Leu Leu Thr Asn Leu Gln His Glu
130          135          140
His Ile Val Lys Phe Tyr Gly Val Cys Val Glu Gly Asp Pro Leu Ile
145          150          155          160
Met Val Phe Glu Tyr Met Lys His Gly Asp Leu Asn Lys Phe Leu Arg
165          170          175
Ala His Gly Pro Asp Ala Val Leu Met Ala Glu Gly Asn Pro Pro Thr
180          185          190
Glu Leu Thr Gln Ser Gln Met Leu His Ile Ala Gln Gln Ile Ala Ala
195          200          205
Gly Met Val Tyr Leu Ala Ser Gln His Phe Val His Arg Asp Leu Ala
210          215          220
Thr Arg Asn Cys Leu Val Gly Glu Asn Leu Leu Val Lys Ile Gly Asp
225          230          235          240
Phe Gly Met Ser Arg Asp Val Tyr Ser Thr Asp Tyr Tyr Arg Val Gly
245          250          255
Gly His Thr Met Leu Pro Ile Arg Trp Met Pro Pro Glu Ser Ile Met
260          265          270
Tyr Arg Lys Phe Thr Thr Glu Ser Asp Val Trp Ser Leu Gly Val Val
275          280          285
Leu Trp Glu Ile Phe Thr Tyr Gly Lys Gln Pro Trp Tyr Gln Leu Ser
290          295          300
Asn Asn Glu Val Ile Glu Cys Ile Thr Gln Gly Arg Val Leu Gln Arg
305          310          315          320
Pro Arg Thr Cys Pro Gln Glu Val Tyr Glu Leu Met Leu Gly Cys Trp
325          330          335
Gln Arg Glu Pro His Thr Arg Lys Asn Ile Lys Asn Ile His Thr Leu
340          345          350
Leu Gln Asn Leu Ala Lys Ala Ser Pro Val Tyr Leu Asp Ile Leu Gly
355          360          365

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<210> SEQ ID NO 59

<211> LENGTH: 31

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 59

Met Ser Ser Trp Ile Arg Trp His Gly Pro Ala Met Ala Arg Leu Trp

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1	5	10	15
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Gly Phe Cys Trp Leu Val Val Gly Phe Trp Arg Ala Ala Phe Ala
 20 25 30

<210> SEQ ID NO 60
 <211> LENGTH: 399
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 60

Cys Pro Thr Ser Cys Lys Cys Ser Ala Ser Arg Ile Trp Cys Ser Asp
 1 5 10 15

Pro Ser Pro Gly Ile Val Ala Phe Pro Arg Leu Glu Pro Asn Ser Val
 20 25 30

Asp Pro Glu Asn Ile Thr Glu Ile Phe Ile Ala Asn Gln Lys Arg Leu
 35 40 45

Glu Ile Ile Asn Glu Asp Asp Val Glu Ala Tyr Val Gly Leu Arg Asn
 50 55 60

Leu Thr Ile Val Asp Ser Gly Leu Lys Phe Val Ala His Lys Ala Phe
 65 70 75 80

Leu Lys Asn Ser Asn Leu Gln His Ile Asn Phe Thr Arg Asn Lys Leu
 85 90 95

Thr Ser Leu Ser Arg Lys His Phe Arg His Leu Asp Leu Ser Glu Leu
 100 105 110

Ile Leu Val Gly Asn Pro Phe Thr Cys Ser Cys Asp Ile Met Trp Ile
 115 120 125

Lys Thr Leu Gln Glu Ala Lys Ser Ser Pro Asp Thr Gln Asp Leu Tyr
 130 135 140

Cys Leu Asn Glu Ser Ser Lys Asn Ile Pro Leu Ala Asn Leu Gln Ile
 145 150 155 160

Pro Asn Cys Gly Leu Pro Ser Ala Asn Leu Ala Ala Pro Asn Leu Thr
 165 170 175

Val Glu Glu Gly Lys Ser Ile Thr Leu Ser Cys Ser Val Ala Gly Asp
 180 185 190

Pro Val Pro Asn Met Tyr Trp Asp Val Gly Asn Leu Val Ser Lys His
 195 200 205

Met Asn Glu Thr Ser His Thr Gln Gly Ser Leu Arg Ile Thr Asn Ile
 210 215 220

Ser Ser Asp Asp Ser Gly Lys Gln Ile Ser Cys Val Ala Glu Asn Leu
 225 230 235 240

Val Gly Glu Asp Gln Asp Ser Val Asn Leu Thr Val His Phe Ala Pro
 245 250 255

Thr Ile Thr Phe Leu Glu Ser Pro Thr Ser Asp His His Trp Cys Ile
 260 265 270

Pro Phe Thr Val Lys Gly Asn Pro Lys Pro Ala Leu Gln Trp Phe Tyr
 275 280 285

Asn Gly Ala Ile Leu Asn Glu Ser Lys Tyr Ile Cys Thr Lys Ile His
 290 295 300

Val Thr Asn His Thr Glu Tyr His Gly Cys Leu Gln Leu Asp Asn Pro
 305 310 315 320

Thr His Met Asn Asn Gly Asp Tyr Thr Leu Ile Ala Lys Asn Glu Tyr
 325 330 335

Gly Lys Asp Glu Lys Gln Ile Ser Ala His Phe Met Gly Trp Pro Gly
 340 345 350

-continued

Ile Asp Asp Gly Ala Asn Pro Asn Tyr Pro Asp Val Ile Tyr Glu Asp
355 360 365

Tyr Gly Thr Ala Ala Asn Asp Ile Gly Asp Thr Thr Asn Arg Ser Asn
370 375 380

Glu Ile Pro Ser Thr Asp Val Thr Asp Lys Thr Gly Arg Glu His
385 390 395

<210> SEQ ID NO 61

<211> LENGTH: 24

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 61

Leu Ser Val Tyr Ala Val Val Val Ile Ala Ser Val Val Gly Phe Cys
1 5 10 15

Leu Leu Val Met Leu Phe Leu Leu
20

<210> SEQ ID NO 62

<211> LENGTH: 384

<212> TYPE: PRT

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 62

Lys Leu Ala Arg His Ser Lys Phe Gly Met Lys Asp Phe Ser Trp Phe
1 5 10 15

Gly Phe Gly Lys Val Lys Ser Arg Gln Gly Val Gly Pro Ala Ser Val
20 25 30

Ile Ser Asn Asp Asp Asp Ser Ala Ser Pro Leu His His Ile Ser Asn
35 40 45

Gly Ser Asn Thr Pro Ser Ser Ser Glu Gly Gly Pro Asp Ala Val Ile
50 55 60

Ile Gly Met Thr Lys Ile Pro Val Ile Glu Asn Pro Gln Tyr Phe Gly
65 70 75 80

Ile Thr Asn Ser Gln Leu Lys Pro Asp Thr Phe Val Gln His Ile Lys
85 90 95

Arg His Asn Ile Val Leu Lys Arg Glu Leu Gly Glu Gly Ala Phe Gly
100 105 110

Lys Val Phe Leu Ala Glu Cys Tyr Asn Leu Cys Pro Glu Gln Asp Lys
115 120 125

Ile Leu Val Ala Val Lys Thr Leu Lys Asp Ala Ser Asp Asn Ala Arg
130 135 140

Lys Asp Phe His Arg Glu Ala Glu Leu Leu Thr Asn Leu Gln His Glu
145 150 155 160

His Ile Val Lys Phe Tyr Gly Val Cys Val Glu Gly Asp Pro Leu Ile
165 170 175

Met Val Phe Glu Tyr Met Lys His Gly Asp Leu Asn Lys Phe Leu Arg
180 185 190

Ala His Gly Pro Asp Ala Val Leu Met Ala Glu Gly Asn Pro Pro Thr
195 200 205

Glu Leu Thr Gln Ser Gln Met Leu His Ile Ala Gln Gln Ile Ala Ala
210 215 220

Gly Met Val Tyr Leu Ala Ser Gln His Phe Val His Arg Asp Leu Ala
225 230 235 240

Thr Arg Asn Cys Leu Val Gly Glu Asn Leu Leu Val Lys Ile Gly Asp
245 250 255

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Phe	Gly	Met	Ser	Arg	Asp	Val	Tyr	Ser	Thr	Asp	Tyr	Tyr	Arg	Val	Gly
			260					265					270		
Gly	His	Thr	Met	Leu	Pro	Ile	Arg	Trp	Met	Pro	Pro	Glu	Ser	Ile	Met
		275					280					285			
Tyr	Arg	Lys	Phe	Thr	Thr	Glu	Ser	Asp	Val	Trp	Ser	Leu	Gly	Val	Val
	290					295					300				
Leu	Trp	Glu	Ile	Phe	Thr	Tyr	Gly	Lys	Gln	Pro	Trp	Tyr	Gln	Leu	Ser
305					310					315				320	
Asn	Asn	Glu	Val	Ile	Glu	Cys	Ile	Thr	Gln	Gly	Arg	Val	Leu	Gln	Arg
			325						330					335	
Pro	Arg	Thr	Cys	Pro	Gln	Glu	Val	Tyr	Glu	Leu	Met	Leu	Gly	Cys	Trp
			340					345					350		
Gln	Arg	Glu	Pro	His	Met	Arg	Lys	Asn	Ile	Lys	Gly	Ile	His	Thr	Leu
	355						360					365			
Leu	Gln	Asn	Leu	Ala	Lys	Ala	Ser	Pro	Val	Tyr	Leu	Asp	Ile	Leu	Gly
370						375					380				

<210> SEQ ID NO 63
 <211> LENGTH: 93
 <212> TYPE: DNA
 <213> ORGANISM: Mus musculus

<400> SEQUENCE: 63

atgtcgccct ggctgaagtg gcatggaccc gccatggcgc ggctctgggg cttatgcctg	60
ctggtcttgg gcttctggag ggctctctc gcc	93

<210> SEQ ID NO 64
 <211> LENGTH: 1194
 <212> TYPE: DNA
 <213> ORGANISM: Mus musculus

<400> SEQUENCE: 64

tgccccgacgt cctgcaaatg cagttccgct aggatttgggt gtactgagcc ttctccaggc	60
atcgtggcat tcccgaggtt ggaacctaac agcgttgacc cggagaacat cacggaaatt	120
ctcattgcaa accagaaaag gctagaaatc atcaatgaag atgacgttga agcttacgtg	180
gggctgagaa accttacaat tgtggattcc ggcttaaagt ttgtggctta caaagcgttt	240
ctgaaaaaca gcaacctgcg gcacataaat ttcacacgaa acaagctgac gagtttgtcc	300
aggagacatt tccgccacct tgacttgtct gacctgatcc tgacgggtaa tccgttcacg	360
tgctcctgcg acatcatgtg gctcaagact ctccaggaga ctaaattccag ccccgacact	420
caggatttgt actgctcaa tgagagcagc aagaacatgc cctggcgaa cctgcagata	480
cccaattgtg gtctgccatc tgcacgtctg gctgctccta acctaccgt ggaggaagga	540
aagtctgtga ccttttctcg cagtgtgggg ggtgaccac tccccacctt gtactgggac	600
gttgggaatt tggtttccaa gcacatgaat gaaacaagcc acacacaggg ctcttaagg	660
ataacgaaca tttcatctga tgacagtga aagcaaactc cttgtgtggc agaaaacctt	720
gtaggagaag atcaagattc tgtgaacctc actgtgcatt ttgcgccaac tatcagttt	780
ctcgagtctc caacctcaga tcaccactgg tgcattccat tcaactgtgag aggcaacccc	840
aagcctgcgc ttcagtgggt ctacaatggg gccatactga atgagtccaa gtacatctgt	900
actaagatcc acgtcaccaa tcacacggag taccatggct gcctccagct ggataacccc	960
actcatatga ataacggaga ctacacctg atggccaaga acgagtatgg gaaggatgag	1020
agacagatct ccgctcactt catgggccgg cctggagtcg actacagagac aaacccaat	1080

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taccctgaag tccttatga agactggacc acgccaaactg acattgggga tactacgaac 1140

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<210> SEQ ID NO 65

<211> LENGTH: 72

<212> TYPE: DNA

<213> ORGANISM: Mus musculus

<400> SEQUENCE: 65

ctctcgggtct atgcggtggt ggtgattgca tctgtggtgg gattctgctt gctggtgatg 60

ttgctctctgc tc 72

<210> SEQ ID NO 66

<211> LENGTH: 1104

<212> TYPE: DNA

<213> ORGANISM: Mus musculus

<400> SEQUENCE: 66

aagttggcga gacattccaa gtttggcatg aaaggcccag cttcgggtcat cagcaacgac 60

gatgactctg ccagcccctt ccaccacatc tccaatggga gtaacactcc atcttcttctg 120

gagggcggtc ccgacgtgtt cattattgga atgaccaaga ttctgttat tgaaaacccc 180

cagtactttg gcatcaccaa cagtcagctc aagccagaca catttgttca gcatatcaag 240

agacacaaca tcgttctgaa gagggaaactt ggggaaggag ccttcgggaa agttttcctt 300

gccgagtgtt acaacctctg ccagagcag gataagatcc tgggtggctgt gaagacgctg 360

aaggacgcca gcgacaatgc acgcaaggac tttcatcggtt aagctgagct gctgaccaac 420

ctccagcagc agcacattgt caagttctac ggtgtctgtg tggagggcga cccactcatc 480

atggtctttg agtacatgaa gcacggggac ctcaacaagt tccttagggc acacggggccc 540

gacgcagtgc tgatggcaga gggtaacccg ccacagagc tgacgcagtc gcagatgctg 600

cacatcgctc agcaaatcgc agcaggtatg gtctacctgg cgtcccaaca ctttgtgcac 660

cgtgacctgg ccccccgaa ctgcctggtg ggagagaacc tgctggtgaa aattggggac 720

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tatcagctat cgaacaatga ggtgatagag tgcacacccc aggggaagagt ccttcagcgg 960

cctcgaacgt gtccccagga ggtgatagag ctcatgcttg gatgctggca gcgggaacca 1020

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cccgtctacc tggatatact aggc 1104

<210> SEQ ID NO 67

<211> LENGTH: 93

<212> TYPE: DNA

<213> ORGANISM: Rattus norvegicus

<400> SEQUENCE: 67

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ctggtcttgg gcttctggag ggcttctctt gcc 93

<210> SEQ ID NO 68

<211> LENGTH: 1194

<212> TYPE: DNA

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<213> ORGANISM: Rattus norvegicus

<400> SEQUENCE: 68

tgcccatgt cctgcaaatg cagcaccact aggatttgggt gtaccgagcc ttctcctggc	60
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ctcattgcaa accagaaaag gttagaaatc atcaatgaag atgatgtcga agcttacgtg	180
gggtgaaaa accttacaat tgtggattcc ggcttaaagt ttgtggctta caaggcgttt	240
ctgaagaacg gcaacctgcg gcacatcaat ttcactcgaa acaagctgac gagtttgtcc	300
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tgttcctgtg acatcatgtg gctcaagact ctccaggaga cgaaatccag ccccgacact	420
caggatttgt attgectcaa tgagagcagc aagaataccc ctctggcgaa cctgcagatt	480
cccaattgtg gtctgcgctc tgcacgtctg gccgctccta acctcacggt ggaggaagg	540
aagtctgtga ccatttctcg cagcgtcggg ggtgaccgcg tccccacett gtactgggac	600
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tacctgaag tcctctatga agactggacc acgccaactg acatcgggga tactacaaac	1140
aaaagtaatg agatcccctc cacggatggt gctgaccaa ccaatcggga gcat	1194

<210> SEQ ID NO 69

<211> LENGTH: 72

<212> TYPE: DNA

<213> ORGANISM: Rattus norvegicus

<400> SEQUENCE: 69

ctctcggctc atgccgtggg ggtgattgcc tctgtggtag gattctgcct gctgggtgatg	60
ctgcttctgc tc	72

<210> SEQ ID NO 70

<211> LENGTH: 1104

<212> TYPE: DNA

<213> ORGANISM: Rattus norvegicus

<400> SEQUENCE: 70

aagttggcga gacattccaa gtttggcatg aaaggcccag cttccgtcat cagcaacgac	60
gatgactctg ccagccctct ccaccacatc tccaacggga gcaacactcc gtcttcttcg	120
gagggcgggc cggatgctgt catcattggg atgaccaaga tccctgtcat tgaaaacccc	180
cagtacttcg gtatcaccaa cagccagctc aagccggaca catttgttca gcacatcaag	240
agacacaaca tcgttctgaa gagggagctt ggagaaggag cctttgggaa agttttccta	300
gctggagtgt ataacctctg ccccgagcag gataagatcc tgggtggcgt gaagacgctg	360
aaggacgcca gcgacaatgc tcgcaaggac tttcatcgcg aagccgagct gctgaccaac	420
ctccagcacg agcacattgt caagttctac ggtgtctgtg tggagggcga cccactcatc	480

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atggtctttg agtacatgaa gcacggggac ctcaacaagt tccttagggc acacggggcca 540
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cacatcgctc agcaaatcgc agcaggcatg gtctacctgg catcccaaca cttcgtgcac 660
cgagacctgg ccacccggaa ctgcttggtg ggagagaacc tgctgggtgaa aattggggac 720
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tatcagctat caaacaacga ggtgatagaa tgcatacccc agggcagagt ccttcagcgg 960
cctcgcacgt gtccccagga ggtgtacgag ctgatgctgg gatgctggca gcgggaacca 1020
cacacaagga agaacatcaa gaacatccac acactccttc agaacttggc gaaggcgtcg 1080
cccgctctacc tggacatcct aggc 1104

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<210> SEQ ID NO 71
<211> LENGTH: 93
<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

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<400> SEQUENCE: 71
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<210> SEQ ID NO 72
<211> LENGTH: 1197
<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

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<400> SEQUENCE: 72
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ttcatcgcaa accagaaaag gttagaaatc atcaacgaag atgatgttga agcttatgtg 180
ggactgagaa atctgacaat tgtggattct ggattaaaat ttgtggctca taaagcattt 240
ctgaaaaaca gcaacctgca gcacatcaat ttacccgaa acaactgac gagtttgtct 300
aggaaacatt tccgtcacc tgaactgtct gaactgatcc tgggtggcaa tccatttaca 360
tgctcctgtg acattatgtg gatcaagact ctccaagagg ctaaatccag tccagacact 420
caggatttgt actgctgaa tgaaagcagc aagaatattc ccctggcaaa cctgcagata 480
cccaattgtg gtttgccatc tgcaaatctg gccgcaccta acctcactgt ggaggaaagg 540
aagtctatca cattatcctg tagtgtggca ggtgatccgg ttctaatat gtattgggat 600
gttggtaacc tggtttccaa acatatgaat gaaacaagcc acacacaggg ctcttaagg 660
ataactaaca tttcatccga tgacagtggg aagcagatct cttgtgtggc ggaaaatctt 720
gtaggagaag atcaagattc tgtcaacctc actgtgcatt ttgcaccaac tatcacattt 780
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actaaaatac atgttaccaa tcacacggag taccacggct gcctcagct ggataatccc 960
actcacatga acaatgggga ctacactcta atagccaaga atgagtattg gaaggatgag 1020
aaacagattt ctgctcactt catgggctgg cctggaattg acgatgggtc aaacccaaat 1080

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tatcctgatg taatttatga agattatgga actgcagcga atgacatcgg ggacaccacg	1140
aacagaagta atgaatcccc ttccacagac gtcactgata aaaccggctcg ggaacat	1197

<210> SEQ ID NO 73
 <211> LENGTH: 72
 <212> TYPE: DNA
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 73

ctctcgggtct atgctgtggt ggtgattgct tctgtggtgg gattttgcct tttggtaatg	60
ctgtttctgc tt	72

<210> SEQ ID NO 74
 <211> LENGTH: 1152
 <212> TYPE: DNA
 <213> ORGANISM: Homo sapiens

<400> SEQUENCE: 74

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agcccactcc atcacatctc caatgggagt aacactccat cttcttcgga aggtggccca	180
gatgctgtca ttattggaat gaccaagatc cctgtcattg aaaatcccca gtactttggc	240
atcaccaaca gtcagctcaa gccagacaca tttgttcagc acatcaagcg acataacatt	300
gttttgaaaa gggagctagg cgaaggagcc tttggaaaag tgttcttagc tgaatgctat	360
aaacctctgtc ctgagcagga caagatcttg gtggcagtga agacctgaa ggatgccagt	420
gacaatgcac gcaaggactt ccaccgtgag gccagactcc tgaccaacct ccagcatgag	480
cacatcgtca agttctatgg cgtctgcgtg gagggcgacc ccctcatcat ggtctttgag	540
tacatgaagc atggggacct caacaagttc ctcagggcac acggccctga tgccgtgctg	600
atggctgagg gcaacccgcc cacggaactg acgcagtcgc agatgctgca tatagcccag	660
cagatcgccg cgggcatggt ctacctggcg tcccagcact tcgtgcaccg cgatttggcc	720
accaggaact gcctggctcg ggagaacttg ctggtgaaaa tcggggactt tgggatgtcc	780
cgggacgtgt acagcactga ctactacagg gtcggtggcc acacaatgct gccatttcgc	840
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aacaatgagg tgatagagtg tatcactcag ggccgagtcc tgcagcgacc ccgcacgtgc	1020
ccccaggagg tgtatgagct gatgctgggg tgctggcagc gagagcccca catgaggaag	1080
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gacattctag gc	1152

<210> SEQ ID NO 75
 <211> LENGTH: 822
 <212> TYPE: PRT
 <213> ORGANISM: Homo sapiens
 <220> FEATURE:
 <221> NAME/KEY: MISC_FEATURE
 <222> LOCATION: (1)..(31)
 <223> OTHER INFORMATION: Signal Peptide
 <220> FEATURE:
 <221> NAME/KEY: MISC_FEATURE
 <222> LOCATION: (32)..(430)
 <223> OTHER INFORMATION: Extracellular Domain
 <220> FEATURE:
 <221> NAME/KEY: MISC_FEATURE

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<222> LOCATION: (431)..(454)
<223> OTHER INFORMATION: Transmembrane Domain
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (455)..(822)
<223> OTHER INFORMATION: Cytoplasmic Domain

<400> SEQUENCE: 75

Met Ser Ser Trp Ile Arg Trp His Gly Pro Ala Met Ala Arg Leu Trp
1          5          10          15

Gly Phe Cys Trp Leu Val Val Gly Phe Trp Arg Ala Ala Phe Ala Cys
20          25          30

Pro Thr Ser Cys Lys Cys Ser Ala Ser Arg Ile Trp Cys Ser Asp Pro
35          40          45

Ser Pro Gly Ile Val Ala Phe Pro Arg Leu Glu Pro Asn Ser Val Asp
50          55          60

Pro Glu Asn Ile Thr Glu Ile Phe Ile Ala Asn Gln Lys Arg Leu Glu
65          70          75          80

Ile Ile Asn Glu Asp Asp Val Glu Ala Tyr Val Gly Leu Arg Asn Leu
85          90          95

Thr Ile Val Asp Ser Gly Leu Lys Phe Val Ala His Lys Ala Phe Leu
100         105         110

Lys Asn Ser Asn Leu Gln His Ile Asn Phe Thr Arg Asn Lys Leu Thr
115         120         125

Ser Leu Ser Arg Lys His Phe Arg His Leu Asp Leu Ser Glu Leu Ile
130         135         140

Leu Val Gly Asn Pro Phe Thr Cys Ser Cys Asp Ile Met Trp Ile Lys
145         150         155         160

Thr Leu Gln Glu Ala Lys Ser Ser Pro Asp Thr Gln Asp Leu Tyr Cys
165         170         175

Leu Asn Glu Ser Ser Lys Asn Ile Pro Leu Ala Asn Leu Gln Ile Pro
180         185         190

Asn Cys Gly Leu Pro Ser Ala Asn Leu Ala Ala Pro Asn Leu Thr Val
195         200         205

Glu Glu Gly Lys Ser Ile Thr Leu Ser Cys Ser Val Ala Gly Asp Pro
210         215         220

Val Pro Asn Met Tyr Trp Asp Val Gly Asn Leu Val Ser Lys His Met
225         230         235         240

Asn Glu Thr Ser His Thr Gln Gly Ser Leu Arg Ile Thr Asn Ile Ser
245         250         255

Ser Asp Asp Ser Gly Lys Gln Ile Ser Cys Val Ala Glu Asn Leu Val
260         265         270

Gly Glu Asp Gln Asp Ser Val Asn Leu Thr Val His Phe Ala Pro Thr
275         280         285

Ile Thr Phe Leu Glu Ser Pro Thr Ser Asp His His Trp Cys Ile Pro
290         295         300

Phe Thr Val Lys Gly Asn Pro Lys Pro Ala Leu Gln Trp Phe Tyr Asn
305         310         315         320

Gly Ala Ile Leu Asn Glu Ser Lys Tyr Ile Cys Thr Lys Ile His Val
325         330         335

Thr Asn His Thr Glu Tyr His Gly Cys Leu Gln Leu Asp Asn Pro Thr
340         345         350

His Met Asn Asn Gly Asp Tyr Thr Leu Ile Ala Lys Asn Glu Tyr Gly
355         360         365

Lys Asp Glu Lys Gln Ile Ser Ala His Phe Met Gly Trp Pro Gly Ile

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370	375	380
Asp Asp Gly Ala Asn Pro Asn Tyr Pro Asp Val Ile Tyr Glu Asp Tyr 385 390 395 400		
Gly Thr Ala Ala Asn Asp Ile Gly Asp Thr Thr Asn Arg Ser Asn Glu 405 410 415		
Ile Pro Ser Thr Asp Val Thr Asp Lys Thr Gly Arg Glu His Leu Ser 420 425 430		
Val Tyr Ala Val Val Val Ile Ala Ser Val Val Gly Phe Cys Leu Leu 435 440 445		
Val Met Leu Phe Leu Leu Lys Leu Ala Arg His Ser Lys Phe Gly Met 450 455 460		
Lys Gly Pro Ala Ser Val Ile Ser Asn Asp Asp Asp Ser Ala Ser Pro 465 470 475 480		
Leu His His Ile Ser Asn Gly Ser Asn Thr Pro Ser Ser Ser Glu Gly 485 490 495		
Gly Pro Asp Ala Val Ile Ile Gly Met Thr Lys Ile Pro Val Ile Glu 500 505 510		
Asn Pro Gln Tyr Phe Gly Ile Thr Asn Ser Gln Leu Lys Pro Asp Thr 515 520 525		
Phe Val Gln His Ile Lys Arg His Asn Ile Val Leu Lys Arg Glu Leu 530 535 540		
Gly Glu Gly Ala Phe Gly Lys Val Phe Leu Ala Glu Cys Tyr Asn Leu 545 550 555 560		
Cys Pro Glu Gln Asp Lys Ile Leu Val Ala Val Lys Thr Leu Lys Asp 565 570 575		
Ala Ser Asp Asn Ala Arg Lys Asp Phe His Arg Glu Ala Glu Leu Leu 580 585 590		
Thr Asn Leu Gln His Glu His Ile Val Lys Phe Tyr Gly Val Cys Val 595 600 605		
Glu Gly Asp Pro Leu Ile Met Val Phe Glu Tyr Met Lys His Gly Asp 610 615 620		
Leu Asn Lys Phe Leu Arg Ala His Gly Pro Asp Ala Val Leu Met Ala 625 630 635 640		
Glu Gly Asn Pro Pro Thr Glu Leu Thr Gln Ser Gln Met Leu His Ile 645 650 655		
Ala Gln Gln Ile Ala Ala Gly Met Val Tyr Leu Ala Ser Gln His Phe 660 665 670		
Val His Arg Asp Leu Ala Thr Arg Asn Cys Leu Val Gly Glu Asn Leu 675 680 685		
Leu Val Lys Ile Gly Asp Phe Gly Met Ser Arg Asp Val Tyr Ser Thr 690 695 700		
Asp Tyr Tyr Arg Val Gly Gly His Thr Met Leu Pro Ile Arg Trp Met 705 710 715 720		
Pro Pro Glu Ser Ile Met Tyr Arg Lys Phe Thr Thr Glu Ser Asp Val 725 730 735		
Trp Ser Leu Gly Val Val Leu Trp Glu Ile Phe Thr Tyr Gly Lys Gln 740 745 750		
Pro Trp Tyr Gln Leu Ser Asn Asn Glu Val Ile Glu Cys Ile Thr Gln 755 760 765		
Gly Arg Val Leu Gln Arg Pro Arg Thr Cys Pro Gln Glu Val Tyr Glu 770 775 780		
Leu Met Leu Gly Cys Trp Gln Arg Glu Pro His Met Arg Lys Asn Ile 785 790 795 800		

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Lys Gly Ile His Thr Leu Leu Gln Asn Leu Ala Lys Ala Ser Pro Val
 805 810 815

Tyr Leu Asp Ile Leu Gly
 820

<210> SEQ ID NO 76

<211> LENGTH: 8498

<212> TYPE: DNA

<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 76

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ggcccagatg ctgtcattat tggaatgacc aagatccctg tcattgaaaa tcccagtac   1920

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cagacgggct gagaggatga acatctttta actgcgctg gaggccacca agctgctctc	2940
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taaaaccaga gagaagaag atttattatg aaccgcaata tgggaggaac aaagacaacc	3540
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agacgtcgtt tccctgatag gtatagcaga tccataaaaa ggtatgactt atacaattag	4140
gggaagctaa tggagtattat tagctgagta tcaatgtctc tgcgtgttac ggtggtgatg	4200
ggttttaatg aatatggacc ctgaagcctg gaaatcctca tccacgtcga acccacagga	4260
ctgtgggaag ggcagaatca atccctaagg gaaaggaaac ctcaccctga gggcatcaca	4320

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tgcactcatg ttcagtgtac acagggtcaag tcccttgctc tgggctctag ttgggagagt	4380
ggtttcattc caagtgtact ccattgtcag tatgtgtttt ttgtttcctt cactccattc	4440
aaaaagtcaa aatacaaaat ttggcacagc atgccaacgg gaggtgtgc ccagaccaag	4500
cactggaagt gtgcttctag gcatagtcac ttgttttgca aaaagagggc tcaaatttaa	4560
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ttgtaatggt ccaggaaagg aacattgctt tctttttgtc tttcagcaca tttgtattat 6720
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ccaaatataa attggaatgt tggcagccca gtggcttgaa ggccaatgat gagcagtgcca 6840
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actaacagta tagaatgctg tccttttcaa agcgtcctaa cagcaggatt acctgggtcaa 6960
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tggccatata tcaactgttat aggaaatctc atgagaggaa tggctagtga cccaactctc 7200
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aaaaaagttt tgtttgtaaa tcactgtgacc agcttctctc aacctgacat ggaaagtctc 8400
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gagactaaaa atgaaaaaaa aaaaaaaaaa aaaaaaaa 8498

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<210> SEQ ID NO 77
<211> LENGTH: 2469
<212> TYPE: DNA
<213> ORGANISM: Homo sapiens
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (1)..(93)
<223> OTHER INFORMATION: Signal Peptide
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (94)..(1290)
<223> OTHER INFORMATION: Extracellular Domain
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (1291)..(1362)
<223> OTHER INFORMATION: Transmembrane Domain

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<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (1363)..(2466)
<223> OTHER INFORMATION: Cytoplasmic Domain

<400> SEQUENCE: 77

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tctcggatct ggtgcagcga ccttctcctt ggcatcgtgg catttccgag attggagcct      180
aacagtgtag atcctgagaa catcacgaa attttcatcg caaaccagaa aaggttagaa      240
atcatcaacg aagatgatgt tgaagcttat gtgggactga gaaatctgac aattgtggat      300
tctggattaa aatttgtggc tcataaagca tttctgaaaa acagcaacct gcagcacatc      360
aattttaccc gaaacaaact gacgagtttg tctaggaaac atttcgctca ccttgacttg      420
tctgaactga tcctgggtgg caatccattt acatgctcct gtgacattat gtggatcaag      480
actctccaag aggctaaatc cagtccagac actcaggatt tgtactgcct gaatgaaagc      540
agcaagaata ttccctgggc aaacctgcag ataccgaatt gtggtttgcc atctgcaaat      600
ctggccgcac ctaacctcac tgtggaggaa ggaaagtcta tcacattatc ctgtagtgtg      660
gcaggtgatc cggttcctaa tatgtattgg gatgttggtt acctggtttc caaacatatg      720
aatgaaacaa gccacacaca gggctcctta aggataacta acatttcate cgatgacagt      780
gggaagcaga tctcttgtgt ggcggaaaat cttgtaggag aagatcaaga ttctgtcaac      840
ctcactgtgc attttgcacc aactatcaca tttctcgaat ctccaacctc agaccaccac      900
tgggtgcattc cattcactgt gaaaggcaac cccaaaccag cgcttcagtg gttctataac      960
ggggcaatat tgaatgagtc caaatacatc tgtactaaaa tacatgttac caatcacacg     1020
gagtaccacg gctgcctcca gctggataat cccactcaca tgaacaatgg ggactacact     1080
ctaatagcc aagaatgagta tgggaaggat gagaacaga tttctgctca cttcatgggc     1140
tggcctggaa ttgacgatgg tgcaaaccca aattatcctg atgtaattta tgaagattat     1200
ggaaatgcag cgaatgacat cggggacacc acgaacagaa gtaatgaaat cccttcacac     1260
gacgtcactg ataaaaccgg tcgggaacat ctctcgggtc atgctgtggg ggtgattgcg     1320
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aacagtcagc tcaagccaga cacatttgtt cagcacatca agcgacataa cattgttctg     1620
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tgtcctgagc aggacaagat cttggtggca gtgaagacct tgaaggatgc cagtgacaat     1740
gcacgcaagg acttcaccg tgaggccgag ctcccgacca acctccagca tgagcacatc     1800
gtcaagttct atggcgtctg cgtggagggc gacccctca tcatggctct tgagtacatg     1860
aagcatgggg acctcaacaa gttcctcagg gcacacggcc ctgatgccgt gctgatggct     1920
gagggcaacc cgccacgga actgacgcag tcgcagatgc tgcatatagc ccagcagatc     1980
gccgcgggca tgggtctacct ggcgtcccag cacttcgtgc accgcgattt ggccaccagg     2040
aactgcctgg tcggggagaa cttgctggtg aaaatcgggg actttgggat gtcccgggac     2100
gtgtacagca ctgactacta cagggtcggg gccacacaa tgctgcccac tcgctggatg     2160

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cctccagaga gcatcatgta caggaaattc acgacggaaa gcgacgtctg gagectgggg 2220
gtcgtgttgt gggagatttt cacctatggc aaacagccct ggtaccagct gtcaaacaat 2280
gaggtgatag agtgtatcac tcagggccga gtctgcagc gacccgcac gtgccccag 2340
gaggtgatg agctgatgct ggggtgctgg cagcgagagc cccacatgag gaagaacatc 2400
aagggcatcc ataccctcct tcagaacttg gccaaaggcat ctccggtcta cctggacatt 2460
ctaggctag 2469

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<210> SEQ ID NO 78
<211> LENGTH: 354
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

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<400> SEQUENCE: 78

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caggtgcagc tgggtggagtc tgggggaggc gtggtccagc ctgggaggtc cctgagactc 60
tcctgtgcag cctctggatt cagcttcagt agctttggca tgcactgggt ccgccaggct 120
ccaggcaagg gactggagtg ggtgtcagtt atatcatatg atggaattaa tacatactat 180
acagactccg tgaaggccgc attcaccatc tccagagaca attccaagaa cacgctgtat 240
ctgcaaatga acggcctgag agctgaggac acggctcttt attactgtgt gcaaggggtca 300
attggaaccg tttttgaata ctggggccag ggaaccctgg tcaccgtctc ctca 354

```

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<210> SEQ ID NO 79
<211> LENGTH: 118
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

```

```

<400> SEQUENCE: 79

```

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Gln Val Gln Leu Val Glu Ser Gly Gly Val Val Gln Pro Gly Arg
1      5      10      15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Ser Phe Ser Ser Phe
20     25     30
Gly Met His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35     40     45
Ser Val Ile Ser Tyr Asp Gly Ile Asn Thr Tyr Tyr Thr Asp Ser Val
50     55     60
Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ser Lys Asn Thr Leu Tyr
65     70     75     80
Leu Gln Met Asn Gly Leu Arg Ala Glu Asp Thr Ala Leu Tyr Tyr Cys
85     90     95
Val Gln Gly Ser Ile Gly Thr Val Phe Glu Tyr Trp Gly Gln Gly Thr
100    105    110
Leu Val Thr Val Ser Ser
115

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```

<210> SEQ ID NO 80
<211> LENGTH: 24
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

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```

<400> SEQUENCE: 80

```

```

ggattcagct tcagtagctt tggc

```

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<210> SEQ ID NO 81
 <211> LENGTH: 8
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 81

Gly Phe Ser Phe Ser Ser Phe Gly
 1 5

<210> SEQ ID NO 82
 <211> LENGTH: 24
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 82

atatcatatg atggaattaa taca 24

<210> SEQ ID NO 83
 <211> LENGTH: 8
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 83

Ile Ser Tyr Asp Gly Ile Asn Thr
 1 5

<210> SEQ ID NO 84
 <211> LENGTH: 33
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 84

gtgcaagggt caattggaac cgtttttgaa tac 33

<210> SEQ ID NO 85
 <211> LENGTH: 11
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 85

Val Gln Gly Ser Ile Gly Thr Val Phe Glu Tyr
 1 5 10

<210> SEQ ID NO 86
 <211> LENGTH: 321
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 86

gacatccaga tgaccagtc tccatcctcc ctgtctgcat ctgtaggaga cagagtcacc 60

atcacttgcc gggcgagtc gggcattagc aattatttag cctgggtatca gcagaaacca 120

gggaaagttc ctaaactcct gatctatgct gcattccact tacaatcagg ggtcccatct 180

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cggttcgggtg gcagtggtatc tgggacagat ttcactctca ccacagcag cctgcagcct    240
gaagatgttg caacttatta ctgtcaaaag tataccagtg ccccatcac ttcgggcct    300
gggaccaaag tggatatcaa a                                           321

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<210> SEQ ID NO 87
<211> LENGTH: 107
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

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```

<400> SEQUENCE: 87

```

```

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1         5             10             15
Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Gly Ile Ser Asn Tyr
20        25             30
Leu Ala Trp Tyr Gln Gln Lys Pro Gly Lys Val Pro Lys Leu Leu Ile
35        40             45
Tyr Ala Ala Ser Thr Leu Gln Ser Gly Val Pro Ser Arg Phe Gly Gly
50        55             60
Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
65        70             75             80
Glu Asp Val Ala Thr Tyr Tyr Cys Gln Lys Tyr Thr Ser Ala Pro Phe
85        90             95
Thr Phe Gly Pro Gly Thr Lys Val Asp Ile Lys
100       105

```

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<210> SEQ ID NO 88
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

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<400> SEQUENCE: 88

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cagggcatta gcaattat                                           18

```

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<210> SEQ ID NO 89
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

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```

<400> SEQUENCE: 89

```

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Gln Gly Ile Ser Asn Tyr
1         5

```

```

<210> SEQ ID NO 90
<211> LENGTH: 9
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

```

```

<400> SEQUENCE: 90

```

```

gctgcatcc                                           9

```

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<210> SEQ ID NO 91
<211> LENGTH: 3
<212> TYPE: PRT

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<213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 91

Ala Ala Ser
 1

<210> SEQ ID NO 92
 <211> LENGTH: 27
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 92

caaaagtata ccagtgcgcc attcact

27

<210> SEQ ID NO 93
 <211> LENGTH: 9
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 93

Gln Lys Tyr Thr Ser Ala Pro Phe Thr
 1 5

<210> SEQ ID NO 94
 <211> LENGTH: 366
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 94

gagggtgcagc tgggtggagtc tggggggaggc ttggtccagc ctgggggggtc cctaaaaactc	60
tccctgtacag cctctggggt caccctcagt ggctctgtta ttcactgggt ccgccaggct	120
tccgggaaag ggctggagtg gattggccgt attagaaaca aggctaacag ttacgcgaca	180
gcatatggtg cgctgggtgac aggcagggtc accatctcca gagatgattc aaagaacacg	240
gcgtatctgc aaatgaacag cctgaaaacc gaggacacgg ccgtttatta ctgtactttc	300
ccgggtgtag tgggacgagg aggttttgac tactggggcc agggcaccct ggtcaccgtc	360
tcctca	366

<210> SEQ ID NO 95
 <211> LENGTH: 122
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 95

Glu Val Gln Leu Val Glu Ser Gly Gly Leu Val Gln Pro Gly Gly	
1 5 10 15	
Ser Leu Lys Leu Ser Cys Thr Ala Ser Gly Phe Thr Phe Ser Gly Ser	
20 25 30	
Val Ile His Trp Val Arg Gln Ala Ser Gly Lys Gly Leu Glu Trp Ile	
35 40 45	
Gly Arg Ile Arg Asn Lys Ala Asn Ser Tyr Ala Thr Ala Tyr Gly Ala	
50 55 60	

-continued

Ser Val Thr Gly Arg Phe Thr Ile Ser Arg Asp Asp Ser Lys Asn Thr
65 70 75 80

Ala Tyr Leu Gln Met Asn Ser Leu Lys Thr Glu Asp Thr Ala Val Tyr
85 90 95

Tyr Cys Thr Phe Pro Gly Val Val Gly Arg Gly Gly Phe Asp Tyr Trp
100 105 110

Gly Gln Gly Thr Leu Val Thr Val Ser Ser
115 120

<210> SEQ ID NO 96
<211> LENGTH: 24
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 96

gggttcacct tcagtggctc tggt

24

<210> SEQ ID NO 97
<211> LENGTH: 8
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 97

Gly Phe Thr Phe Ser Gly Ser Val
1 5

<210> SEQ ID NO 98
<211> LENGTH: 30
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 98

attagaaaca aggctaacag ttacgcgaca

30

<210> SEQ ID NO 99
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 99

Ile Arg Asn Lys Ala Asn Ser Tyr Ala Thr
1 5 10

<210> SEQ ID NO 100
<211> LENGTH: 39
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 100

actttcccggtgtgtagtggg acgaggaggt tttgactac

39

<210> SEQ ID NO 101
<211> LENGTH: 13
<212> TYPE: PRT

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<213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 101

Thr Phe Pro Gly Val Val Gly Arg Gly Gly Phe Asp Tyr
 1 5 10

<210> SEQ ID NO 102
 <211> LENGTH: 339
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 102

gacatcgtga tgacccagtc tccagactcc ctggctgtgt ctctgggcga gggggccacc 60
 atcaactgca tgtccagcca gagtgtttta ttcagctcca acaataagaa ctacttagct 120
 tggtagcaac agaaaccagg acagcctcct aagttgctct tttactgggc atctaccgg 180
 gaatccgggg tcctgaccg attcgggtgc agcgggtctg ggacagattt ctctctcacc 240
 atcaacagcc tgcagactga agatgtggca gtttattact gtctccaata ttatagtatt 300
 ccgtggacgt tcggccaagg gaccaagggtg gaaatcaaa 339

<210> SEQ ID NO 103
 <211> LENGTH: 113
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 103

Asp Ile Val Met Thr Gln Ser Pro Asp Ser Leu Ala Val Ser Leu Gly
 1 5 10 15
 Glu Gly Ala Thr Ile Asn Cys Met Ser Ser Gln Ser Val Leu Phe Ser
 20 25 30
 Ser Asn Asn Lys Asn Tyr Leu Ala Trp Tyr Gln Gln Lys Pro Gly Gln
 35 40 45
 Pro Pro Lys Leu Leu Phe Tyr Trp Ala Ser Thr Arg Glu Ser Gly Val
 50 55 60
 Pro Asp Arg Phe Gly Gly Ser Gly Ser Gly Thr Asp Phe Ser Leu Thr
 65 70 75 80
 Ile Asn Ser Leu Gln Thr Glu Asp Val Ala Val Tyr Tyr Cys Leu Gln
 85 90 95
 Tyr Tyr Ser Ile Pro Trp Thr Phe Gly Gln Gly Thr Lys Val Glu Ile
 100 105 110

Lys

<210> SEQ ID NO 104
 <211> LENGTH: 36
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 104

cagagtgttt tattcagctc caacaataag aactac 36

<210> SEQ ID NO 105
 <211> LENGTH: 12

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<212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 105

Gln Ser Val Leu Phe Ser Ser Asn Asn Lys Asn Tyr
 1 5 10

<210> SEQ ID NO 106
 <211> LENGTH: 9
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 106

tgggcatct

9

<210> SEQ ID NO 107
 <211> LENGTH: 3
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 107

Trp Ala Ser
 1

<210> SEQ ID NO 108
 <211> LENGTH: 27
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 108

ctccaatatt atagtattcc gtggacg

27

<210> SEQ ID NO 109
 <211> LENGTH: 9
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 109

Leu Gln Tyr Tyr Ser Ile Pro Trp Thr
 1 5

<210> SEQ ID NO 110
 <211> LENGTH: 381
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 110

gagggtgcagc tgggtggagtc tgggggaggc ttggtacagc ctggagggtc cctgacactc 60

tcctgtgcag cctctggatt caatttccgt gattatgaaa tgatctgggt ccgccagact 120

ccagggaagg ggctggagtg gatttcatac attagtaata gtggttatac catatactac 180

gcagactctg tgaagggccg attcaccatc tccagagaca acgccaggaa ctcaatatat 240

ctgcaagtga acagcctgag agccgaggac acggctgttt attactgttc gagacgtact 300

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actatgattc ggggcattag ggcgtactac tattacggtc tggacgtctg gggccaaggg 360

accacgggtca ccgtctcctc a 381

<210> SEQ ID NO 111
 <211> LENGTH: 127
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 111

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
 1 5 10 15

Ser Leu Thr Leu Ser Cys Ala Ala Ser Gly Phe Asn Phe Arg Asp Tyr
 20 25 30

Glu Met Ile Trp Val Arg Gln Thr Pro Gly Lys Gly Leu Glu Trp Ile
 35 40 45

Ser Tyr Ile Ser Asn Ser Gly Tyr Thr Ile Tyr Tyr Ala Asp Ser Val
 50 55 60

Lys Gly Arg Phe Thr Ile Ser Arg Asp Asn Ala Arg Asn Ser Ile Tyr
 65 70 75 80

Leu Gln Val Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
 85 90 95

Ser Arg Arg Thr Thr Met Ile Arg Gly Ile Arg Ala Tyr Tyr Tyr Tyr
 100 105 110

Gly Leu Asp Val Trp Gly Gln Gly Thr Thr Val Thr Val Ser Ser
 115 120 125

<210> SEQ ID NO 112
 <211> LENGTH: 24
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 112

ggattcaatt tccgtgatta tgaa 24

<210> SEQ ID NO 113
 <211> LENGTH: 8
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 113

Gly Phe Asn Phe Arg Asp Tyr Glu
 1 5

<210> SEQ ID NO 114
 <211> LENGTH: 24
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 114

attagtaata gtgggttatac cata 24

<210> SEQ ID NO 115
 <211> LENGTH: 8

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<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 115

Ile Ser Asn Ser Gly Tyr Thr Ile
1 5

<210> SEQ ID NO 116
<211> LENGTH: 60
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 116

tcgagacgta ctactatgat tcggggcatt agggcgctact actattacgg tctggacgtc 60

<210> SEQ ID NO 117
<211> LENGTH: 20
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 117

Ser Arg Arg Thr Thr Met Ile Arg Gly Ile Arg Ala Tyr Tyr Tyr Tyr
1 5 10 15

Gly Leu Asp Val
20

<210> SEQ ID NO 118
<211> LENGTH: 324
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 118

gacatccaga tgacccagtc tccatcctcc ctgtctgcat ctgtaggaga cagagtcacc 60
atcacttgcc gggcaagtca gagcattagc agctatttaa attggtatca gcagaaacca 120
gggaaagccc ctaagctcct gatctatgct gcattccagtt tgcaaagtgg ggtcccgta 180
aggttcagtg gcagtggatc tgggacagat ttcactctca ccacagcag tctgcaacct 240
gaagatttgg caacttacta ctgtcaacag agttacagta cccctccgat caccttcggc 300
caagggacac gactggagat taaa 324

<210> SEQ ID NO 119
<211> LENGTH: 108
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 119

Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala Ser Val Gly
1 5 10 15

Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Ser Ile Ser Ser Tyr
20 25 30

Leu Asn Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys Leu Leu Ile
35 40 45

-continued

Tyr Ala Ala Ser Ser Leu Gln Ser Gly Val Pro Ser Arg Phe Ser Gly
50 55 60

Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
65 70 75 80

Glu Asp Phe Ala Thr Tyr Tyr Cys Gln Gln Ser Tyr Ser Thr Pro Pro
85 90 95

Ile Thr Phe Gly Gln Gly Thr Arg Leu Glu Ile Lys
100 105

<210> SEQ ID NO 120
<211> LENGTH: 18
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 120

cagagcatta gcagctat

18

<210> SEQ ID NO 121
<211> LENGTH: 6
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 121

Gln Ser Ile Ser Ser Tyr
1 5

<210> SEQ ID NO 122
<211> LENGTH: 9
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 122

gctgcatcc

9

<210> SEQ ID NO 123
<211> LENGTH: 3
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 123

Ala Ala Ser
1

<210> SEQ ID NO 124
<211> LENGTH: 30
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 124

caacagagtt acagtacccc tccgatcacc

30

<210> SEQ ID NO 125
<211> LENGTH: 10
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence

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<220> FEATURE:

<223> OTHER INFORMATION: Synthetic

<400> SEQUENCE: 125

Gln Gln Ser Tyr Ser Thr Pro Pro Ile Thr
 1 5 10

We claim:

1. A genetically modified mouse whose genome comprises a genetically modified endogenous mouse TrkB locus comprising a humanized TrkB gene encoding a humanized tropomyosin receptor kinase B (TRKB) protein comprising:
 - an endogenous mouse TRKB protein signal peptide;
 - a human TRKB protein extracellular domain;
 - an endogenous mouse TRKB protein transmembrane domain; and
 - an endogenous mouse TRKB protein cytoplasmic domain,
 wherein the humanized TrkB gene is under the control of an endogenous mouse TrkB promoter, wherein the humanized TRKB protein comprises the amino acid sequence as set forth in SEQ ID NO: 4,
 wherein the humanized TrkB gene comprises a replacement of an endogenous mouse TrkB genomic sequence comprising a region starting from the codon in exon 2 encoding amino acid 32 through exon 10, including all introns between exons 2 and 10, with a corresponding human TrkB genomic sequence comprising a region starting from the codon in exon 2 encoding amino acid 32 through exon 10, including all introns between exons 2 and 10,
 wherein the genetically modified mouse expresses the humanized TRKB protein,
 wherein the genetically modified mouse is homozygous for the genetically modified endogenous mouse TrkB locus, and
 wherein the genetically modified mouse exhibits reduced (a) body weight and (b) total body fat mass upon treatment with a human TRKB agonist antibody relative to a control genetically modified mouse not treated with the antibody.
2. The genetically modified mouse of claim 1, wherein the human TRKB protein extracellular domain comprises the sequence set forth in SEQ ID NO: 60.
3. The genetically modified mouse of claim 1, wherein the human TRKB protein extracellular domain is encoded by the nucleotide sequence set forth in SEQ ID NO: 72.
4. The genetically modified mouse of claim 1, wherein the endogenous mouse TRKB protein signal peptide comprises the sequence set forth in SEQ ID NO: 51.
5. The genetically modified mouse of claim 1, wherein the endogenous mouse TRKB protein signal peptide is encoded by the nucleotide sequence set forth in SEQ ID NO: 63.

6. The genetically modified mouse of claim 1, wherein the endogenous mouse TRKB protein transmembrane domain comprises the sequence set forth in SEQ ID NO: 53.

7. The genetically modified mouse of claim 1, wherein the endogenous mouse TRKB protein transmembrane domain is encoded by the nucleotide sequence set forth in SEQ ID NO: 65.

8. The genetically modified mouse of claim 1, wherein the endogenous mouse TRKB protein cytoplasmic domain comprises the sequence set forth in SEQ ID NO: 54.

9. The genetically modified mouse of claim 1, wherein the endogenous mouse TRKB protein cytoplasmic domain is encoded by the nucleotide sequence set forth in SEQ ID NO: 66.

10. The genetically modified mouse of claim 1, wherein the endogenous mouse TRKB protein signal peptide comprises the sequence set forth in SEQ ID NO: 51, the endogenous mouse TRKB protein transmembrane domain comprises the sequence set forth in SEQ ID NO: 53, and the endogenous mouse TRKB protein cytoplasmic domain comprises the sequence set forth in SEQ ID NO: 54.

11. The genetically modified mouse of claim 1, wherein the endogenous mouse TRKB protein signal peptide is encoded by the nucleotide sequence set forth in SEQ ID NO: 63, the endogenous mouse TRKB protein transmembrane domain is encoded by the nucleotide sequence set forth in SEQ ID NO: 65, and the endogenous mouse TRKB protein cytoplasmic domain is encoded by the nucleotide sequence set forth in SEQ ID NO: 66.

12. The genetically modified mouse of claim 1, wherein the humanized TRKB protein is encoded by the nucleotide sequence set forth in SEQ ID NO: 12.

13. A method of assessing activity of a human-TRKB-targeting reagent in vivo, comprising:

- (a) administering the human-TRKB-targeting reagent to the genetically modified mouse of claim 1; and
- (b) assessing the activity of the human-TRKB-targeting reagent in the genetically modified mouse of step (a) as compared to an untreated control mouse.

14. A mouse cell isolated from the genetically modified mouse of claim 1, wherein the mouse cell's genome comprises the genetically modified endogenous mouse TrkB locus comprising the humanized TrkB gene encoding the humanized TRKB protein comprising the amino acid sequence as set forth in SEQ ID NO: 4, wherein the mouse cell expresses the humanized protein.

* * * * *